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**Senior Project Report**

# Nway's Love Vet Clinic Management System

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## Senior Project Approval

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The Senior Project committee's cooperation between the Department of Computer Science and Information Technology, Vincent Mary School of Engineering, Science and Technology, Assumption University had approved this Senior Project. The Senior Project in partial fulfilment of the requirement for the degree of Bachelor of Science in Computer Science and Information technology.

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# Abstract

Nway's Love Vet Clinic – Pet Clinic Management Application is a proposed digital management system designed to improve the daily operations of Nway's Love Vet Clinic ( ). The clinic has provided veterinary healthcare services for more than seven years and officially reached its seventh anniversary on 25 April 2026. As clinic services expand, manual processes for appointments, medical records, vaccinations, inpatient care, laboratory tests, grooming services, product sales, and inventory management can become difficult to maintain consistently.

The proposed system centralizes major clinic activities into web application. Pet owners can use the client-facing application to book appointments, manage pet profiles, review relevant information, and receive reminders. Veterinarians can access medical records, document diagnoses and treatments, and manage digital prescriptions. Clinic staff can manage queues, grooming and spa services, inventory, product sales, records, and reports. Administrators are responsible for user management and overall system access.

The proposed technology stack consists of Flutter for the application interface, Node.js for backend services and API logic, MySQL and Neon for database management, and Figma for UI/UX design. The system is intended to reduce paperwork, organize information, minimize manual errors, improve communication, and provide a more structured workflow for both clinic personnel and pet owners.

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# Chapter 1: Project Overview

## 1.1 Introduction

The Pet Clinic Management Application is a digital veterinary management system developed for Nway's Love Vet Clinic ( ). The purpose of the system is to improve clinic operations and provide better healthcare service management for pets.

Nway's Love Vet Clinic has been providing veterinary healthcare services for more than seven years and officially reached its seventh anniversary on 25 April 2026. As the clinic continues to expand its services, managing appointments, doctor schedules, pet medical records, vaccination information, inpatient care, laboratory results, grooming services, products, and inventory through manual processes can require considerable time and effort.

The proposed application digitalizes these activities and provides a centralized platform for pet owners, veterinarians, clinic staff, and administrators. The system is designed to support appointment booking, pet profile management, medical record management, queue management, emergency handling, home visits, products, inventory management, and reporting.

The system also aims to improve communication between the clinic and pet owners. Instead of depending mainly on paper records and repeated manual communication, important information can be organized digitally and accessed according to the user's role.

## 1.2 Clinic Information

*Table : 1 Clinic information*

| **Information** | **Details** |
|---|---|
| Client Name | Nway's Love Vet Clinic<br>( ) |
| Business Type | Veterinary Clinic |
| System Type | Web Application |
| Clinic Experience | 7 Years |
| Opening Hours | 9:00 AM – 12:00 PM / 4:00 PM – 7:00 PM |
| Operating Days | Open Daily |
| Official Facebook | -Vet Clinic<br>https://www.facebook.com/share/1D8rTb62Ke/?mibextid=wwXIfr |
| Location | Chinwin Street, Mingalar Dipa Quarter, Pobba Thiri Township, Naypyidaw, Myanmar |

The operating schedule and service information above are based on the client information provided for this project.

## 1.3 Problem Statement

The clinic currently faces several operational challenges associated with manual management processes. These challenges can affect the speed of service, consistency of records, and the ability of staff to coordinate different clinic activities.

- Manual appointment booking process.
- Difficulty managing doctor schedules and available time slots.
- Long waiting times for customers.
- Paper-based pet medical records.
- Risk of losing pet medical history and treatment records.
- Difficulty tracking vaccination schedules.
- No centralized reminder and notification system.
- Difficulty managing inpatient and day-care services.
- Manual recording of laboratory test results.
- Increased workload for clinic staff.
- Difficulty managing home-visit services.
- Manual inventory and product management.
- Limited communication between clinic staffs and pet owners.

## 1.4 Scope of the Project

The scope of the Nway's Love Vet Clinic Management System covers the development of a mobile and web-based application designed to support the main operational activities of the veterinary clinic. The system focuses on organizing clinic information, reducing manual work, and providing different functions according to the responsibilities of each user.

The system includes the following major areas:

- Appointment management, including appointment booking, cancellation, and rescheduling.
- Doctor schedule management and assignment based on treatment requirements.
- Queue management for normal and emergency cases.
- Emergency case handling, including priority over the normal queue.
- Home-visit management for eligible treatments.
- Pet profile management and digital medical records.
- Vaccination tracking and vaccination due-date management.
- Appointment and vaccination reminders.
- Billing and digital invoice management.
- Laboratory result recording.
- Grooming and spa service management.
- Product sales management for medicine, pet food, and accessories.
- Inventory management, including stock levels, expiry dates, and low-stock alerts.
- Reporting, including appointment, emergency, vaccination, service, sales, and inventory reports.
- Excel export for supported reports.
- Announcement and information-sharing functions.
- Role-based access for Pet Owners, Veterinarians, Clinic Staff, and Administrators.

The system is intended to provide a centralized platform where clinic personnel can manage operational information while pet owners can access relevant services and pet information. The proposed scope is based on the modules and functions identified in the project requirements. Senior Project

## 1.5 Objectives

- Digitalize clinic management operations.
- Simplify appointment booking processes.
- Manage pet medical records systematically.
- Improve communication between clinic staff and pet owners.
- Provide vaccination and appointment reminder systems.
- Reduce paperwork and manual errors.
- Improve doctor schedule management.
- Support inpatient and treatment management.
- Manage pet products efficiently.
- Improve overall clinic service quality and customer satisfaction.

# Chapter 2: Related Work and System Context

## 2.1 Current Operational Context

Nway's Love Vet Clinic currently handles several important clinic activities that require coordination between clinic staff, veterinarians, and pet owners. These activities include appointment booking, doctor scheduling, queue management, medical records, vaccination tracking, home visits, grooming and spa services, product showcase, and inventory management.

Some of these activities can require manual recording and communication. For example, appointments may initially be handled through phone calls or walk-in requests, while medical information and other clinic records may require staff to enter and update information manually. The clinic also needs to coordinate doctor availability because doctors have rotational off days and assignments may depend on the treatment or operation required.

The clinic also needs to manage different types of cases. Normal cases follow a first-come-first-served queue, while emergency cases can receive priority. Home-visit services are also available for certain treatments during the defined service period from 12:00  to 14:00.

In addition, pet healthcare information needs to be maintained over time. Medical records contain information such as breed, age, weight, and medical history, while vaccination records include vaccination information and the next vaccination date. The system therefore needs to provide authorized clinic personnel with organized access to this information.

## 2.2 Need for a Centralized System

The existing operational challenges demonstrate the need for a centralized management system. Managing appointments, medical records, vaccination information, inventory, and other services separately can increase the amount of manual work required by clinic staff.

A centralized system can bring these activities together and allow information to be stored in a structured manner. Appointment information can be connected with pet and veterinarian information, while medical records, vaccination records, invoices, products, and inventory can be managed through related system modules.

The proposed system also uses role-based access to separate the responsibilities of different users. Pet owners can access their pet-related services, veterinarians can work with clinical information, clinic staff can manage operational activities, and

administrators can manage users and system-level access. This helps ensure that each user can access the functions required for their responsibilities. Senior Project

By centralizing these activities, the system is expected to reduce paperwork, reduce repetitive manual data entry, improve information retrieval, and provide a more organized workflow for the clinic.

# Chapter 3: Proposed Methodology

## 3.1 Methodology

The development methodology for the Nway's Love Vet Clinic Management System is based on identifying the clinic's operational requirements and designing the system around those requirements. The development process focuses on understanding the existing workflow, identifying problems, defining system requirements, designing the system structure, and implementing the required functions.

The proposed development process can be divided into the following stages:

1. **Requirement Analysis**
   Identify the clinic's current activities, problems, stakeholders, and system requirements.
2. **System Design**
   Design the system architecture, database structure, user roles, system workflow, and main modules.
3. **UI/UX Design**
   Use Figma to design the interfaces, navigation flows, prototypes, and screen layouts before implementation.
4. **System Development**
   Develop the mobile and web interfaces, backend services, database, and required system functions.
5. **Testing and Evaluation**
   Test the implemented functions against the defined functional and non-functional requirements.

The methodology focuses on creating a system that is usable, reliable, secure, and capable of supporting the clinic's future expansion. The project requirements identify security, performance, usability, scalability, and reliability as important non-functional requirements.

## 3.2 Solving the Problem

The proposed system solves the clinic's operational problems by replacing fragmented manual activities with centralized digital processes. Instead of maintaining different records and

handling different activities separately, the system provides connected modules for appointments, pet information, medical records, vaccinations, billing, inventory, and reporting.For appointment management, the system provides digital appointment information and supports cancellation and rescheduling. Doctor schedules can also be managed according to availability and treatment requirements. This can help staff organize available time slots and reduce scheduling difficulties. Senior Project

For medical information, the system provides digital pet profiles and medical records. Authorized clinic personnel can record and access medical information, while pet owners cannot directly modify official medical records. Vaccination information can also be recorded and used to identify upcoming vaccination requirements. Senior Project

The system also addresses operational activities such as inventory and billing. Product information can include price, stock, and expiry date, while stock levels can be monitored and low-stock alerts can be supported. Digital invoices can include consultation fees, treatment charges, and product purchases. Senior Project

Through these functions, the system aims to reduce paperwork and manual errors while improving the organization and accessibility of clinic information.

## 3.3 Building the System – Key Features

The Nway's Love Vet Clinic Management System is organized into several modules. Each module focuses on a specific clinic activity while sharing information with other parts of the system.

The main features include:

- **Appointment Management:** Allows appointments to be booked, cancelled, and rescheduled according to clinic rules.
- **Doctor Management:** Supports doctor schedules and assignment based on availability and treatment requirements.
- **Queue and Emergency Management:** Maintains the normal queue while allowing emergency cases to receive priority.
- **Home Visit Management:** Manages eligible treatments provided at the pet owner's location.
- **Pet Management:** Stores pet profiles including breed, age, weight, and other relevant information.
- **Medical Record Management:** Allows authorized personnel to record diagnoses, treatments, medical history, and related clinical information.
- **Vaccination Management:** Records vaccination information and upcoming vaccination dates.
- **Reminder System:** Provides appointment and vaccination reminders.
- **Billing Management:** Generates digital invoices for consultation, treatment, and product charges.

- **Laboratory Management:** Records laboratory test results.
- **Grooming and Spa Management:** Supports the management of grooming and spa services.
- **Product Management:** Manages products such as medicine, pet food, and accessories.
- **Inventory Management:** Tracks stock, expiry dates, sales, and low-stock conditions.
- **Reporting:** Provides appointment, emergency, vaccination, service, sales, and inventory reports, with Excel export support.
- **Announcement System:** Provides a centralized method for sharing clinic announcements.

These features are designed to work together as one centralized system rather than as separate manual processes.

## 3.4 Technology and Architecture

The proposed system uses a client-server architecture consisting of the application interface, backend services, and database. This structure separates the user interface from the application logic and data storage, making the system easier to organize and maintain.

## 3.5 ROI Analysis

The Return on Investment (ROI) of the proposed Nway's Love Vet Clinic Management System is mainly based on improving operational efficiency and reducing manual work. The system can reduce paperwork, improve appointment and queue management, organize medical records, provide better inventory control, and make billing and reporting more efficient.

Although the system requires costs for development, maintenance, database services, staff training, and equipment, these costs can be balanced by the long-term benefits of reducing administrative workload and improving clinic operations.Overall, the system is expected to provide long-term value by saving time, reducing manual errors, improving information management, and increasing the quality and efficiency of services provided by the clinic.

# Chapter 4: Design of the System

## 4.1 Functional Requirements Specification

The functional requirements describe the main functions that the Nway's Love Vet Clinic Management System should provide. The system is designed to support pet owners, veterinarians, clinic staff, and administrators by organizing the clinic's daily operations in one centralized system.

### 4.1.1 Stakeholders

*Table : 2 Stakeholders*

| Stakeholder | Description |
|---|---|
| Pet Owners | Book appointments, view pet information, receive reminders, check invoices, and access available clinic services. |
| Veterinarians | Manage appointments, view pet medical records, update treatment information, and handle consultations and emergencies. |
| Clinic Staff | Manage appointments, queues, billing, products, inventory, vaccination information, and daily clinic operations. |
| Administrator | Manage system users, clinic information, system data, reports, and overall system operations. |

### 4.1.2 Actors and Responsibilities

*Table : 3 Actors and Responsibilities*

| Actor | Responsibilities |
|---|---|
| Pet Owner | Login, manage pet profiles, book or cancel appointments, view medical information, receive reminders, and browse products. |
| Veterinarian | View assigned appointments, access medical records, update treatment records, manage vaccination information, handle emergency cases and announcements. |
| Clinic Staff | Manage appointments, queues, home visits, products and inventory, vaccination records. |
| Administrator | Manage/register users, system information, staff access, reports, and overall system management. |

### 4.1.3 Use Cases

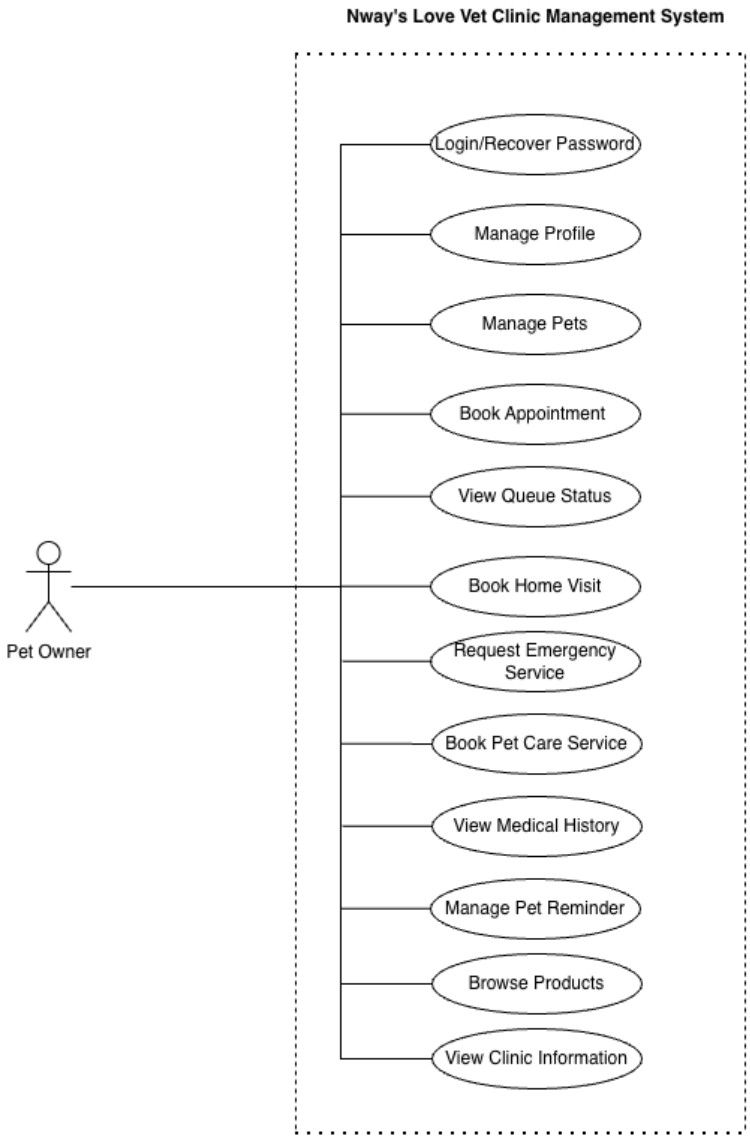



*Fig : 1 Use Case Diagram (Pet Owner)*

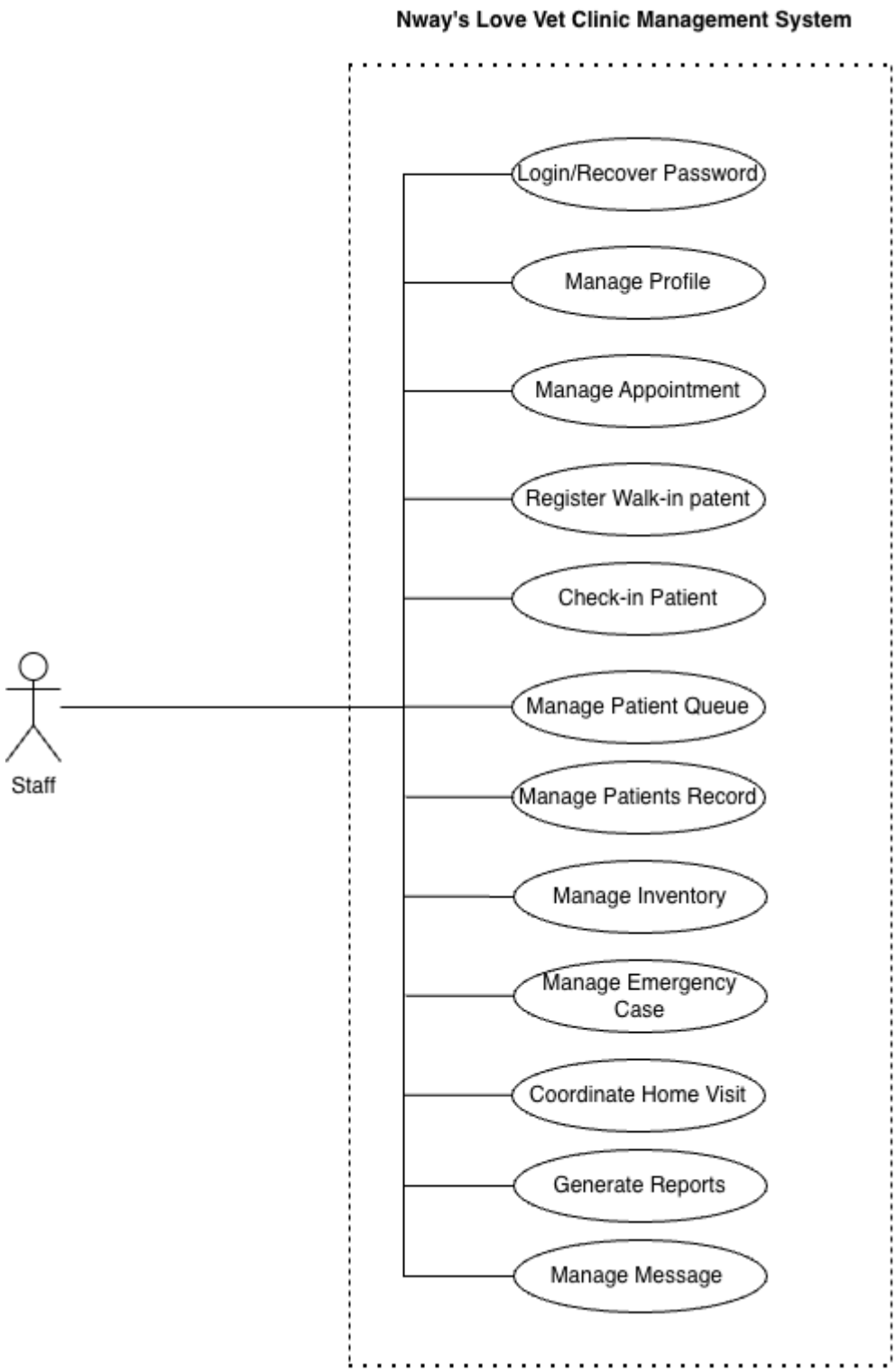



*Fig : 2 Use Case Diagram (Staff)*

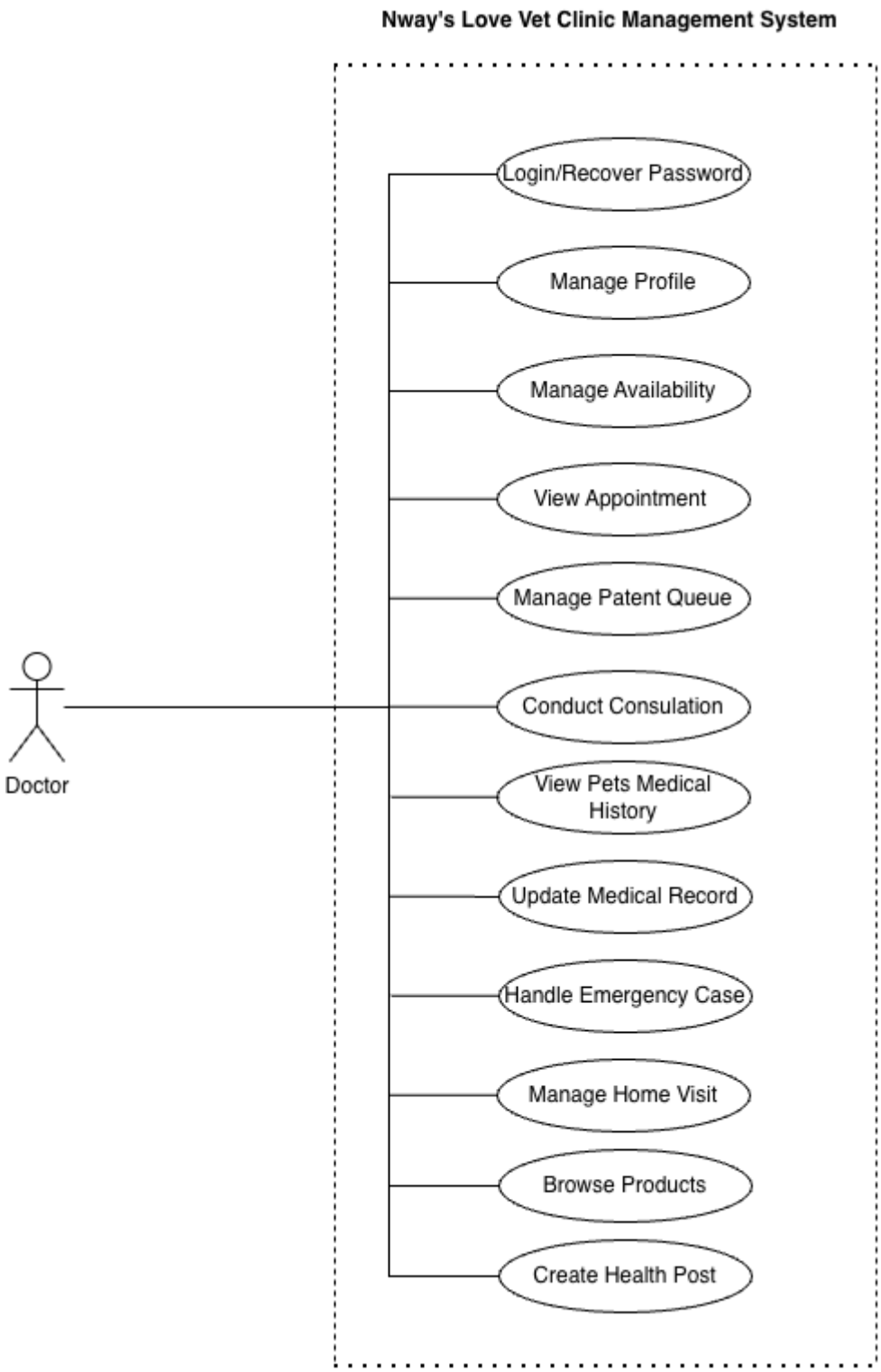



*Fig : 3 Use Case Diagram (Doctor)*

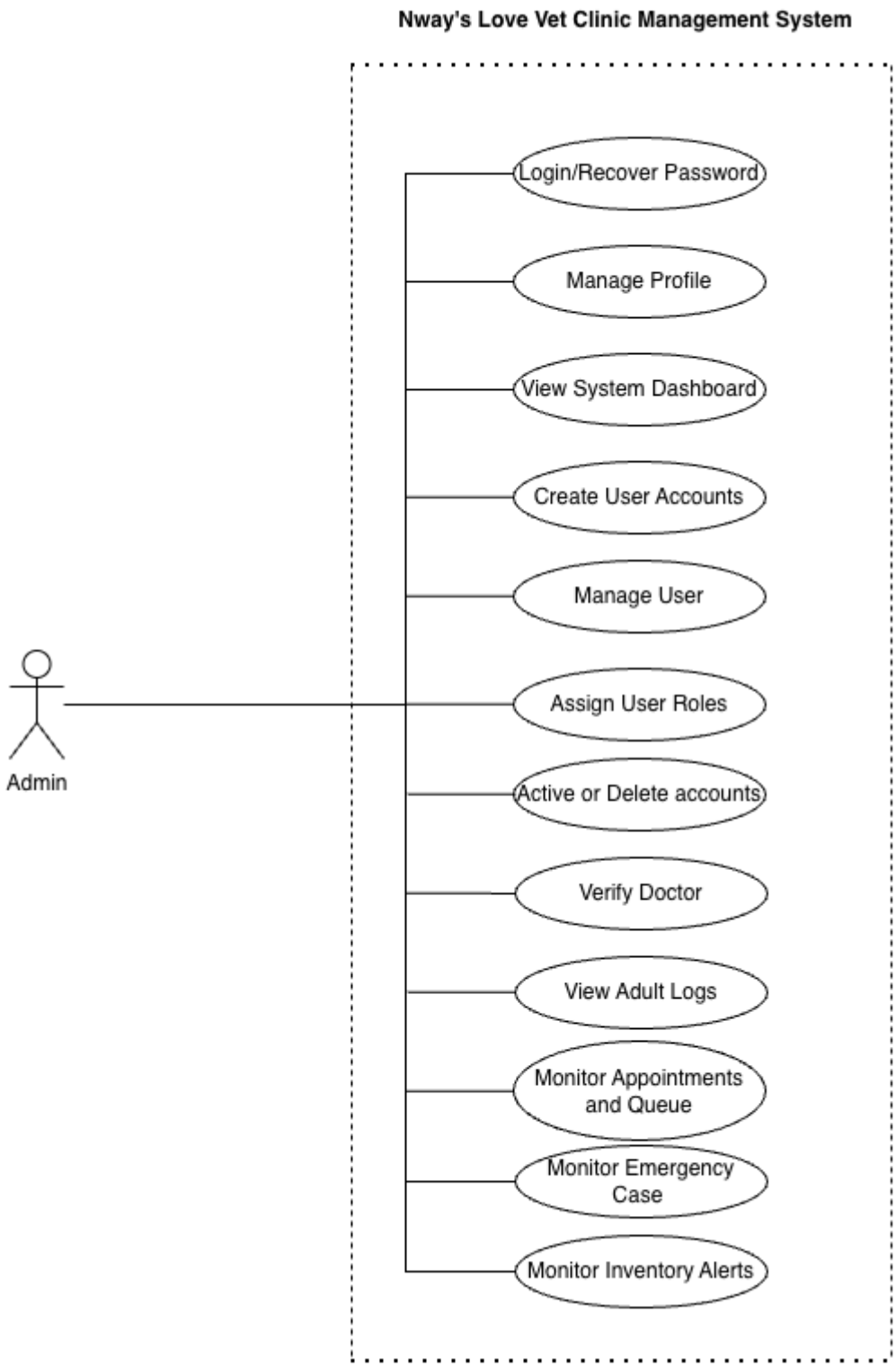



*Fig : 4 Use Case Diagram (Admin)*

## 4.2 System Design

The system design describes the structure, workflow, database, and interaction between the different components of the Nway's Love Vet Clinic Management System.

### 4.2.1 Data Flow / Workflow

The general workflow of the system begins when a user logs into the application. Based on the user's role, the system provides access to the appropriate functions.

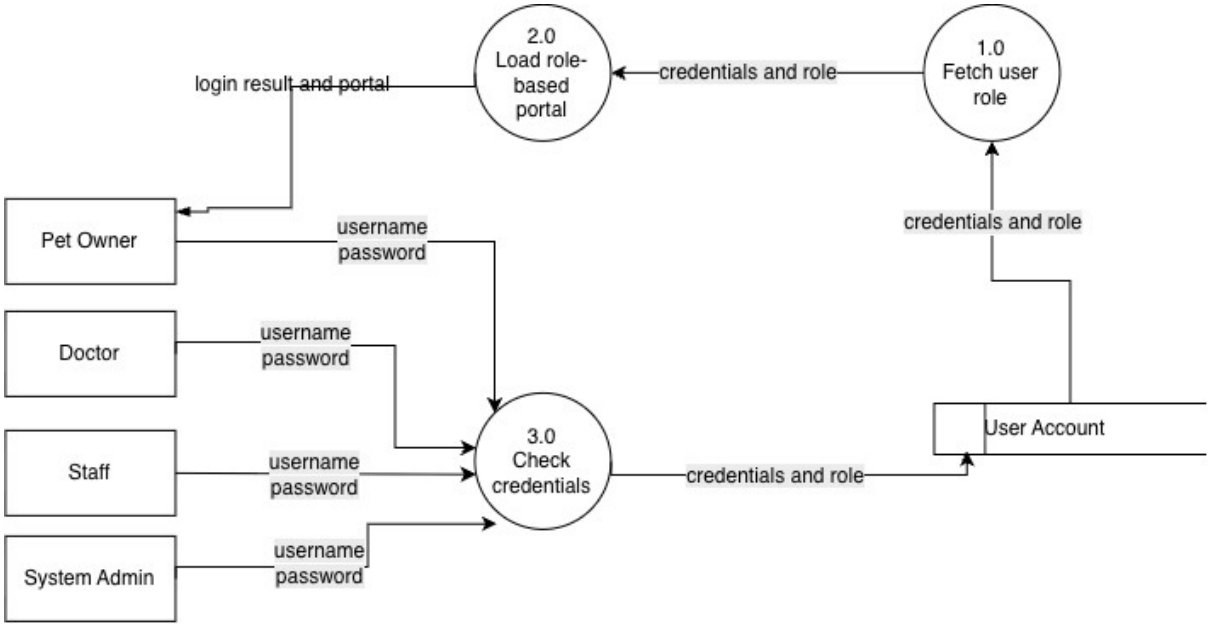



*Fig : 5 DFD for User Login and Role-Based Access*

A user enters a username, email, or phone number and password. The application validates the credentials against the user account data, identifies the account role and status, and opens the correct Pet Owner,Doctor, Staff, or System Admin portal.

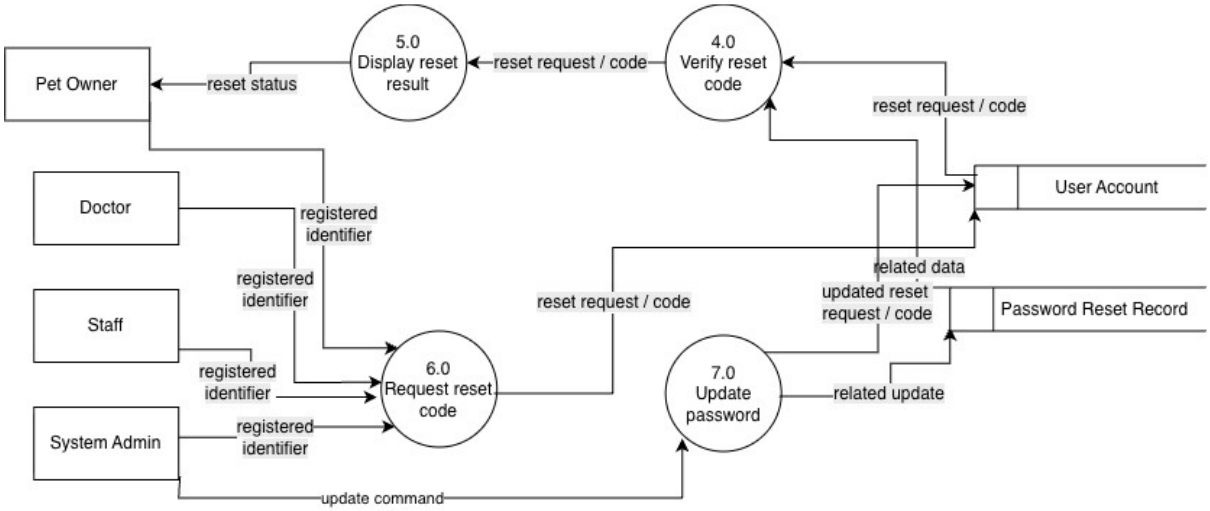



*Fig : 6 DFD for Password Recovery*

A user requests password recovery using a registered identifier. The system locates the account, creates and sends a reset code, verifies the submitted code, and stores the new password after successful validation.

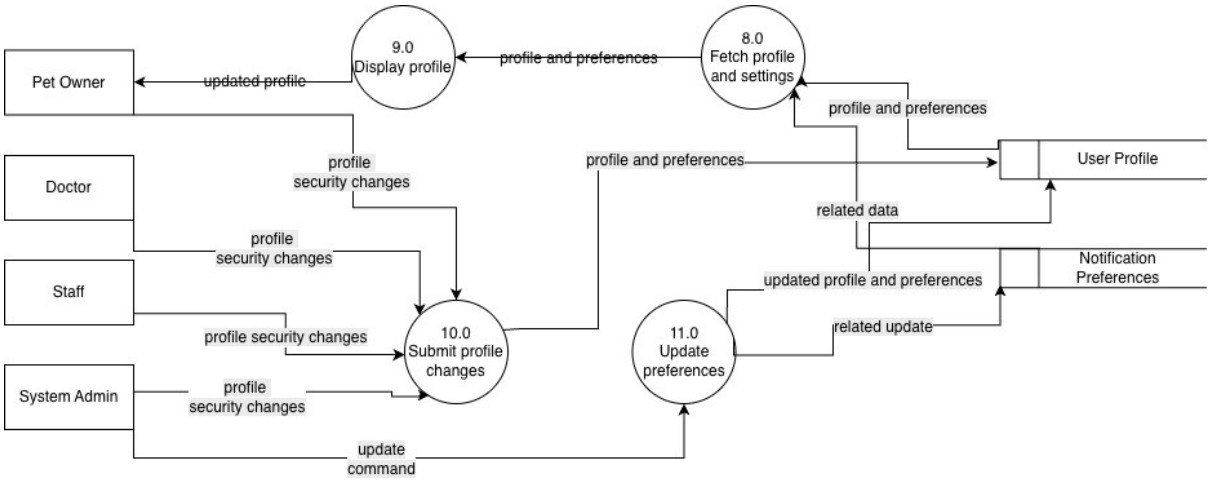



*Fig : 7 DFD for Profile, Security, and Notification Settings*

Authenticated users view and update their profile, photo, contact information, password, security options, and role-specific notification preferences. Validated changes are saved and returned to the account interface.

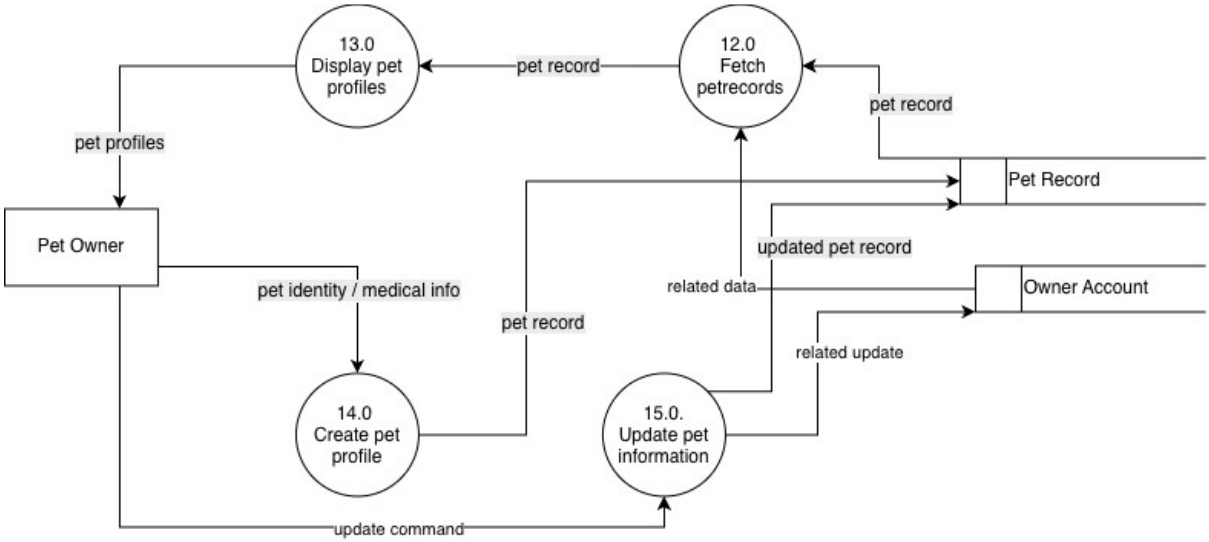



*Fig : 8 DFD forPet Profile Management (Pet Owner)*

The pet owner creates, views, and updates pet profiles containing the pet's name, species, breed, sex, weight, photo, vaccine information, and other medical details. Every pet record remains linked to the authenticated owner account.

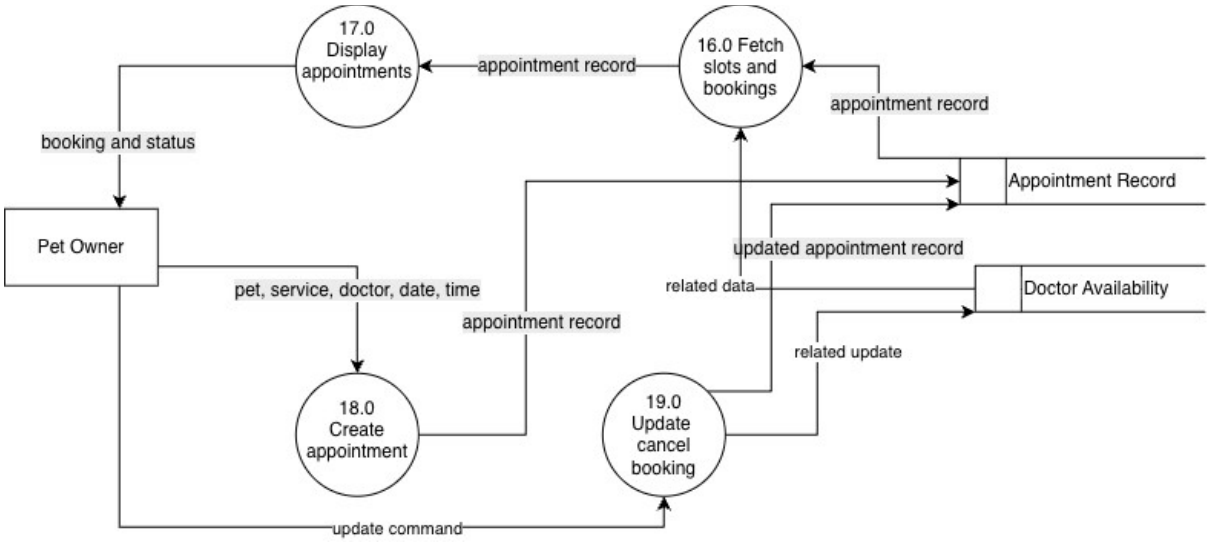



*Fig : 9 DFD for Appointment Booking and Tracking (Pet Owner)*

A pet owner selects a pet, service, veterinarian, date, time, and reason. The application checks doctor availability, stores the booking, and displays appointment details and subsequent confirmation, rescheduling, cancellation, or completion status.

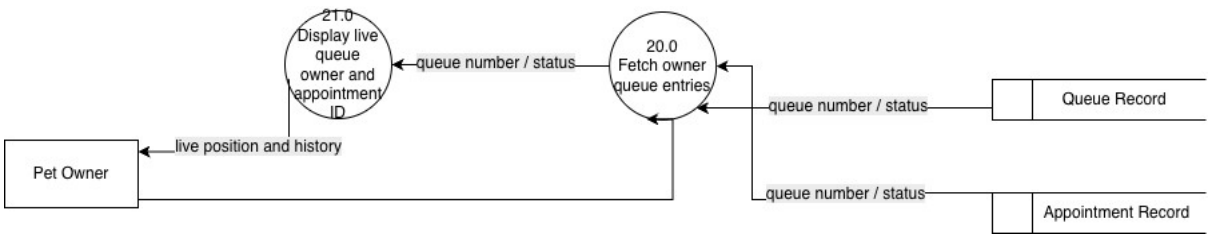


*Fig : 10 DFD for Queue Status and Queue History (Pet Owner)*

The pet owner opens the queue screen to retrieve active and historical queue entries linked to the owner's appointments. The application displays the queue number, service group, doctor, current status, and previous queue activity.

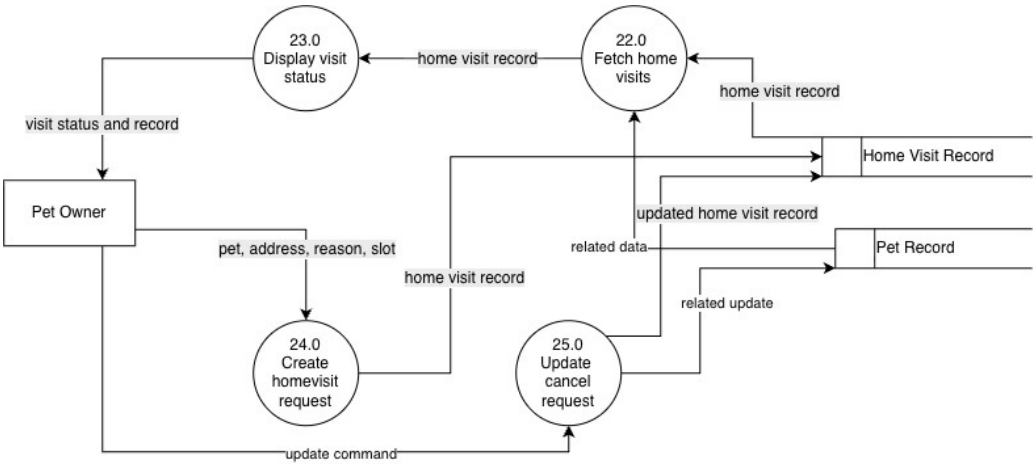



*Fig : 11 DFD for Home Visit Booking and Tracking (Pet Owner)*

The pet owner submits a home-visit request with the selected pet, address, contact details, reason, preferred date, and time. The system stores the request and displays assignment, progress, completion, and the resulting home-visit medical record.

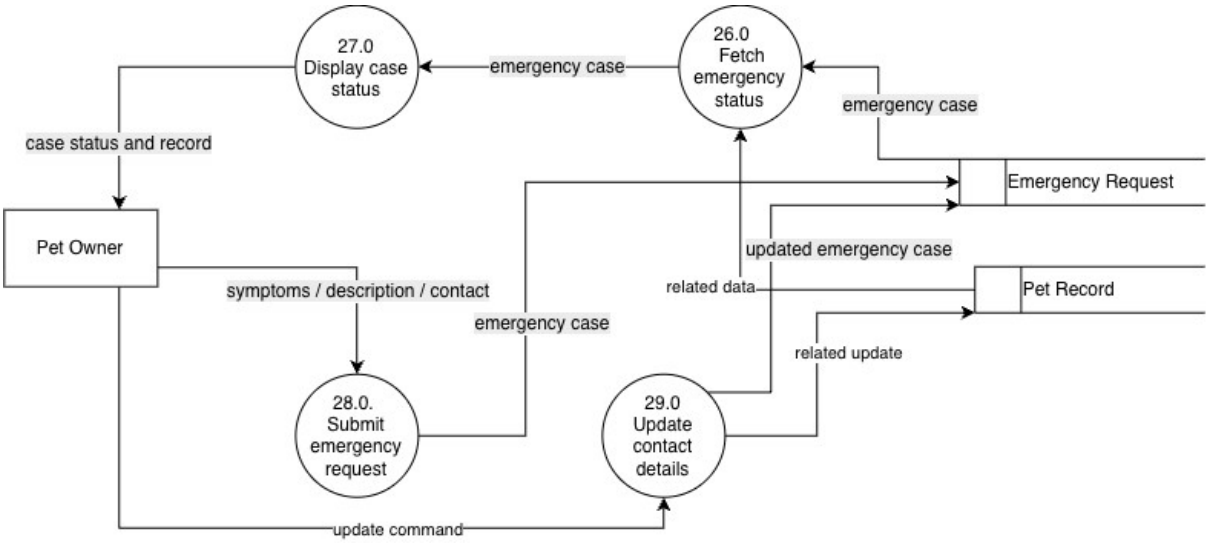



*Fig : 12 DFD for Emergency Service Request and Tracking (Pet Owner)*

A pet owner reports an emergency by selecting a pet and providing symptoms, a description, and contact details. The application stores the case and displays staff review, acceptance, arrival, treatment, and the completed emergency medical record.

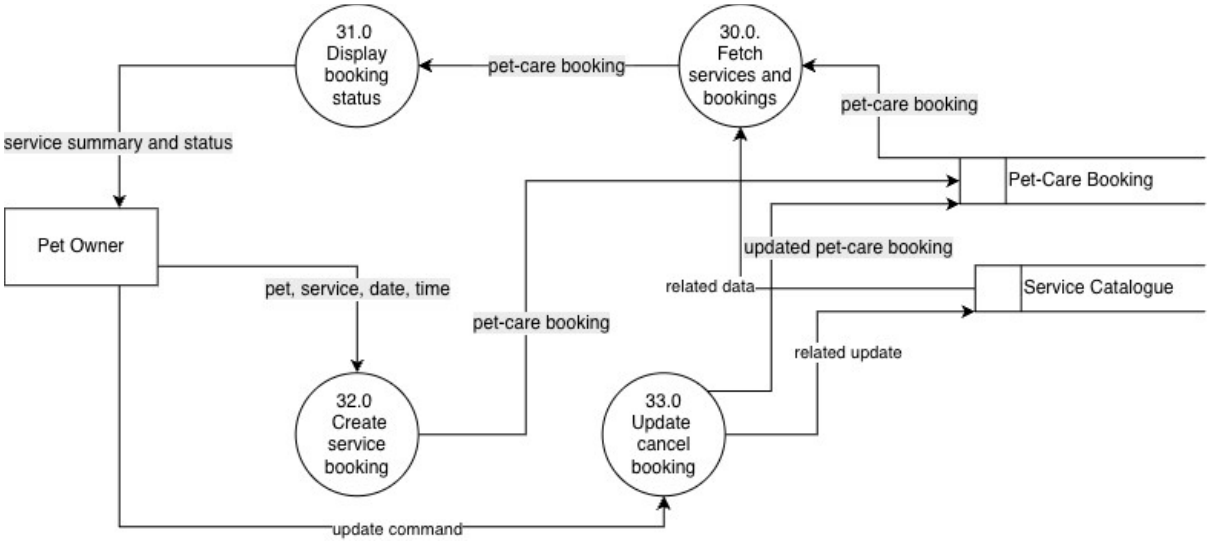



*Fig : 13 DFD for Pet-Care Service Booking (Pet Owner)*

The pet owner selects a grooming, bathing, nail-trimming, boarding, or wellness service and submits the pet, date, and time information. The system stores the service booking and returns its summary and current status.

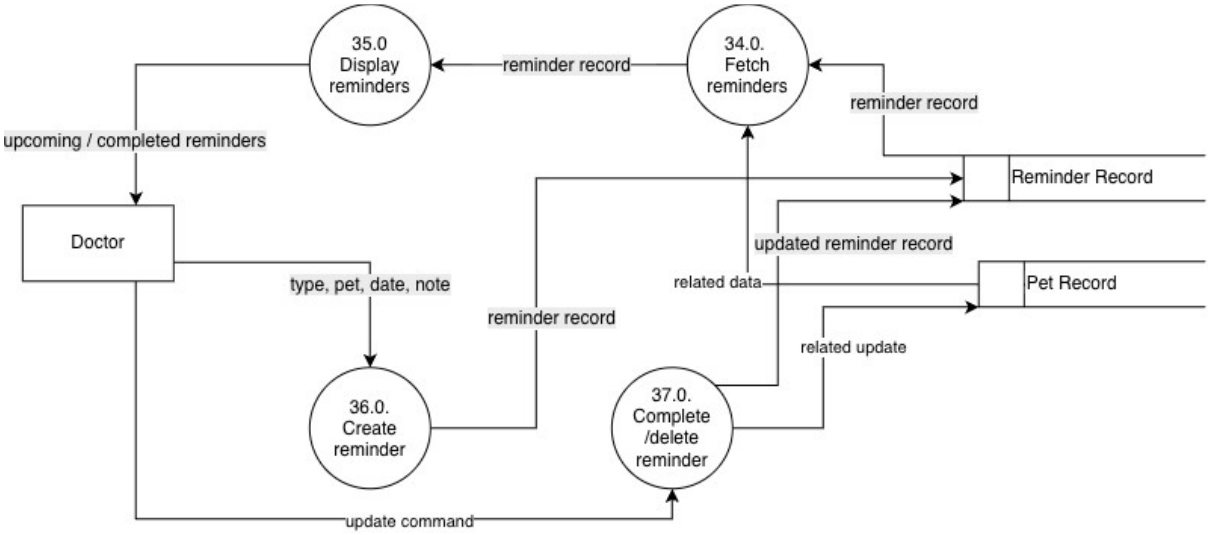



*Fig : 14 DFD for Pet Reminder Management (Pet Owner)*

The pet owner creates reminders for medicine, vaccines, checkups, or other care tasks. The application stores each reminder, retrieves upcoming and completed items, and records completion or deletion commands.

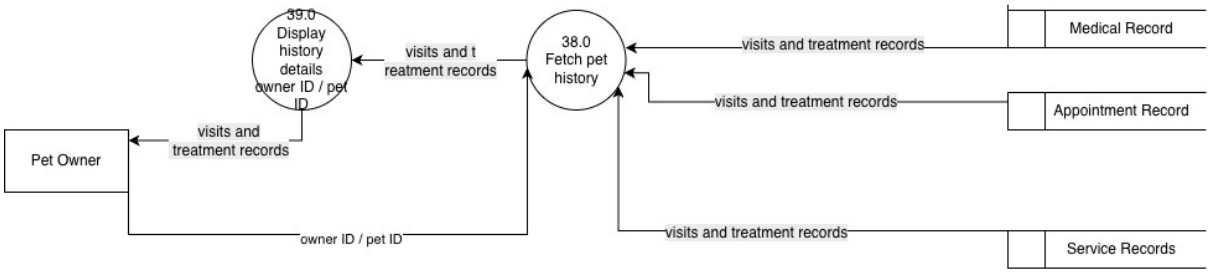



*Fig : 15 DFD for Medical and Treatment History Viewing (Pet Owner)*

The pet owner requests a selected pet's history. The application combines clinic appointments, diagnoses, prescriptions, vaccinations, home visits, emergencies, pet-care services, and queue records and displays detailed historical information.

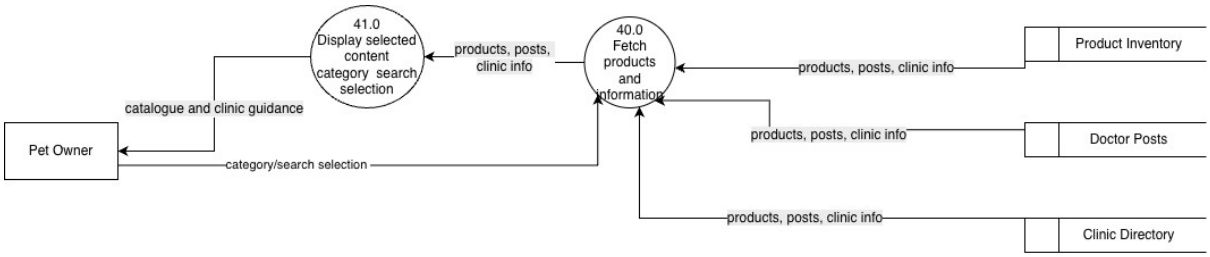



*Fig : 16 DFD for Products and Clinic Information Viewing (Pet Owner)*

The pet owner browses products, doctor profiles, medical services, health posts, first-aid guides, clinic contact information, and available service categories. The application retrieves and displays the selected information without changing operational records.

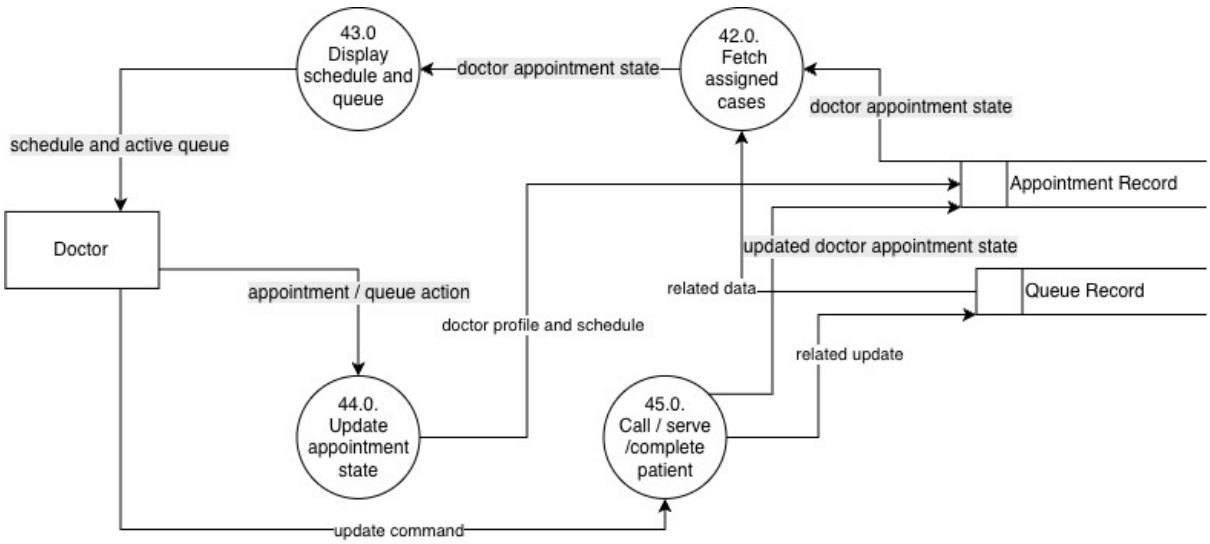


*Fig : 17 DFD for Doctor Appointment and Queue Management*

The doctor retrieves assigned appointments and the live patient queue, opens patient details, and records workflow actions such as calling the next patient, starting a consultation, marking a patient missed, or completing the visit.

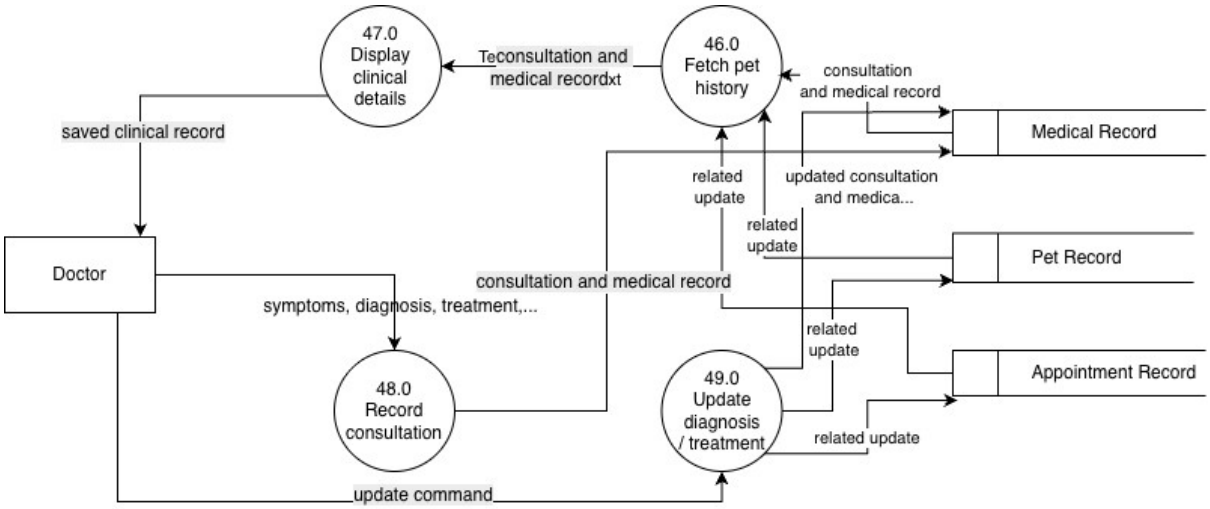



*Fig : 18 DFD for Doctor Consultation and Medical Record Update*

During a consultation, the doctor reviews the pet profile and previous history, records symptoms and findings, enters diagnosis, treatment, prescription, vaccination, and follow-up details, and stores the completed medical record.

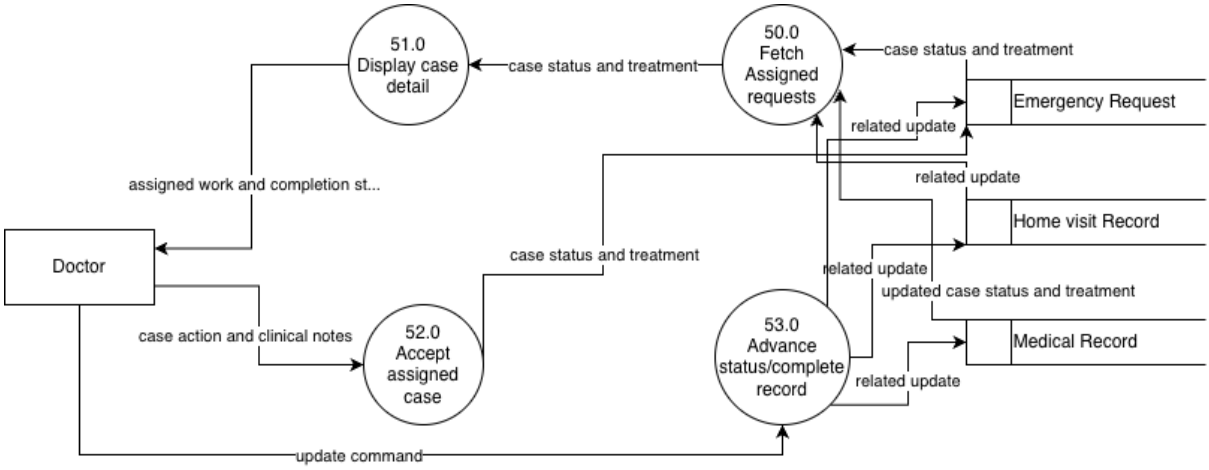



*Fig : 19 DFD for Doctor Emergency and Home-Visit Handling*

The doctor views assigned emergency cases and home visits, reviews owner and pet details, advances the service status, contacts the owner when needed, and records treatment and completion information for the final medical record.

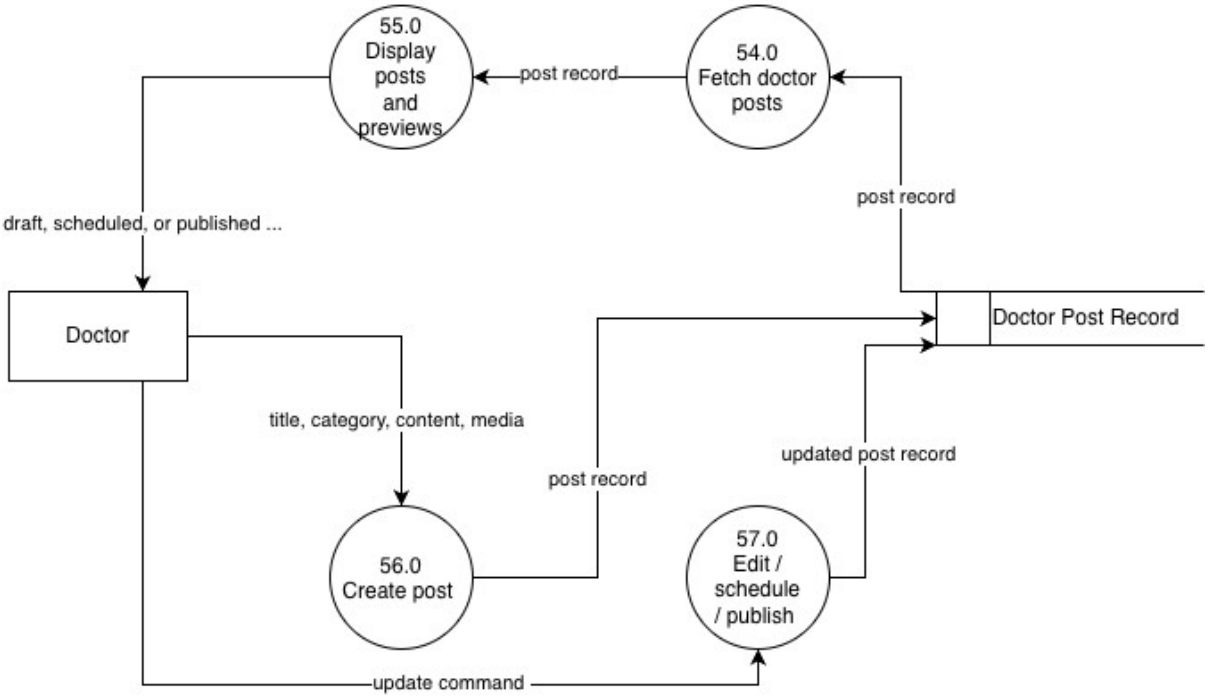



*Fig : 20 DFD for Doctor Health-Post Management*

The doctor creates health-information posts containing a title, category, content, cover image, and attachments. Drafts can be previewed, edited, scheduled, published, or deleted, and published posts become available to pet owners and staff.

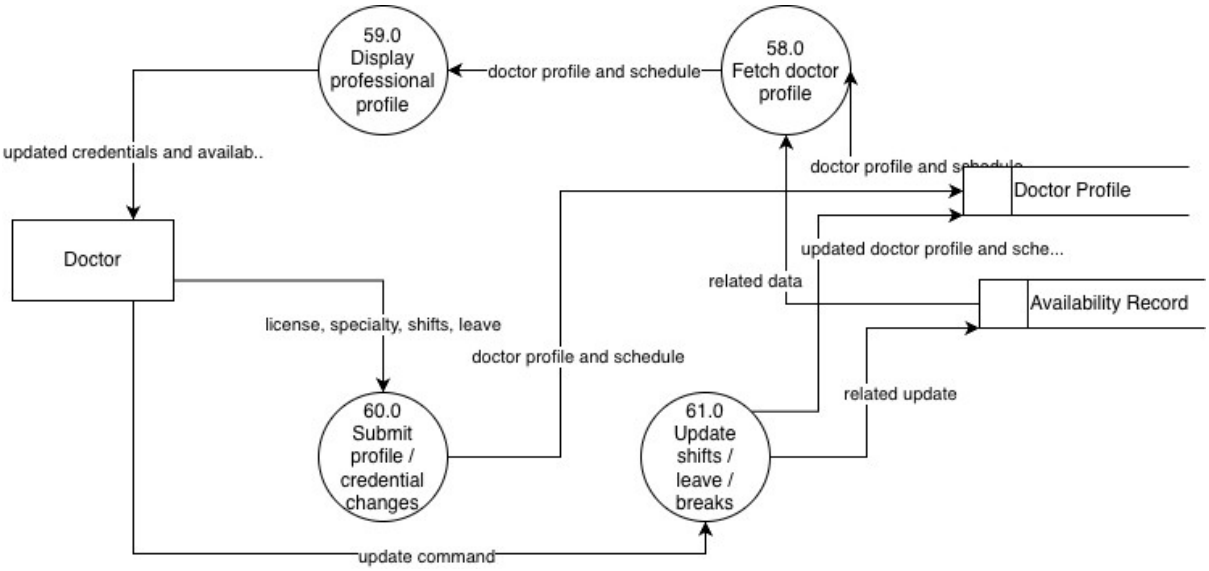



*Fig : 21 DFD for Doctor Profile, Credentials, and Availability*

The doctor maintains contact details, specialty, qualifications, license information, weekly availability, shifts, breaks, and leaves. The application validates and stores these details for appointment scheduling and public doctor profiles.

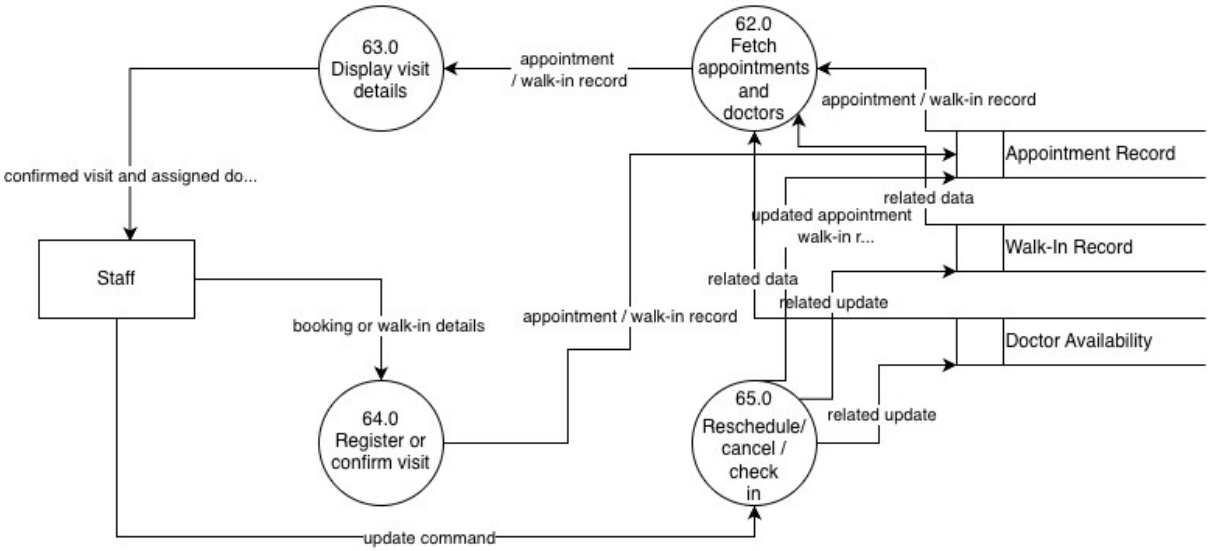



*Fig : 22 DFD for Staff Appointment and Walk-In Management*

Staff retrieve appointment requests, confirm or reschedule bookings, assign doctors, contact owners, cancel visits, check in scheduled patients, and register walk-in patients before sending them to the live queue.

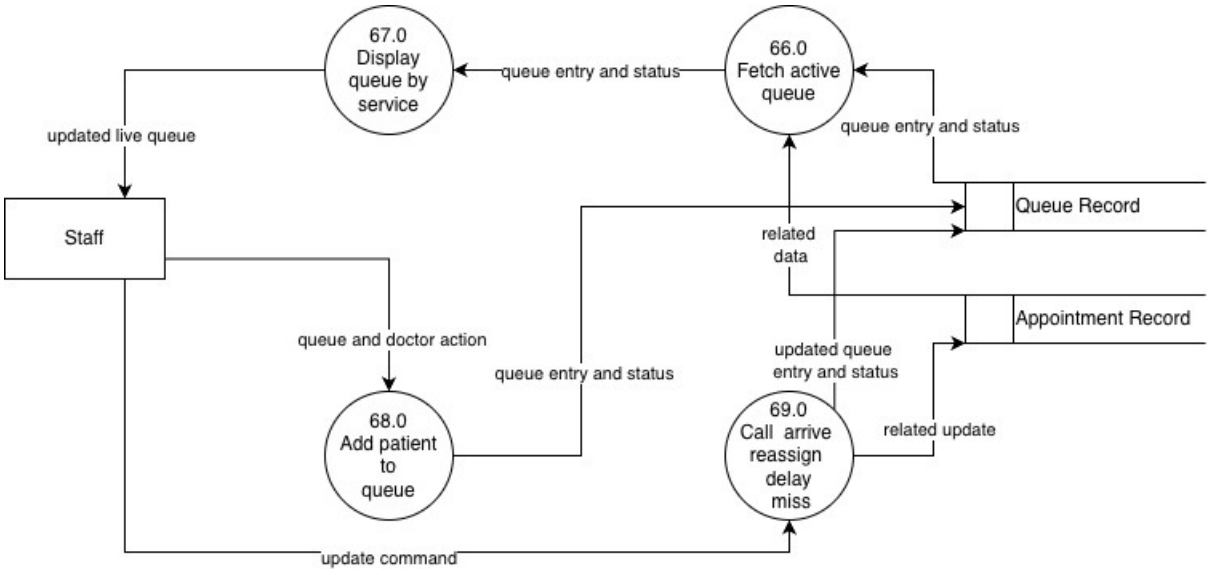



*Fig : 23 DFD for Staff Live Queue Management*

Staff manage the real-time patient queue by service group and status. They add checked-in patients, call patients, mark arrival, reassign doctors, announce delays, mark missed visits, and monitor completion.

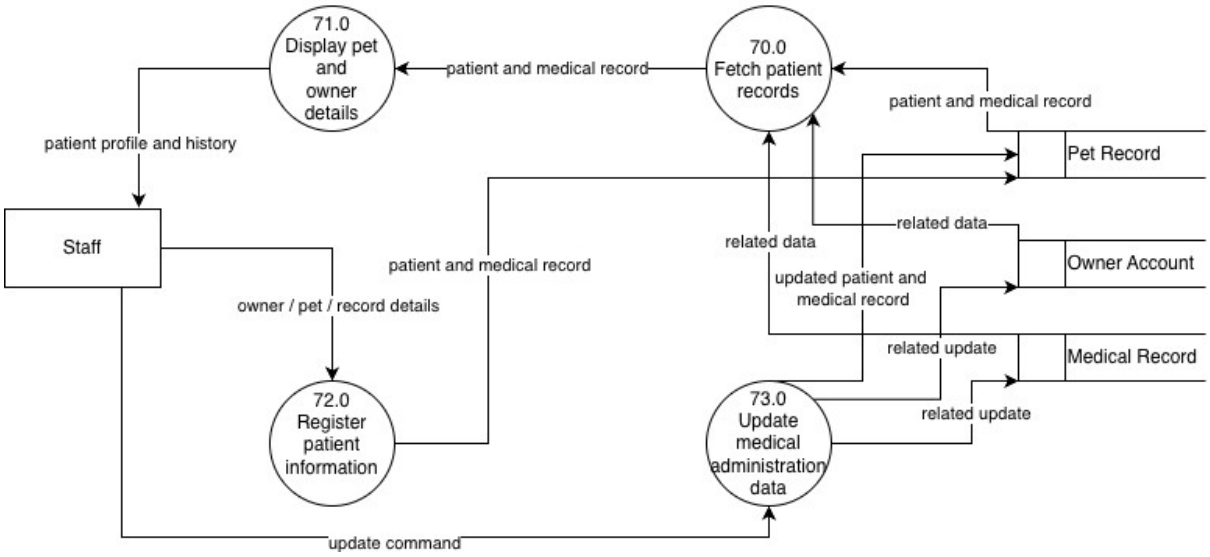



*Fig : 24 DFD for Staff Patient and Medical Record Management*

Staff search pet owners and patients, open pet profiles, review treatment history, and maintain administrative medical-record details needed for clinic operations while preserving the doctor's clinical entries.

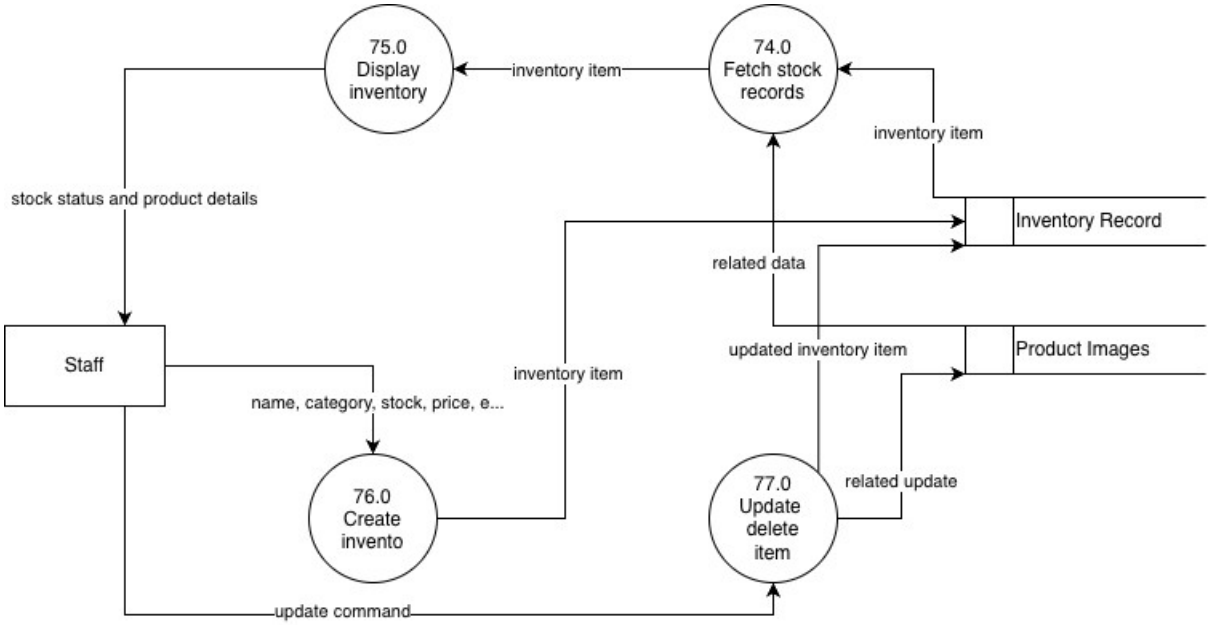



*Fig : 25 DFD for Staff Inventory Management*

Staff create, view, search, update, and remove inventory items. Each record includes product images, category, brand, pet type, price, stock, reorder level, expiry date, and description, allowing the system to identify low-stock and expired items.

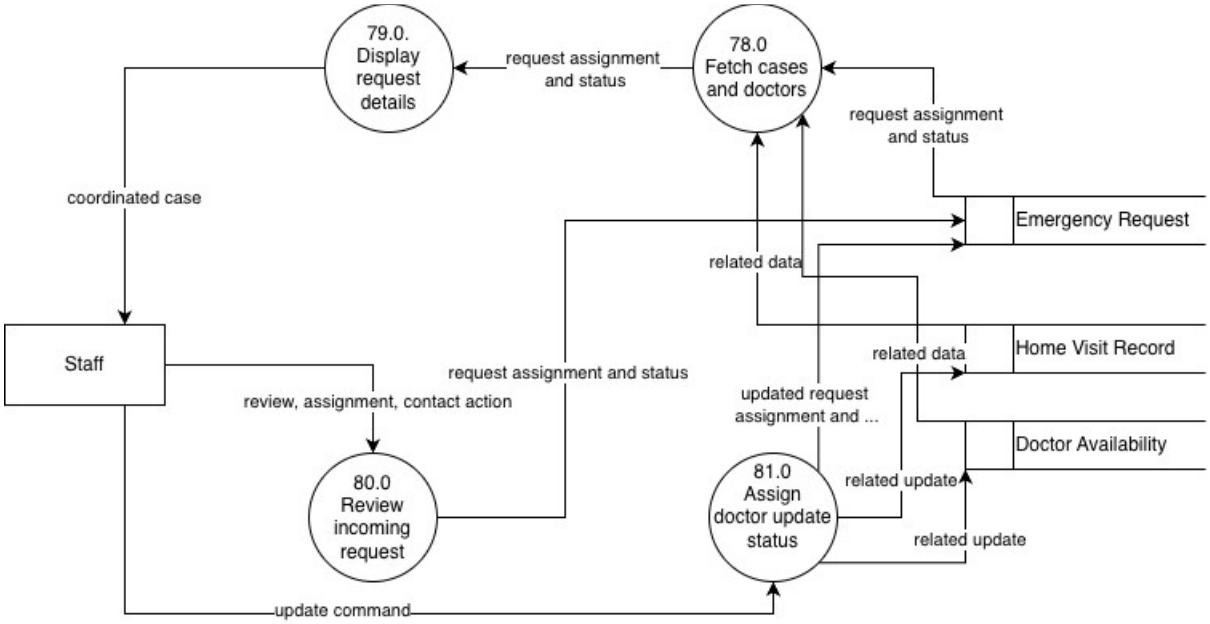



*Fig : 26 DFD for Staff Emergency and Home-Visit Coordination*

Staff review incoming emergency and home-visit requests, contact the owner, select an available doctor, assign the case, and update acceptance, arrival, scheduling, and completion-related administrative status.

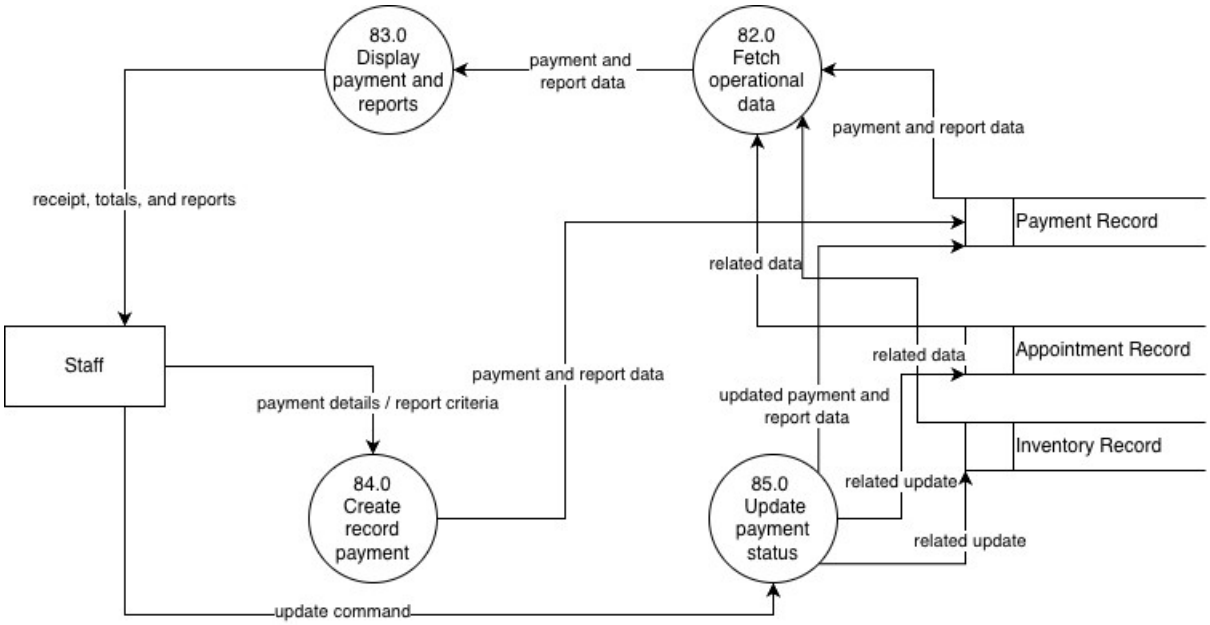



*Fig : 27 DFD for Staff Payment Management and Reporting*

Staff record and review payments and generate operational reports from appointment, service, payment, and inventory information. The system returns report results and stores the latest payment status for clinic accounting.

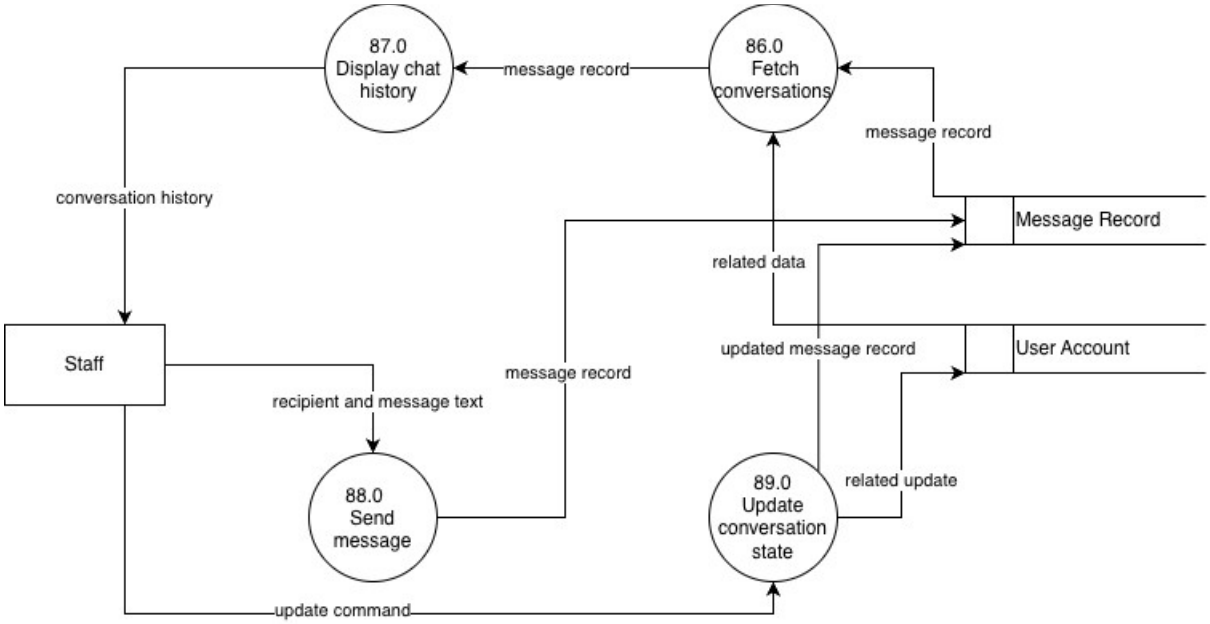



*Fig : 28 DFD for Staff Messaging*

Staff retrieve user conversations, open a selected thread, and send messages. Each message is stored with

sender, recipient, conversation, and time information so the complete communication history can be

displayed and traced.

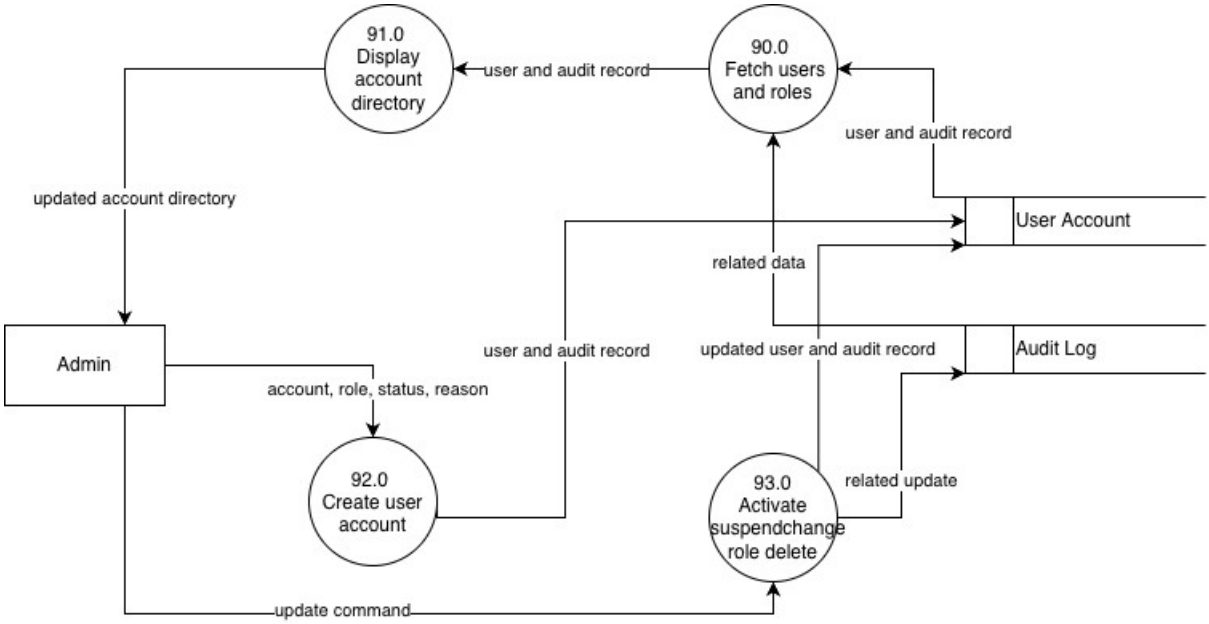



*Fig : 29 DFD for System Administrator User and Role Management*

The System Administrator creates accounts, searches and views users, assigns roles, activates or suspends accounts, changes account roles, and deletes accounts. Every sensitive action is recorded in the audit log.

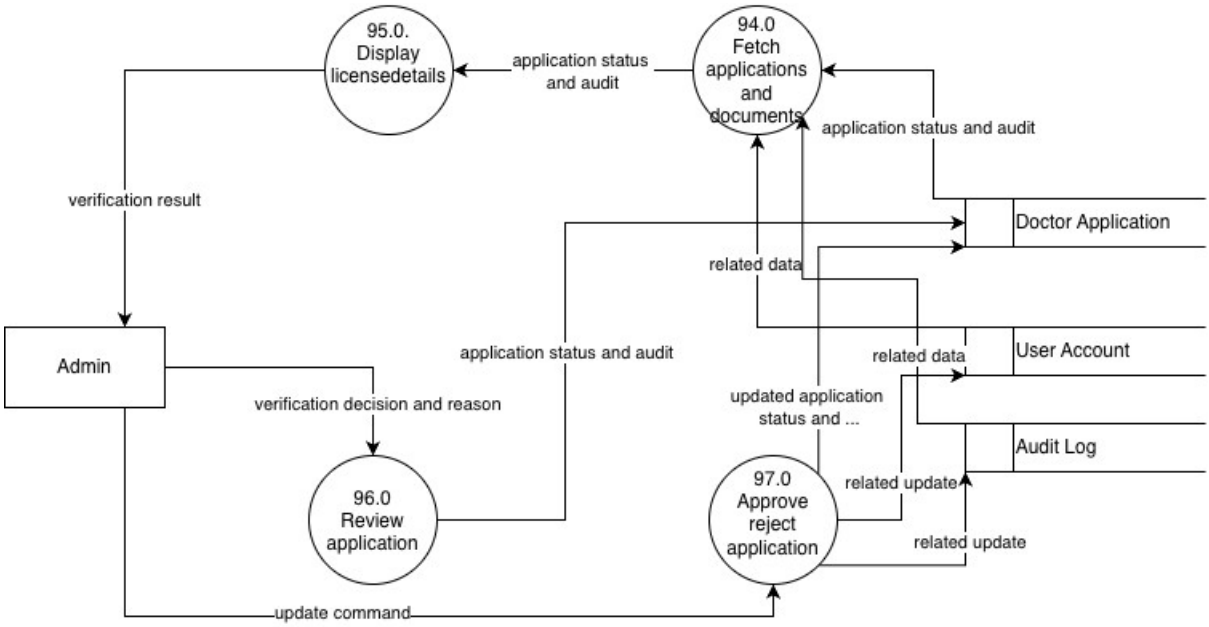



*Fig : 30 DFD for System Administrator Doctor Verification*

The System Administrator retrieves pending doctor applications, reviews contact details, qualifications, license number, expiry date, specialty, and submitted documents, and then approves or rejects the application with a recorded reason.

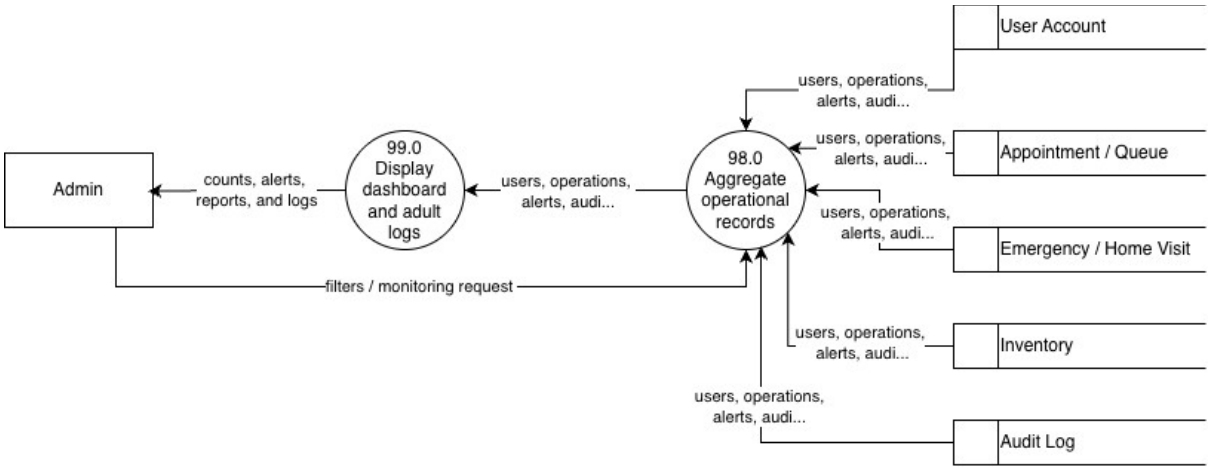


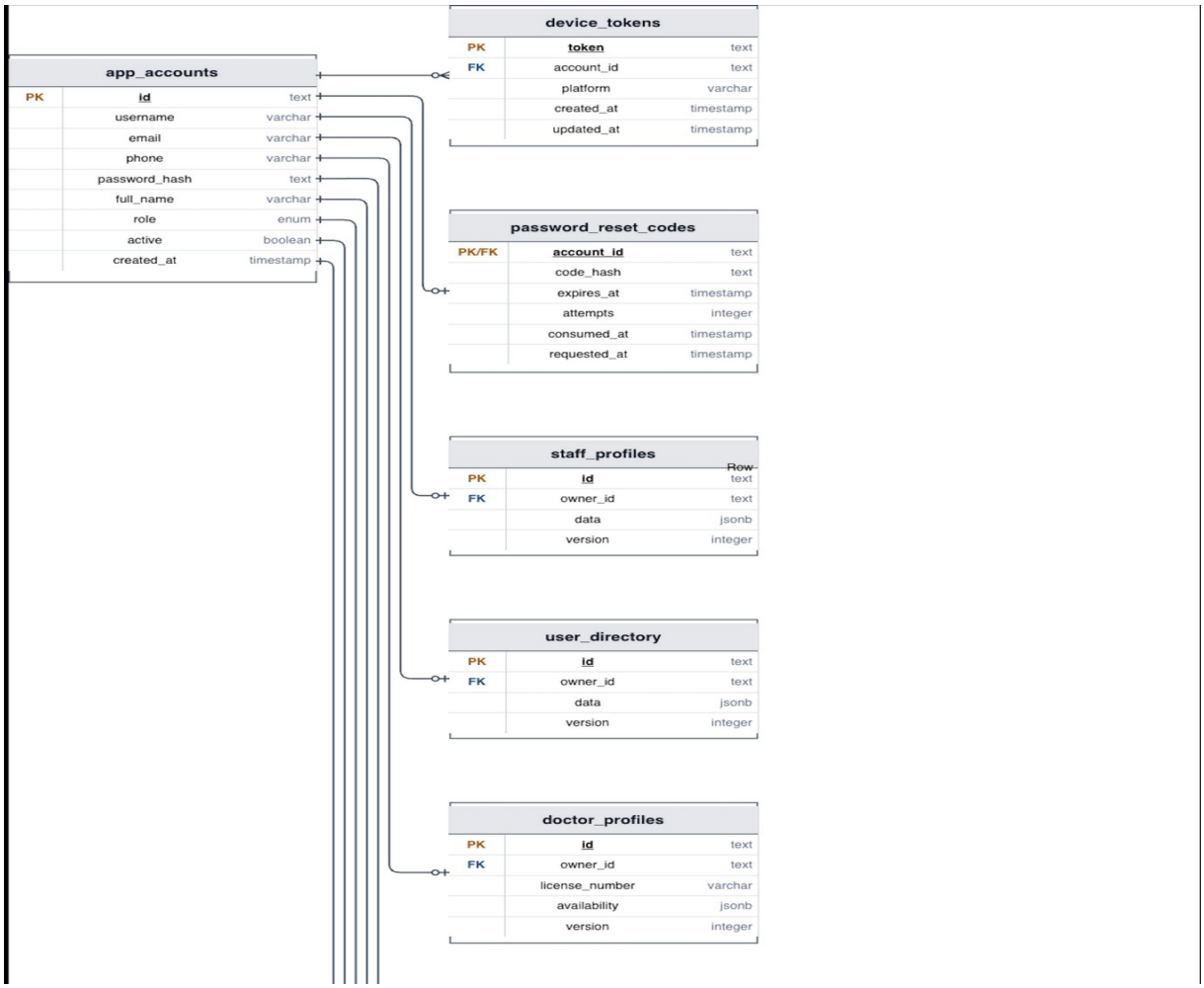

*Fig : 31 DFD for System Dashboard, Alerts, Reports, and Audit Logs*

The System Administrator monitors totals for users, doctors, staff, owners, appointments, and queue entries; reviews doctor-verification, emergency, low-stock, and expiry alerts; and searches the append-only audit trail of sensitive administrative actions.

## 4.2.2 Entity Relationship Diagram

The Entity Relationship Diagram (ERD) represents the relationships between the main data entities used by the system.

### 1. Accounts and Access



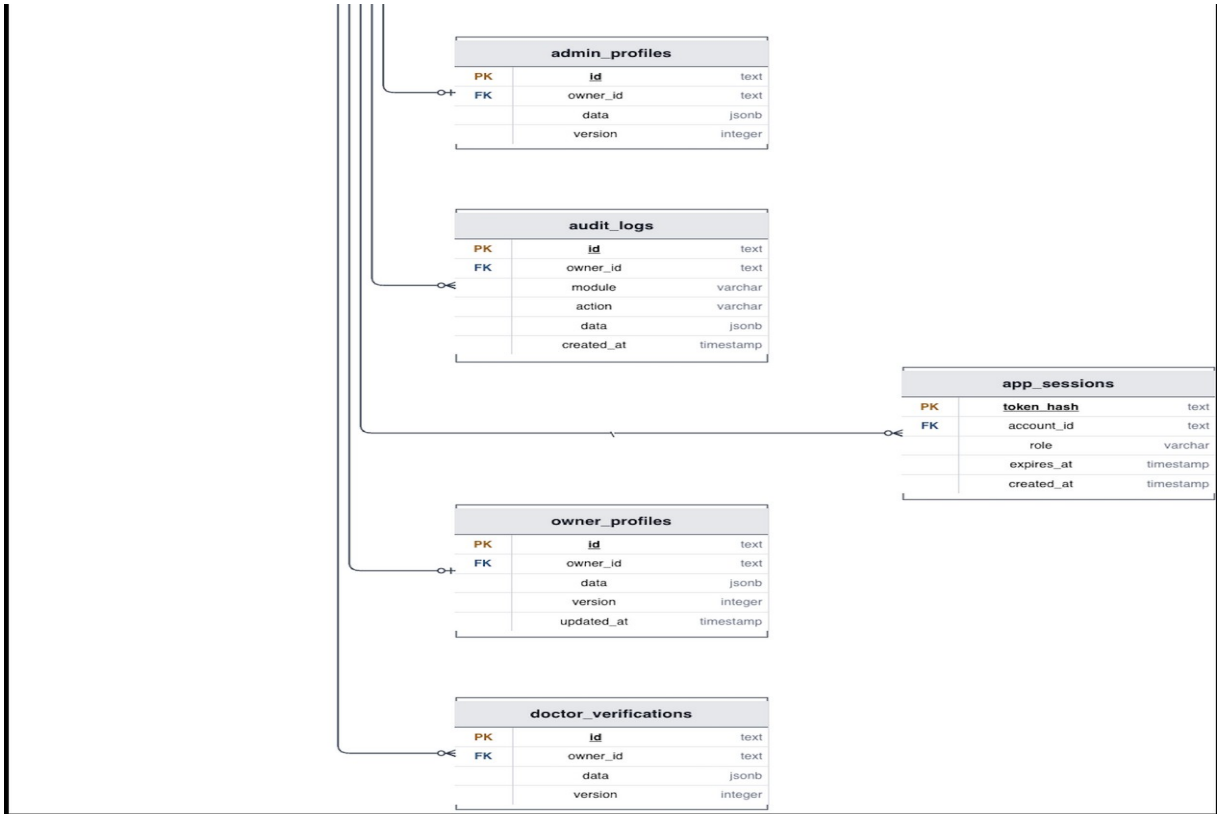



*Fig : 32 ERD for Accounts and Access Diagram*

The Accounts and Access section manages authentication, authorization, profiles, and security information. The `app_accounts` table is the main table containing usernames, email addresses, phone numbers, password hashes, full names, roles, account statuses, and creation dates. One account can have multiple sessions, password-reset requests, device tokens, and audit-log entries. Depending on the user's role, the account can also have an owner, doctor, staff, or administrator profile. The `doctor_verifications` table stores doctor-verification information, while the `user_directory` table provides centralized user information.

## 2. Pets and Bookings

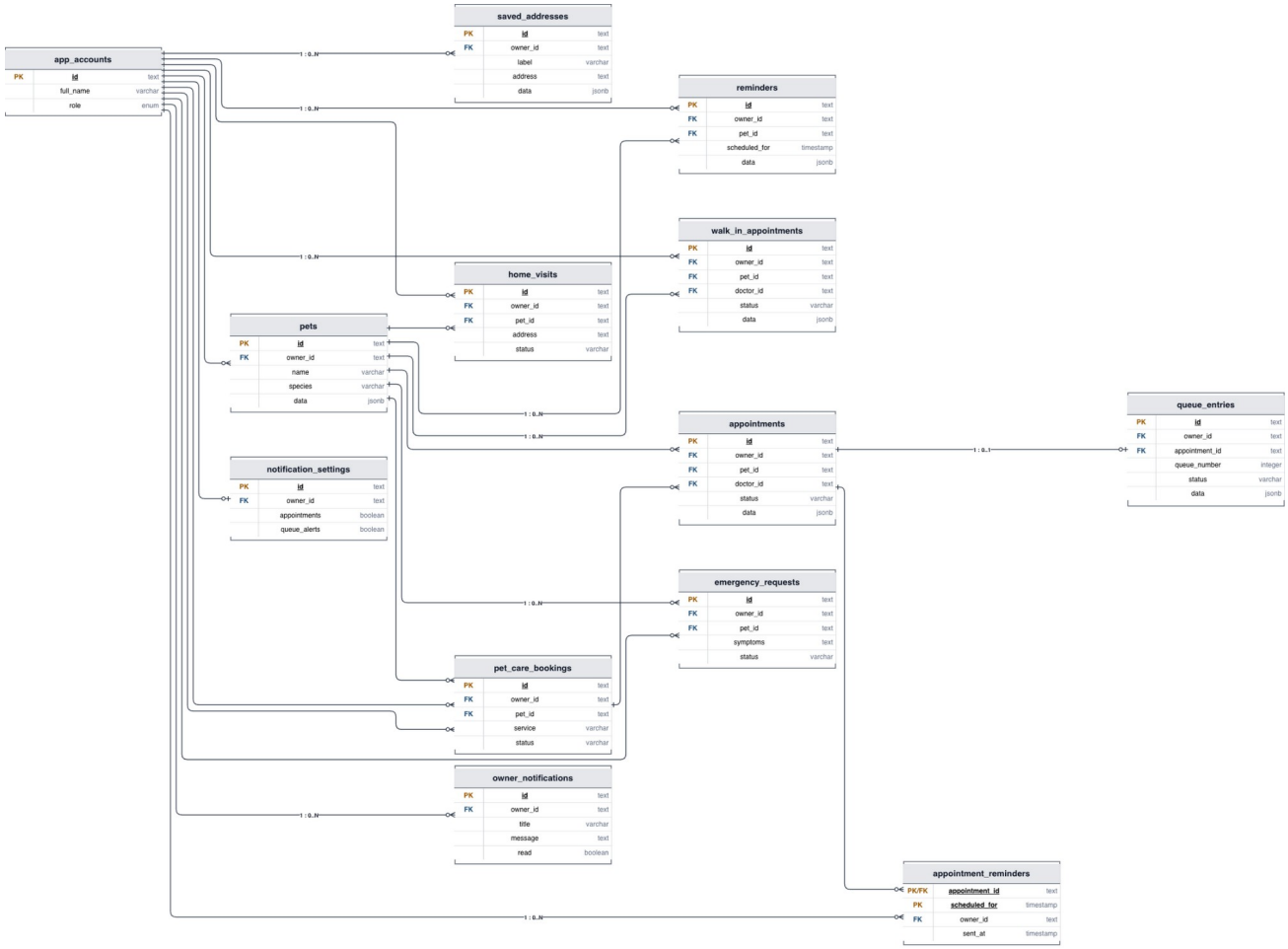



*Fig : 33 ERD for Pets and Booking Diagram*

The Pets and Bookings section manages pets and the services requested by their owners. One account can own multiple pets, while each pet belongs to one owner. A pet can have multiple scheduled appointments, walk-in appointments, home visits, emergency requests, care-service bookings, and reminders. The `appointments` table connects an owner, pet, and doctor. The `queue_entries` table assigns queue numbers to appointments, while `appointment_reminders` schedules appointment notifications. Saved addresses, notification settings, and owner notifications support the owner's booking and communication experience.

## 3. Clinical Operations

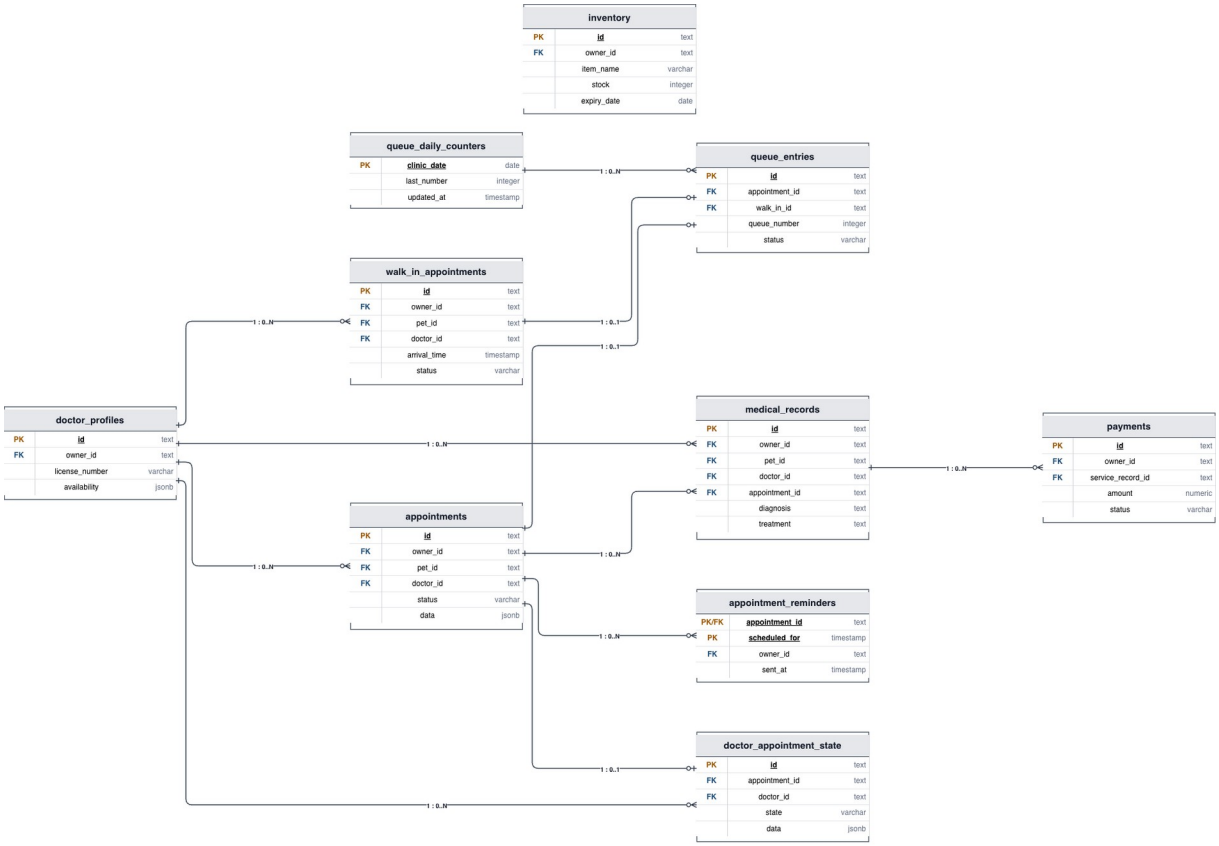



*Fig : 34 ERD for Clinical Operations Diagram*

The Clinical Operations section represents the clinic's daily medical workflow. Doctors are assigned to appointments through the `doctor_profiles` and `appointments` tables. After a consultation, the clinic creates a `medical_records` entry containing the diagnosis and treatment. A related payment can then be stored in the `payments` table. Walk-in patients are recorded in `walk_in_appointments` and placed into the clinic queue through `queue_entries`. The `doctor_appointment_state` table tracks the current condition of an appointment, while `queue_daily_counters` maintains daily queue numbers. The `inventory` table stores clinic items, stock quantities, and expiration dates.

## 4. Content

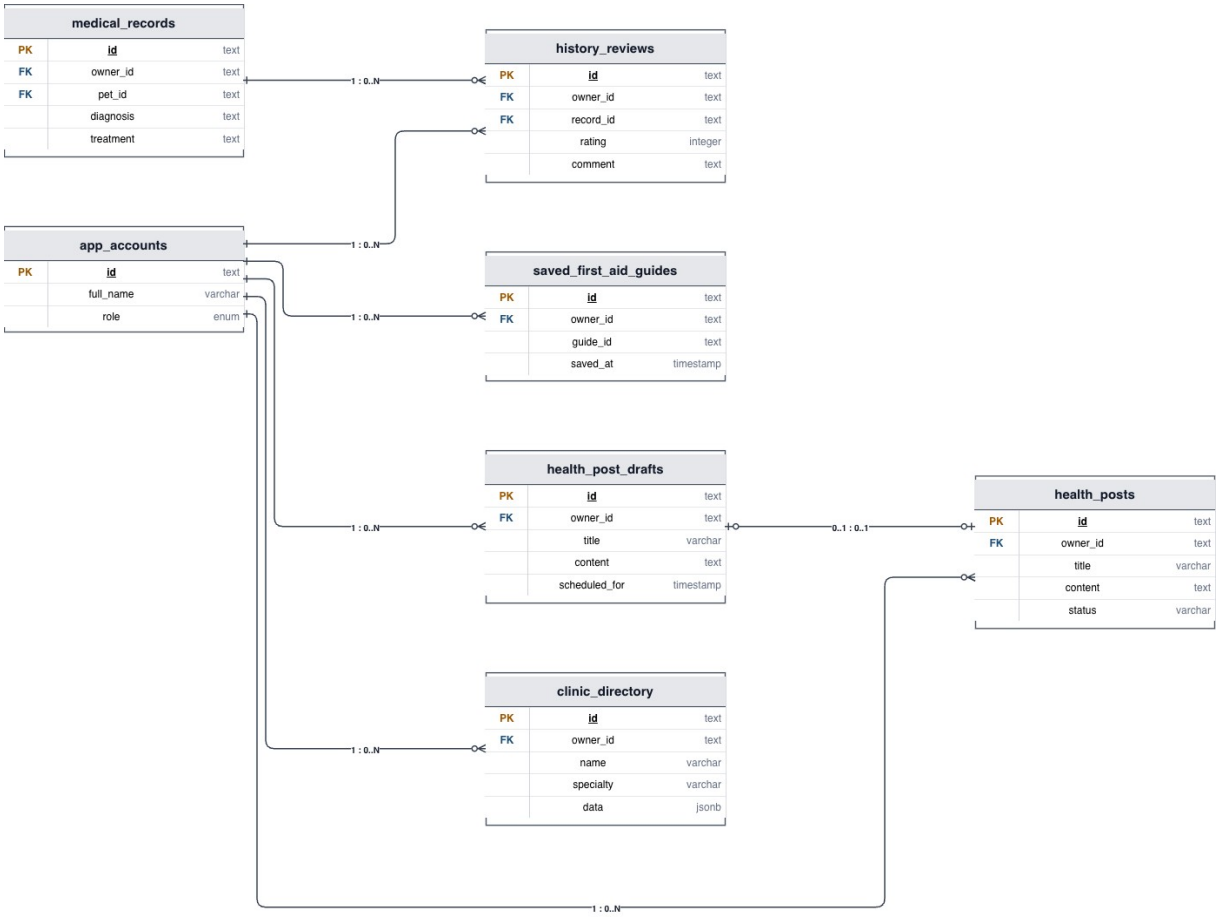



*Fig : 35 ERD for Content*

The Content section manages educational and informational material. The `health_posts` table stores published health articles, while `health_post_drafts` stores unfinished or scheduled articles. Users can save first-aid guides through `saved_first_aid_guides` and write ratings and comments through `history_reviews`. The `clinic_directory` table stores clinic names, specialties, and other clinic information. Medical records are also referenced in this section so reviews can be associated with completed services or treatment histories.

## 5.Administration

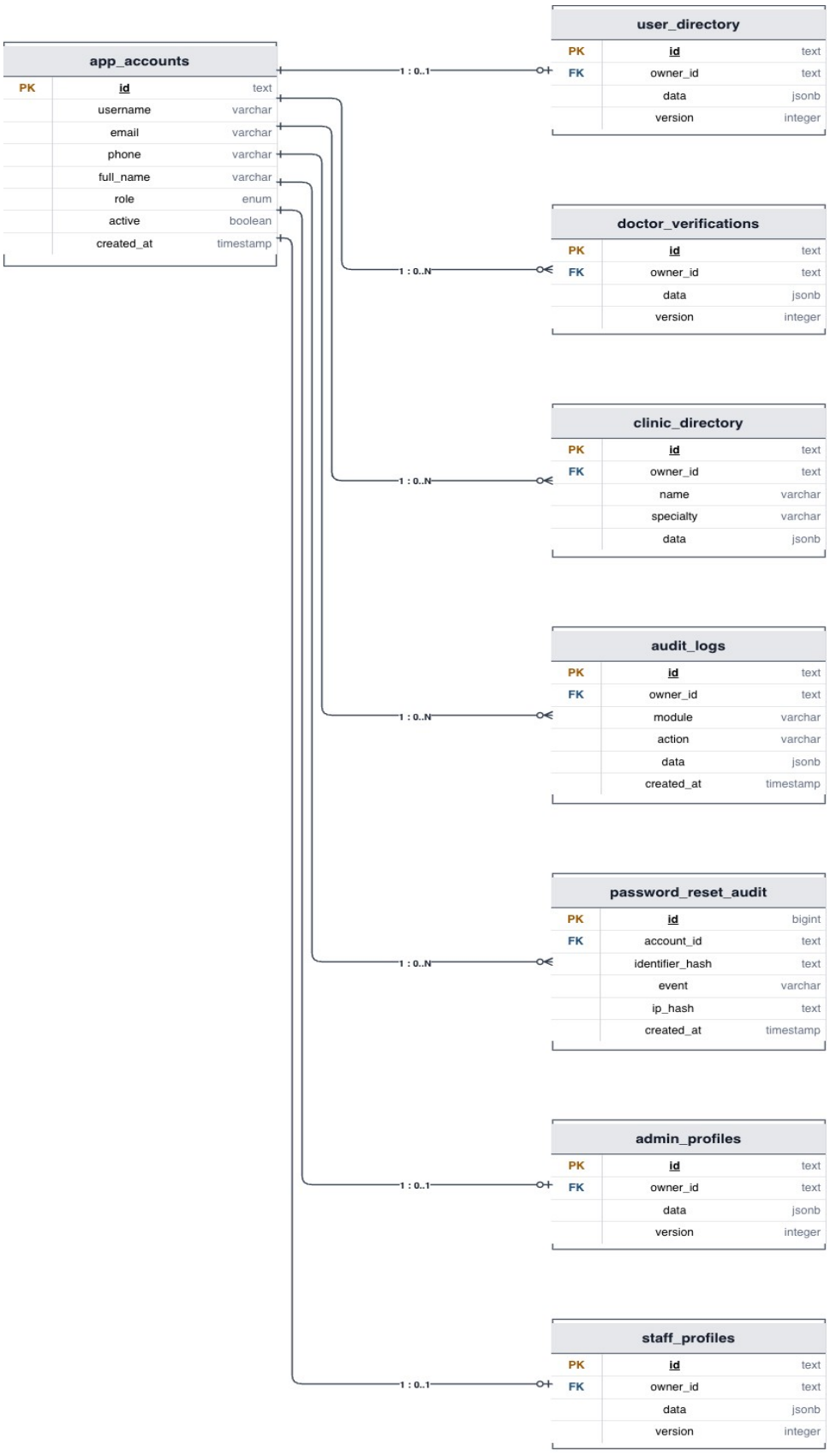



*Fig : 36 ERD for Administration*

The Administration section supports system management and governance. Administrators use `app_accounts`, `admin_profiles`, and `staff_profiles` to manage users and employees. The `user_directory` provides a centralized account listing, while `doctor_verifications` helps administrators confirm doctor credentials. The `clinic_directory` allows administrative management of clinic information and specialties. The `audit_logs` table records actions performed throughout the system, and `password_reset_audit` records password-reset activity for security monitoring.

| Entity | Description | Relationships |
|---|---|---|
| App_Accounts | Main user account and access information for pet owners, veterinarians, staff, and administrators. | App_Sessions, Password_Reset_Codes, Device_Tokens, Owner_Profiles, Doctor_Profiles, Staff_Profiles, Admin_Profiles, Audit_Logs, Pets, Appointments |
| Owner_Profiles | Stores pet owner profile information. | App_Accounts, Pets, Appointments, Home_Visits, Emergency_Requests, Notifications |
| Doctor_Profiles | Stores veterinarian profile and verification information. | App_Accounts, Appointments, Medical_Records, Doctor_Appointment_State |
| Staff_Profiles | Stores clinic staff profile information. | App_Accounts |
| Admin_Profiles | Stores administrator profile information. | App_Accounts |
| Pets | Stores information about pets registered by owners. | Owner_Profiles, Appointments, Home_Visits, Emergency_Requests, Pet_Care_Bookings, Medical_Records, Reminders |
| Appointments | Stores scheduled veterinary appointments. | Pets, App_Accounts, Doctor_Profiles, Queue_Entries, Reminders, Appointment_Reminders, Doctor_Appointment_State |
| Queue_Entries | Stores clinic queue information and appointment queue status. | Appointments |
| Walk_In_Appointments | Stores walk-in appointment information. | Pets, Doctor_Profiles, Queue_Entries |
| Home_Visits | Stores veterinary home visit information. | Owner_Profiles, Pets |
| Emergency_Requests | Stores emergency requests for pets. | Owner_Profiles, Pets |
| Pet_Care_Bookings | Stores additional pet-care service bookings. | Owner_Profiles, Pets |
| Medical_Records | Stores pet medical history, diagnosis, and treatment information. | Pets, Doctor_Profiles |
| Payments | Stores payment information for clinic services. | App_Accounts, Service information |
| Inventory | Stores clinic product and stock information. | App_Accounts |

| | | |
|---|---|---|
| Reminders | Stores appointment and pet-care reminders. | App_Accounts, Pets, Appointments |
| Notification_Settings | Stores user notification preferences. | App_Accounts |
| Owner_Notifications | Stores notifications received by pet owners. | App_Accounts |
| Appointment_Reminders | Stores reminders related to appointments. | Appointments, App_Accounts |
| Health_Posts | Stores health-related posts published by the clinic. | App_Accounts |
| Health_Post_Drafts | Stores health-related posts before publication. | App_Accounts |
| Clinic_Messages | Stores clinic communication messages. | App_Accounts |
| Doctor_Notifications | Stores notifications for veterinarians. | App_Accounts |
| Support_Tickets | Stores user support requests. | App_Accounts |
| History_Reviews | Stores user reviews and ratings. | App_Accounts |
| Saved_First_Aid_Guides | Stores first-aid guides saved by users. | App_Accounts |
| Clinic_Directory | Stores clinic directory information. | App_Accounts |
| Password_Reset_Codes | Stores temporary password reset codes. | App_Accounts |
| Password_Reset_Audit | Records password reset activities. | App_Accounts |
| Device_Tokens | Stores device tokens used for notifications. | App_Accounts |
| App_Sessions | Stores user login sessions. | App_Accounts |
| Doctor_Verifications | Stores veterinarian verification information. | Doctor_Profiles / App_Accounts |
| Audit_Logs | Records important system activities for auditing. | App_Accounts |
| Doctor_Appointment_State | Stores veterinarian-related appointment state information. | Appointments, Doctor_Profiles |
| Queue_Daily_Counters | Stores daily queue numbering information. | Clinic date / queue management |

*Table : 4 ERD Diagram Table*

| **Relationship** | **Type** | **Explanation** |
|---|---|---|
| App_Accounts → App_Sessions | One-to-Many | One account can have multiple login sessions. |
| App_Accounts → Password_Reset_Codes | One-to-Many | One account can have multiple password reset codes. |
| App_Accounts → Device_Tokens | One-to-Many | One account can have multiple registered devices. |
| App_Accounts → Owner_Profiles | One-to-One | Each owner profile belongs to one account. |
| App_Accounts → | One-to-One | Each doctor profile belongs to one |

| | | |
|---|---|---|
| Doctor_Profiles | | account. |
| App_Accounts → Staff_Profiles | One-to-One | Each staff profile belongs to one account. |
| App_Accounts → Admin_Profiles | One-to-One | Each administrator profile belongs to one account. |
| App_Accounts → Audit_Logs | One-to-Many | One account can generate multiple audit records. |
| App_Accounts → Pets | One-to-Many | One owner account can register multiple pets. |
| Pets → Appointments | One-to-Many | One pet can have multiple appointments. |
| Doctor_Profiles → Appointments | One-to-Many | One veterinarian can manage multiple appointments. |
| App_Accounts → Appointments | One-to-Many | One owner can make multiple appointments. |
| Appointments → Queue_Entries | One-to-One | An appointment can have a queue entry. |
| Pets → Home_Visits | One-to-Many | One pet can have multiple home visits. |
| Pets → Emergency_Requests | One-to-Many | One pet can have multiple emergency requests. |
| Pets → Pet_Care_Bookings | One-to-Many | One pet can have multiple care bookings. |
| Pets → Medical_Records | One-to-Many | One pet can have multiple medical records. |
| Doctor_Profiles → Medical_Records | One-to-Many | One veterinarian can manage multiple medical records. |
| App_Accounts → Reminders | One-to-Many | One owner can receive multiple reminders. |
| Pets → Reminders | One-to-Many | One pet can have multiple reminders. |
| Appointments → Reminders | One-to-Many | One appointment can have reminder records. |
| App_Accounts → Notification_Settings | One-to-One | Each owner has notification settings. |
| App_Accounts → Owner_Notifications | One-to-Many | One owner can receive multiple notifications. |
| Appointments → Appointment_Reminders | One-to-Many | One appointment can have multiple reminder records. |
| Appointments → Doctor_Appointment_State | One-to-Many | An appointment can have appointment-state records. |
| Doctor_Profiles → Doctor_Appointment_State | One-to-Many | One doctor can have multiple appointment-state records. |
| App_Accounts → Health_Posts | One-to-Many | One account can manage multiple health posts. |
| App_Accounts → Health_Post_Drafts | One-to-Many | One account can create multiple draft posts. |
| App_Accounts → Clinic_Messages | One-to-Many | One account can be associated with multiple clinic messages. |
| App_Accounts → Doctor_Notifications | One-to-Many | One doctor account can receive multiple notifications. |
| App_Accounts → Support_Tickets | One-to-Many | One account can submit multiple support tickets. |
| App_Accounts → History_Reviews | One-to-Many | One account can create multiple reviews. |
| App_Accounts → Saved_First_Aid_Guides | One-to-Many | One account can save multiple first-aid guides. |
| App_Accounts → Password_Reset_Audit | One-to-Many | One account can have multiple password-reset audit records. |

*Table : 5 ERD Diagram Relationship Table*

## 4.2.3 Class Diagram

The Class Diagram illustrates the main classes and their relationships within the Nway's Love Vet Clinic Management System. It represents users, pets, appointments, clinical operations, communication, and administration. The diagram is organized into Supporting Classes, Appointments and Clinical Operations, Users, Pets and Bookings, and Communication and Administration, showing how these system components interact.

### 4.2.3.1 Supporting Classes



*Fig : 37 Supporting Class Diagram*




*Fig : 38 Supporting Class Diagram Detail 1*




*Fig : 39 Supporting Class Diagram Detail 2*

## 4.2.3.2 Appointments and Clinical Operations

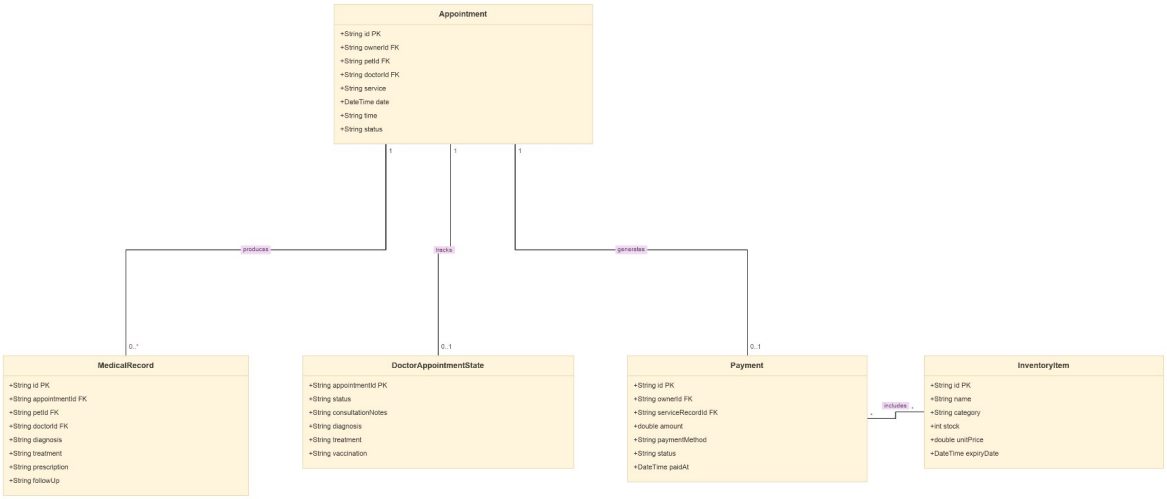



*Fig : 40 Appointments and Clinical Operations Class Diagram*

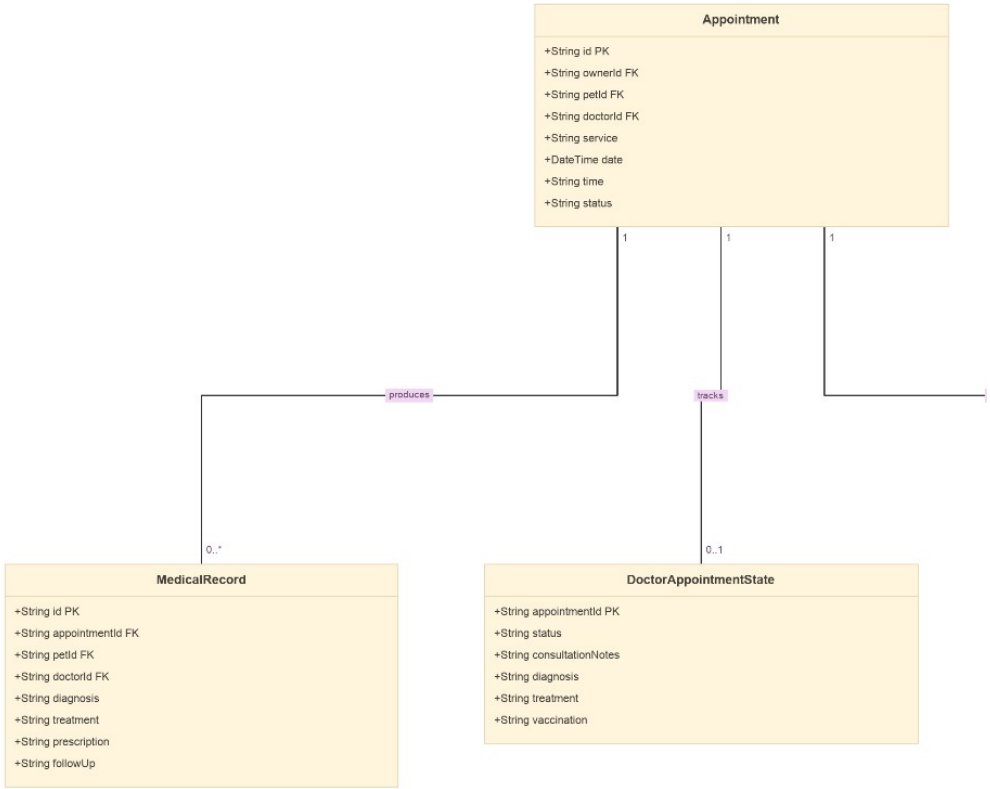



*Fig : 41 Appointments and Clinical Operations Class Diagram Detail 1*

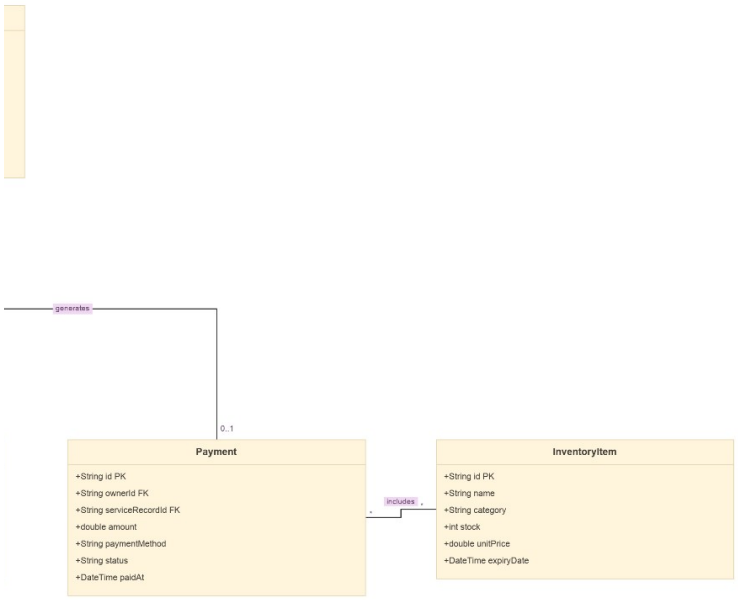



*Fig : 42 Appointments and Clinical Operations Class Diagram Detail 2*

## 4.2.3.3 Users, Pets and Booking

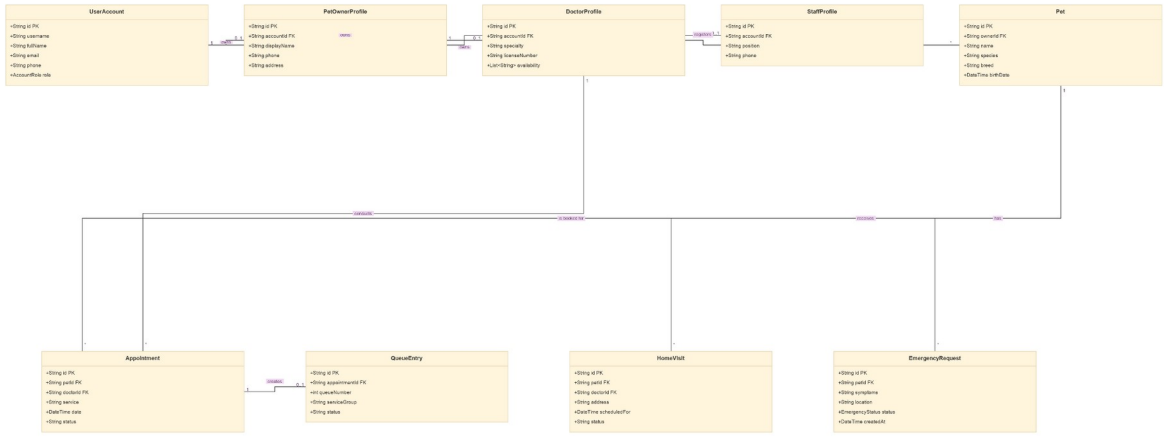



*Fig : 43 Users, Pets and Booking Class Diagram*

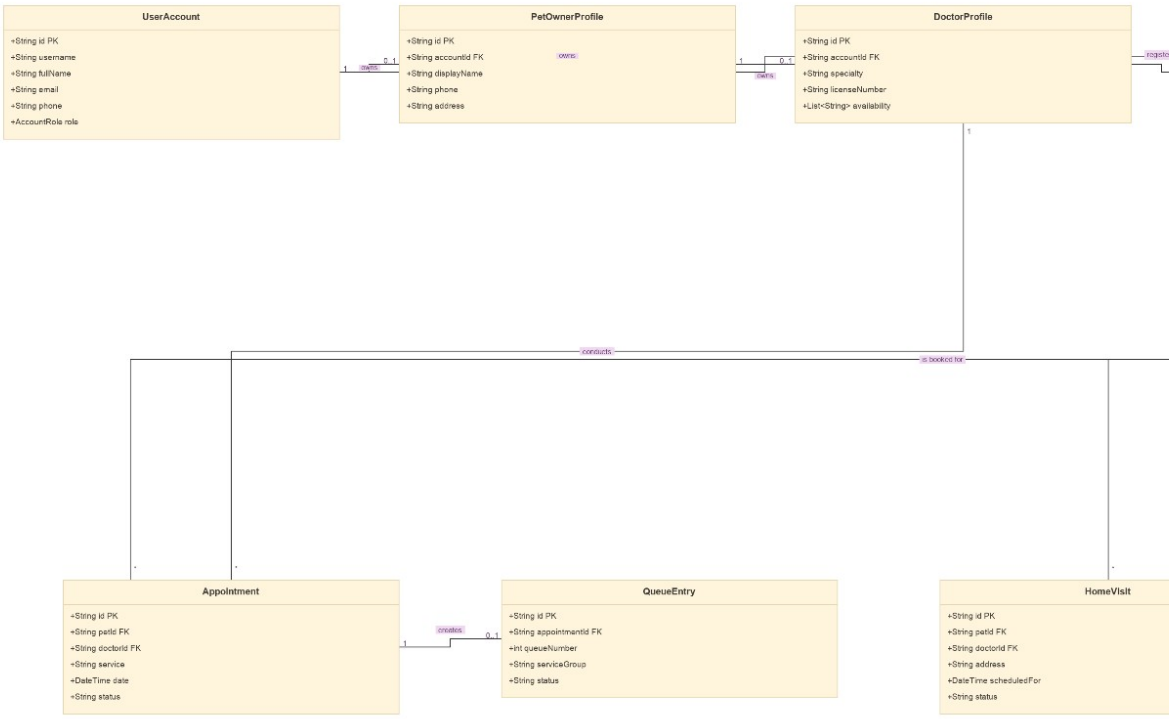



*Fig : 44 Users, Pets and Booking Class Diagram Detail 1*

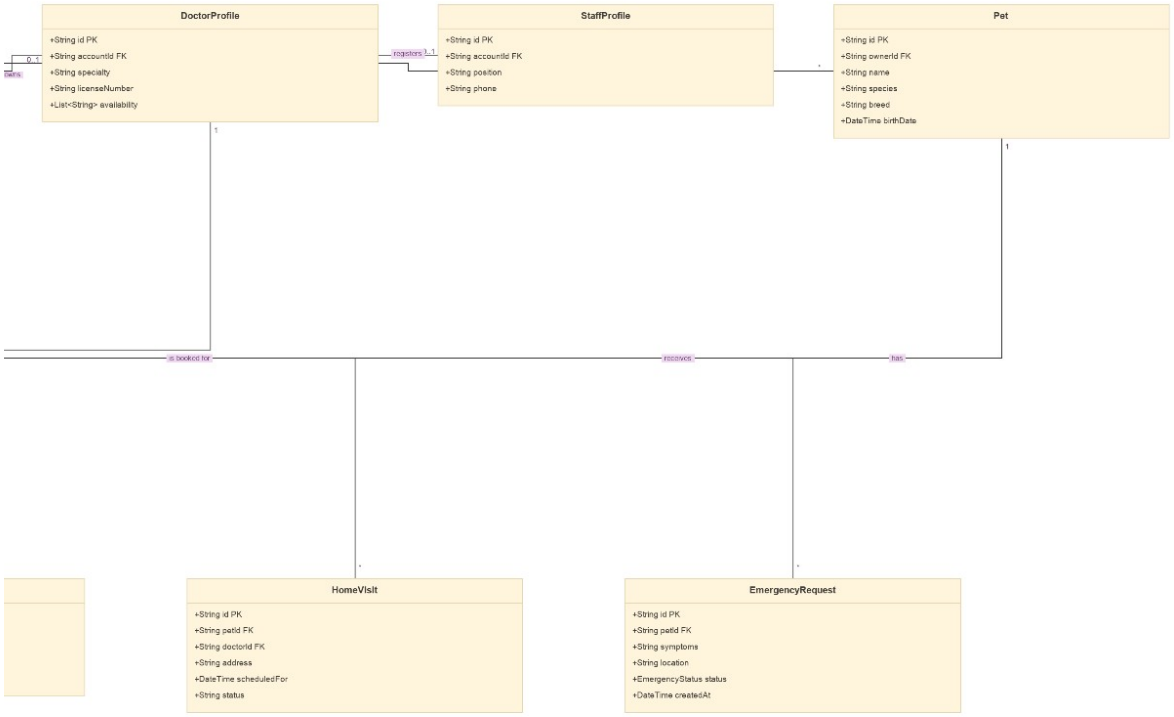



*Fig : 45 Users, Pets and Booking Class Diagram Detail 2*

## 4.2.3.4 Communication and Administration

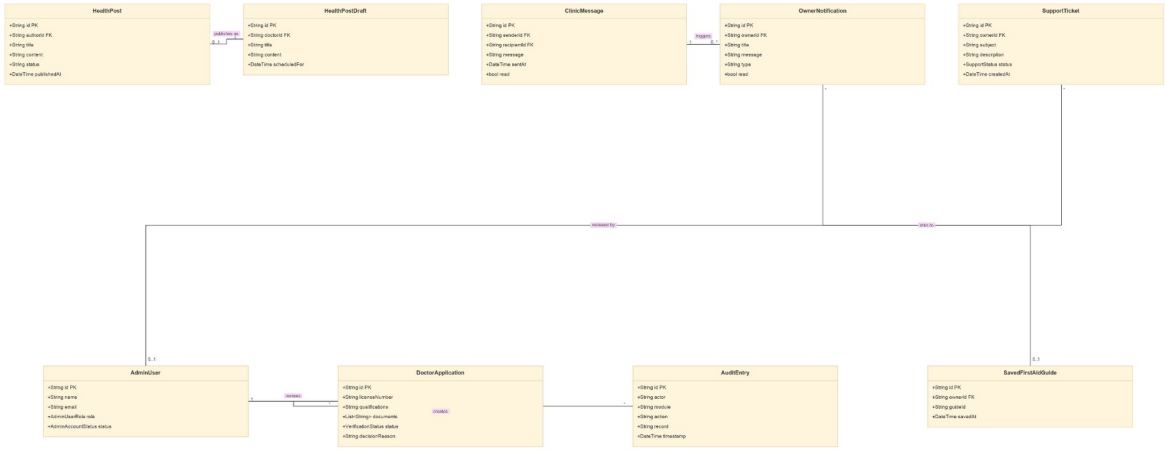



*Fig : 46 Communication and Administration Class Diagram*

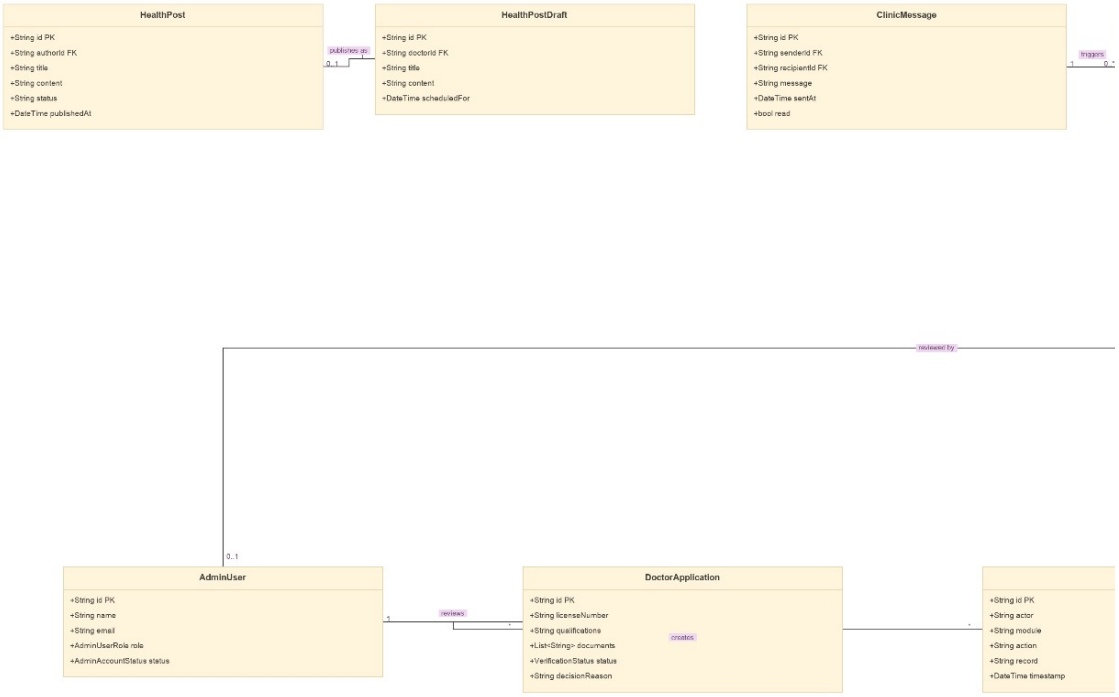



*Fig : 47 Communication and Administration Class Diagram Detail 1*

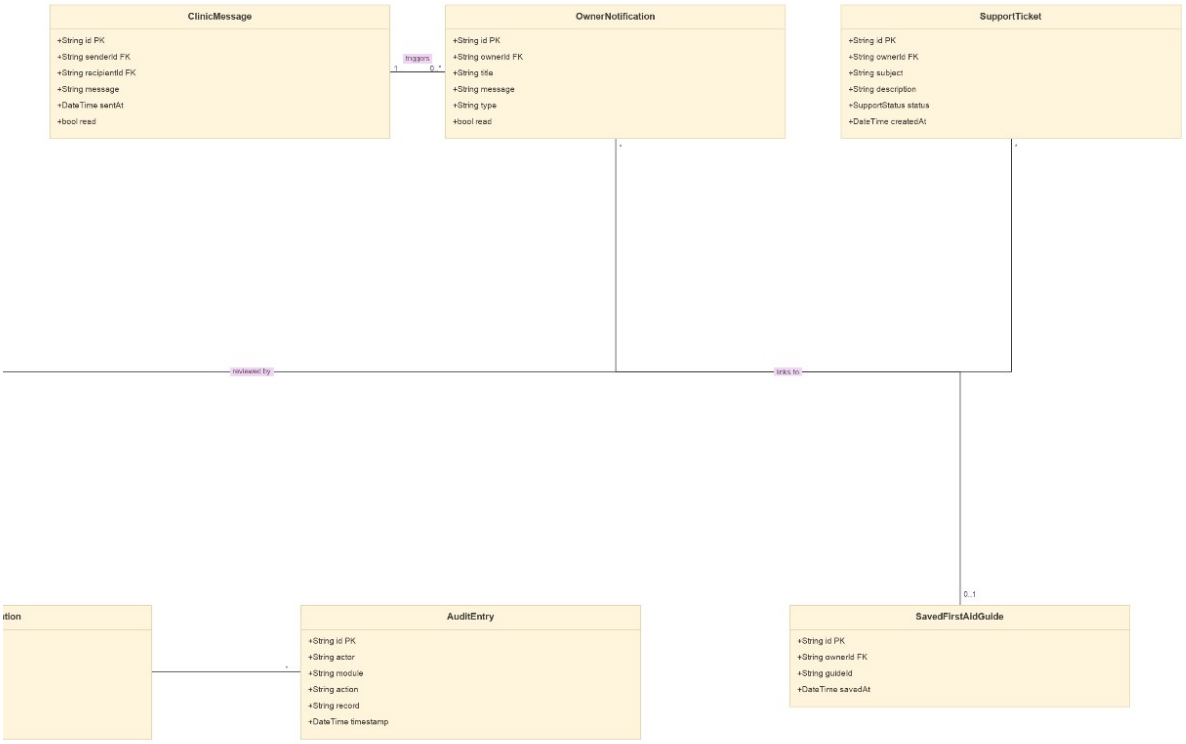



*Fig : 48 Communication and Administration Class Diagram Detail 2*

| Class | Description | Relationship / Interaction |
|---|---|---|
| UserAccount | Represents the main user account and stores username, name, email, phone, and role information. | Connects with PetOwnerProfile, DoctorProfile, and StaffProfile. |
| PetOwnerProfile | Stores the personal information of a pet owner, including phone and address. | Belongs to UserAccount and owns Pet records. |
| DoctorProfile | Represents a veterinarian and stores specialization, license number, and availability. | Belongs to UserAccount and manages Appointments and MedicalRecords. |
| StaffProfile | Represents clinic staff and stores staff position and contact information. | Belongs to UserAccount and supports clinic operations. |
| Pet | Represents a pet registered in the system, including name, species, breed, and birth date. | Belongs to a PetOwnerProfile and is involved in Appointments, HomeVisits, and EmergencyRequests. |
| Appointment | Represents a scheduled veterinary appointment with date, time, service, and status. | Connects Pet, PetOwnerProfile, DoctorProfile, |

| | | |
|---|---|---|
| | | MedicalRecord, DoctorAppointmentState, and Payment. |
| QueueEntry | Represents a pet's position and status in the clinic queue. | Works with Appointment to manage the clinic waiting queue. |
| HomeVisit | Represents a veterinary service provided at the pet owner's location. | Connects a Pet Owner and Pet with a home-visit service. |
| EmergencyRequest | Represents an emergency request submitted for a pet, including symptoms and status. | Connects Pet and PetOwnerProfile with emergency service handling. |
| MedicalRecord | Stores diagnosis, treatment, prescription, and follow-up information. | Connected to Appointment, Pet, and DoctorProfile. |
| DoctorAppointmentState | Stores the veterinarian's status, consultation notes, diagnosis, treatment, and vaccination information for an appointment. | Connected to Appointment and supports veterinarian appointment management. |
| Payment | Stores payment amount, payment method, status, and payment date. | Generated from an Appointment and can include InventoryItem purchases. |
| InventoryItem | Represents products managed by the clinic, including category, stock, unit price, and expiry date. | Connected to Payment when products are included in a transaction. |
| AppSession | Stores active user session information, including token and expiry details. | Connected to UserAccount for session management. |
| DeviceToken | Stores device tokens and platform information for notifications. | Connected to UserAccount for notification delivery. |
| SavedAddress | Stores addresses saved by pet owners. | Connected to PetOwnerProfile. |
| NotificationPreferences | Stores the user's notification preferences, including appointment and emergency alerts. | Connected to PetOwnerProfile and controls notification settings. |
| ClinicPerson | Represents clinic personnel information such as name, role, specialty, and photo. | Supports clinic personnel information and operations. |
| QueueDailyCounter | Stores the daily queue date and the latest queue number. | Supports QueueEntry in generating daily queue numbers. |
| HealthPost | Represents health-related information published by the clinic. | Connected to HealthPostDraft and managed through clinic content functions. |
| HealthPostDraft | Stores health-post content before it is published. | Used to prepare content for HealthPost. |

| | | |
|---|---|---|
| ClinicMessage | Represents messages sent between users through the clinic communication system. | Interacts with OwnerNotification and user accounts. |
| OwnerNotification | Stores notifications sent to pet owners, including title, message, and read status. | Receives information from ClinicMessage and connects with pet-owner communication. |
| SupportTicket | Represents a support request submitted by a user, including subject, description, status, and creation date. | Connected to administrative/support functions. |
| AdminUser | Represents an administrator who manages administrative functions and system access. | Interacts with DoctorApplication and administrative records. |
| DoctorApplication | Stores veterinarian application information, qualifications, documents, verification status, and decision reason. | Managed by AdminUser during veterinarian verification. |
| AuditEntry | Records system activities such as actor, module, action, and timestamp. | Supports administrative monitoring and auditing. |
| SavedFirstAidGuide | Stores first-aid guides saved by users for later access. | Connected to the user who saves the guide. |

*Table : 6 Class Diagram Table*

## 4.2.4 Data Dictionary

The data dictionary describes the main database entities used by the Nway's Love Vet Clinic Management System. It lists the column name, data type, constraints, and purpose of each field. The entity names follow the ERD presented in the project report. Because the source diagram does not specify every SQL data type and constraint, the data types and constraints below are proposed database definitions for implementation.

**App_Accounts**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| account_id | INTEGER | PK, AUTO_INCREMENT | Unique account identifier |
| username | VARCHAR(100) | NOT NULL, UNIQUE | Login username |
| email | VARCHAR(150) | NULLABLE, UNIQUE | Registered email address |
| phone | VARCHAR(30) | NULLABLE, UNIQUE | Registered phone number |
| password_hash | VARCHAR(255) | NOT NULL | Encrypted password value |
| role | VARCHAR(30) | NOT NULL | User role such as owner, doctor, staff, or admin |
| status | VARCHAR(20) | NOT NULL | Account status |
| created_at | TIMESTAMP | NOT NULL, DEFAULT CURRENT_TIMESTAMP | Account creation timestamp |

*Table : 7 App_Accounts Data Dictionary Table*

**Owner_Profiles**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| owner_id | INTEGER | PK, FK → App_Accounts.account_id | Owner account identifier |
| name | VARCHAR(150) | NOT NULL | Pet owner's name |
| phone | VARCHAR(30) | NOT NULL | Owner contact number |
| address | VARCHAR(255) | NULLABLE | Owner address |
| profile_photo | VARCHAR(255) | NULLABLE | Profile photo reference |

*Table : 8 Owner_Profiles Data Dictionary Table*

**Doctor_Profiles**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| doctor_id | INTEGER | PK, FK → App_Accounts.account_id | Doctor account identifier |
| specialization | VARCHAR(150) | NOT NULL | Veterinary specialization |
| license_no | VARCHAR(100) | NOT NULL, UNIQUE | Veterinary license number |
| availability | VARCHAR(255) | NULLABLE | Doctor availability information |
| verification_status | VARCHAR(30) | NOT NULL | Doctor verification status |

*Table : 9 Doctor_Proflies Data Dictionary Table*

**Staff_Profiles**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| staff_id | INTEGER | PK, FK → App_Accounts.account_id | Staff account identifier |
| position | VARCHAR(100) | NOT NULL | Clinic staff position |
| phone | VARCHAR(30) | NULLABLE | Staff contact number |

*Table : 10 Staff_Profiles Data Dictionary Table*

**Admin_Profiles**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| admin_id | INTEGER | PK, FK → | Administrator account identifier |

| | | | |
|---|---|---|---|
| | | App_Accounts.account_id | |
| admin_level | VARCHAR(50) | NOT NULL | Administrator access level |

*Table : 11 Admin_Profiles Data Dictionary Table*

**Pets**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| pet_id | INTEGER | PK, AUTO_INCREMENT | Unique pet identifier |
| owner_id | INTEGER | FK→Owner_Profiles.owner_id, NOT NULL | Registered pet owner |
| name | VARCHAR(100) | NOT NULL | Pet name |
| species | VARCHAR(50) | NOT NULL | Animal species |
| breed | VARCHAR(100) | NULLABLE | Pet breed |
| sex | VARCHAR(20) | NULLABLE | Pet sex |
| birth_date | DATE | NULLABLE | Pet date of birth |
| weight | DECIMAL(6,2) | NULLABLE | Pet weight |
| photo | VARCHAR(255) | NULLABLE | Pet photo reference |
| vaccine_info | TEXT | NULLABLE | Vaccination information |
| medical_notes | TEXT | NULLABLE | Additional medical information |

*Table : 12 Pets Data Dictionary Table*

**Appointments**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| appointment_id | INTEGER | PK, AUTO_INCREMENT | Unique appointment identifier |
| pet_id | INTEGER | FK → Pets.pet_id, NOT NULL | Pet receiving the appointment |
| owner_account_id | INTEGER | FK → App_Accounts.account_id, NOT NULL | Owner who makes the appointment |
| doctor_id | INTEGER | FK → Doctor_Profiles.doctor_id, NULLABLE | Assigned veterinarian |
| service | VARCHAR(100) | NOT NULL | Requested clinic service |
| appointment_date | DATE | NOT NULL | Appointment date |
| appointment_time | TIME | NOT NULL | Appointment time |

| | | | |
|---|---|---|---|
| reason | TEXT | NULLABLE | Reason for the appointment |
| status | VARCHAR(30) | NOT NULL | Appointment status |

*Table : 13 Appointments Data Dictionary Table*

**Queue_Entries**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| queue_id | INTEGER | PK, AUTO_INCREMENT | Unique queue entry identifier |
| appointment_id | INTEGER | FK → Appointments.appointment_id, UNIQUE, NOT NULL | Related appointment |
| queue_number | INTEGER | NOT NULL | Queue number |
| service_group | VARCHAR(100) | NULLABLE | Queue service group |
| status | VARCHAR(30) | NOT NULL | Current queue status |
| priority | VARCHAR(20) | NOT NULL | Normal or emergency priority |

*Table : 14 Queue_Entries Data Dictionary Table*

**Walk_In_Appointments**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| walk_in_id | INTEGER | PK, AUTO_INCREMENT | Unique walk-in record identifier |
| pet_id | INTEGER | FK → Pets.pet_id, NOT NULL | Pet for the walk-in visit |
| doctor_id | INTEGER | FK → Doctor_Profiles.doctor_id, NULLABLE | Assigned veterinarian |
| queue_id | INTEGER | FK → Queue_Entries.queue_id, NULLABLE | Related queue entry |
| visit_date | DATE | NOT NULL | Walk-in visit date |
| reason | TEXT | NULLABLE | Reason for walk-in visit |

*Table : 15 Walk_In Data Dictionary Table*

**Home_Visits**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| home_visit_id | INTEGER | PK, AUTO_INCREMENT | Unique home-visit identifier |

| | | | |
|---|---|---|---|
| owner_id | INTEGER | FK → Owner_Profiles.owner_id, NOT NULL | Pet owner requesting the visit |
| pet_id | INTEGER | FK → Pets.pet_id, NOT NULL | Pet receiving the visit |
| address | VARCHAR(255) | NOT NULL | Home-visit address |
| reason | TEXT | NULLABLE | Reason for home visit |
| visit_date | DATE | NOT NULL | Scheduled visit date |
| visit_time | TIME | NOT NULL | Scheduled visit time |
| status | VARCHAR(30) | NOT NULL | Home-visit status |

*Table : 16 Home Visit Data Dictionary Table*

**Emergency_Requests**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| emergency_id | INTEGER | PK, AUTO_INCREMENT | Unique emergency request identifier |
| owner_id | INTEGER | FK → Owner_Profiles.owner_id, NOT NULL | Owner submitting the request |
| pet_id | INTEGER | FK → Pets.pet_id, NOT NULL | Pet requiring emergency service |
| symptoms | TEXT | NOT NULL | Reported symptoms |
| description | TEXT | NULLABLE | Emergency description |
| priority | VARCHAR(20) | NOT NULL | Emergency priority |
| status | VARCHAR(30) | NOT NULL | Emergency request status |

*Table : 17 Emergency_Requests Data Dictionary Table*

**Pet_Care_Bookings**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| booking_id | INTEGER | PK, AUTO_INCREMENT | Unique pet-care booking identifier |
| owner_id | INTEGER | FK → Owner_Profiles.owner_id, NOT NULL | Owner making the booking |
| pet_id | INTEGER | FK → Pets.pet_id, NOT NULL | Pet receiving the service |
| service | VARCHAR(100) | NOT NULL | Grooming, bathing, nail trimming, boarding, or |

| | | | |
|---|---|---|---|
| | | | wellness service |
| booking_date | DATE | NOT NULL | Service date |
| booking_time | TIME | NOT NULL | Service time |
| status | VARCHAR(30) | NOT NULL | Booking status |

*Table : 18 Pet_Care_Bookings Data Dictionary Table*

**Medical_Records**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| record_id | INTEGER | PK, AUTO_INCREMENT | Unique medical record identifier |
| pet_id | INTEGER | FK → Pets.pet_id, NOT NULL | Pet associated with the record |
| doctor_id | INTEGER | FK → Doctor_Profiles.doctor_id, NOT NULL | Veterinarian managing the record |
| appointment_id | INTEGER | FK → Appointments.appointment_id, NULLABLE | Related appointment |
| diagnosis | TEXT | NULLABLE | Veterinarian diagnosis |
| treatment | TEXT | NULLABLE | Treatment provided |
| prescription | TEXT | NULLABLE | Prescription information |
| follow_up | TEXT | NULLABLE | Follow-up instructions |
| created_at | TIMESTAMP | NOT NULL, DEFAULT CURRENT_TIMESTAMP | Record creation timestamp |

*Table : 19 Medical_Records Dictionary Table*

**Payments**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| payment_id | INTEGER | PK, AUTO_INCREMENT | Unique payment identifier |
| account_id | INTEGER | FK → App_Accounts.account_id, NOT NULL | Account associated with payment |
| appointment_id | INTEGER | FK → Appointments.appointment_id, NULLABLE | Related appointment |
| amount | DECIMAL(10,2) | NOT NULL | Payment amount |
| payment_method | VARCHAR(30) | NOT NULL | Cash or digital payment |

| | | | |
|---|---|---|---|
| | | | method |
| status | VARCHAR(30) | NOT NULL | Payment status |
| payment_date | TIMESTAMP | NOT NULL, DEFAULT CURRENT_TIMESTAMP | Payment date and time |

*Table : 20 Payments Data Dictionary Table*

**Inventory**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| inventory_id | INTEGER | PK, AUTO_INCREMENT | Unique inventory record identifier |
| product_name | VARCHAR(150) | NOT NULL | Product name |
| category | VARCHAR(100) | NOT NULL | Product category |
| brand | VARCHAR(100) | NULLABLE | Product brand |
| pet_type | VARCHAR(50) | NULLABLE | Target pet type |
| price | DECIMAL(10,2) | NOT NULL | Product unit price |
| stock | INTEGER | NOT NULL | Current stock quantity |
| reorder_level | INTEGER | NOT NULL | Minimum stock threshold |
| expiry_date | DATE | NULLABLE | Product expiry date |
| description | TEXT | NULLABLE | Product description |

*Table : 21 Inventory Data Dictionary Table*

**Reminders**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| reminder_id | INTEGER | PK, AUTO_INCREMENT | Unique reminder identifier |
| account_id | INTEGER | FK → App_Accounts.account_id, NOT NULL | Account receiving the reminder |
| pet_id | INTEGER | FK → Pets.pet_id, NULLABLE | Related pet |
| appointment_id | INTEGER | FK → Appointments.appointment_id, NULLABLE | Related appointment |
| type | VARCHAR(50) | NOT NULL | Reminder type |
| reminder_date | DATETIME | NOT NULL | Reminder date and time |
| status | VARCHAR(30) | NOT NULL | Reminder status |

*Table : 22 Reminders Data Dictionary Table*

**Notification_Settings**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| setting_id | INTEGER | PK, AUTO_INCREMENT | Unique notification |

| | | | |
|---|---|---|---|
| | | | setting identifier |
| account_id | INTEGER | FK → App_Accounts.account_id, UNIQUE, NOT NULL | Account owning the settings |
| appointment_alert | BOOLEAN | NOT NULL, DEFAULT TRUE | Appointment notification preference |
| emergency_alert | BOOLEAN | NOT NULL, DEFAULT TRUE | Emergency notification preference |
| vaccination_alert | BOOLEAN | NOT NULL, DEFAULT TRUE | Vaccination notification preference |

*Table : 23 Notification_Settings Data Dictionary Table*

**Owner_Notifications**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| notification_id | INTEGER | PK, AUTO_INCREMENT | Unique notification identifier |
| account_id | INTEGER | FK → App_Accounts.account_id, NOT NULL | Owner receiving notification |
| title | VARCHAR(150) | NOT NULL | Notification title |
| message | TEXT | NOT NULL | Notification message |
| is_read | BOOLEAN | NOT NULL, DEFAULT FALSE | Read status |
| created_at | TIMESTAMP | NOT NULL, DEFAULT CURRENT_TIMESTAMP | Notification creation time |

*Table : 24 Owner_Notifications Data Dictionary Table*

**Appointment_Reminders**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| appointment_reminder_id | INTEGER | PK, AUTO_INCREMENT | Unique appointment reminder identifier |
| appointment_id | INTEGER | FK → Appointments.appointment_id, NOT NULL | Related appointment |
| account_id | INTEGER | FK → App_Accounts.account_id, NOT NULL | Account receiving reminder |

| | | | |
|---|---|---|---|
| reminder_time | DATETIME | NOT NULL | Scheduled reminder time |
| status | VARCHAR(30) | NOT NULL | Reminder status |

*Table : 25 Appointment_Reminders Data Dictionary Table*

**Health_Posts**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| post_id | INTEGER | PK, AUTO_INCREMENT | Unique health-post identifier |
| account_id | INTEGER | FK → App_Accounts.account_id, NOT NULL | Account managing the post |
| title | VARCHAR(200) | NOT NULL | Health-post title |
| category | VARCHAR(100) | NOT NULL | Post category |
| content | TEXT | NOT NULL | Post content |
| cover_image | VARCHAR(255) | NULLABLE | Cover image reference |
| published_at | DATETIME | NULLABLE | Publication date and time |

*Table : 26 Health_Posts Data Dictionary Table*

**Health_Post_Drafts**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| draft_id | INTEGER | PK, AUTO_INCREMENT | Unique draft identifier |
| account_id | INTEGER | FK → App_Accounts.account_id, NOT NULL | Account creating the draft |
| title | VARCHAR(200) | NOT NULL | Draft title |
| content | TEXT | NULLABLE | Draft content |
| status | VARCHAR(30) | NOT NULL | Draft status |
| updated_at | TIMESTAMP | NOT NULL, DEFAULT CURRENT_TIMESTAMP | Last update timestamp |

*Table : 27 Health_Posts_Drafts Data Dictionary Table*

**Clinic_Messages**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| message_id | INTEGER | PK, AUTO_INCREMENT | Unique clinic message identifier |
| account_id | INTEGER | FK → App_Accounts.account_id, NOT NULL | Associated user account |

| | | | |
|---|---|---|---|
| message | TEXT | NOT NULL | Message content |
| created_at | TIMESTAMP | NOT NULL, DEFAULT CURRENT_TIMESTAMP | Message creation time |

*Table : 28 Clinic_Messages Data Dictionary Table*

**Doctor_Notifications**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| notification_id | INTEGER | PK, AUTO_INCREMENT | Unique doctor notification identifier |
| account_id | INTEGER | FK → App_Accounts.account_id, NOT NULL | Doctor account receiving notification |
| title | VARCHAR(150) | NOT NULL | Notification title |
| message | TEXT | NOT NULL | Notification message |
| is_read | BOOLEAN | NOT NULL, DEFAULT FALSE | Read status |

*Table : 29 Doctor_Notifications Data Dictionary Table*

**Support_Tickets**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| ticket_id | INTEGER | PK, AUTO_INCREMENT | Unique support ticket identifier |
| account_id | INTEGER | FK → App_Accounts.account_id, NOT NULL | Account submitting the ticket |
| subject | VARCHAR(200) | NOT NULL | Support ticket subject |
| description | TEXT | NOT NULL | Support request description |
| status | VARCHAR(30) | NOT NULL | Ticket status |
| created_at | TIMESTAMP | NOT NULL, DEFAULT CURRENT_TIMESTAMP | Ticket creation time |

*Table : 30 Support_Tickets Data Dictionary Table*

**History_Reviews**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| review_id | INTEGER | PK, AUTO_INCREMENT | Unique review identifier |
| account_id | INTEGER | FK → App_Accounts.account_id, NOT NULL | Account creating the review |
| rating | INTEGER | NOT NULL | Review rating |

| | | | |
|---|---|---|---|
| comment | TEXT | NULLABLE | Review comment |
| created_at | TIMESTAMP | NOT NULL, DEFAULT CURRENT_TIMESTAMP | Review creation time |

*Table : 31 History_Reviews Data Dictionary Table*

**Saved_First_Aid_Guides**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| saved_guide_id | INTEGER | PK, AUTO_INCREMENT | Unique saved-guide identifier |
| account_id | INTEGER | FK → App_Accounts.account_id, NOT NULL | User who saved the guide |
| guide_id | INTEGER | NOT NULL | First-aid guide identifier |
| saved_at | TIMESTAMP | NOT NULL, DEFAULT CURRENT_TIMESTAMP | Date and time saved |

*Table : 32 Saved_First_Aid_Guides Data Dictionary Table*

**Clinic_Directory**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| directory_id | INTEGER | PK, AUTO_INCREMENT | Unique directory record identifier |
| account_id | INTEGER | FK → App_Accounts.account_id, NOT NULL | Associated account |
| clinic_name | VARCHAR(200) | NOT NULL | Clinic name |
| address | VARCHAR(255) | NULLABLE | Clinic address |
| phone | VARCHAR(30) | NULLABLE | Clinic contact number |
| opening_hours | VARCHAR(150) | NULLABLE | Clinic opening hours |

*Table : 33 Clinic_Directory Data Dictionary Table*

**Password_Reset_Codes**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| reset_code_id | INTEGER | PK, AUTO_INCREMENT | Unique reset-code identifier |
| account_id | INTEGER | FK → App_Accounts.account_id, NOT NULL | Account requesting password reset |
| code | VARCHAR(20) | NOT NULL | Temporary password reset code |

| | | | |
|---|---|---|---|
| expires_at | DATETIME | NOT NULL | Code expiry date and time |
| used_at | DATETIME | NULLABLE | Time the code was used |

*Table : 34 Password_Reset_Codes Data Dictionary Table*

**Password_Reset_Audit**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| audit_id | INTEGER | PK, AUTO_INCREMENT | Unique password-reset audit identifier |
| account_id | INTEGER | FK → App_Accounts.account_id, NOT NULL | Account involved in reset activity |
| action | VARCHAR(50) | NOT NULL | Password reset action |
| created_at | TIMESTAMP | NOT NULL, DEFAULT CURRENT_TIMESTAMP | Audit timestamp |

*Table : 35 Password_Reset_Audit Data Dictionary Table*

**Device_Tokens**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| device_token_id | INTEGER | PK, AUTO_INCREMENT | Unique device-token identifier |
| account_id | INTEGER | FK → App_Accounts.account_id, NOT NULL | Account owning the device |
| token | VARCHAR(255) | NOT NULL, UNIQUE | Device notification token |
| platform | VARCHAR(30) | NOT NULL | Device platform |
| created_at | TIMESTAMP | NOT NULL, DEFAULT CURRENT_TIMESTAMP | Token registration time |

*Table : 36 Device_Tokens Data Dictionary Table*

**App_Sessions**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| session_id | INTEGER | PK, AUTO_INCREMENT | Unique session identifier |
| account_id | INTEGER | FK → App_Accounts.account_id, NOT NULL | Account owning the session |
| token | VARCHAR(255) | NOT NULL, UNIQUE | Session token |
| expires_at | DATETIME | NOT NULL | Session expiry time |

| | | | |
|---|---|---|---|
| created_at | TIMESTAMP | NOT NULL, DEFAULT CURRENT_TIMESTAMP | Session creation time |

*Table : 37 App_Sessions Data Dictionary Table*

**Doctor_Verifications**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| verification_id | INTEGER | PK, AUTO_INCREMENT | Unique doctor verification identifier |
| doctor_id | INTEGER | FK → Doctor_Profiles.doctor_id, NOT NULL | Doctor being verified |
| status | VARCHAR(30) | NOT NULL | Verification status |
| license_expiry | DATE | NULLABLE | License expiry date |
| documents | VARCHAR(255) | NULLABLE | Submitted document reference |
| decision_reason | TEXT | NULLABLE | Reason for approval or rejection |

*Table : 38 Doctor_Verifications Data Dictionary Table*

**Audit_Logs**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| audit_log_id | INTEGER | PK, AUTO_INCREMENT | Unique audit-log identifier |
| account_id | INTEGER | FK → App_Accounts.account_id, NOT NULL | Account performing the action |
| module | VARCHAR(100) | NOT NULL | System module affected |
| action | VARCHAR(100) | NOT NULL | Action performed |
| timestamp | TIMESTAMP | NOT NULL, DEFAULT CURRENT_TIMESTAMP | Audit timestamp |

*Table : 39 Audit_Logs Data Dictionary Table*

**Doctor_Appointment_State**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| state_id | INTEGER | PK, AUTO_INCREMENT | Unique appointment-state identifier |
| appointment_id | INTEGER | FK → Appointments.appointment_id, NOT NULL | Related appointment |

| | | | |
|---|---|---|---|
| doctor_id | INTEGER | FK → Doctor_Profiles.doctor_id, NOT NULL | Doctor managing the appointment |
| status | VARCHAR(30) | NOT NULL | Doctor appointment state |
| consultation_notes | TEXT | NULLABLE | Consultation notes |
| diagnosis | TEXT | NULLABLE | Diagnosis information |
| treatment | TEXT | NULLABLE | Treatment information |
| vaccination | TEXT | NULLABLE | Vaccination information |

*Table : 40 Doctor_Appointment_State Data Dictionary Table*

**Queue_Daily_Counters**

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| counter_id | INTEGER | PK, AUTO_INCREMENT | Unique daily counter identifier |
| queue_date | DATE | NOT NULL, UNIQUE | Date for queue numbering |
| last_queue_number | INTEGER | NOT NULL | Latest queue number generated |

*Table : 41 Queue_Daily_Counters Data Dictionary Table*

## 4.3. System Architecture

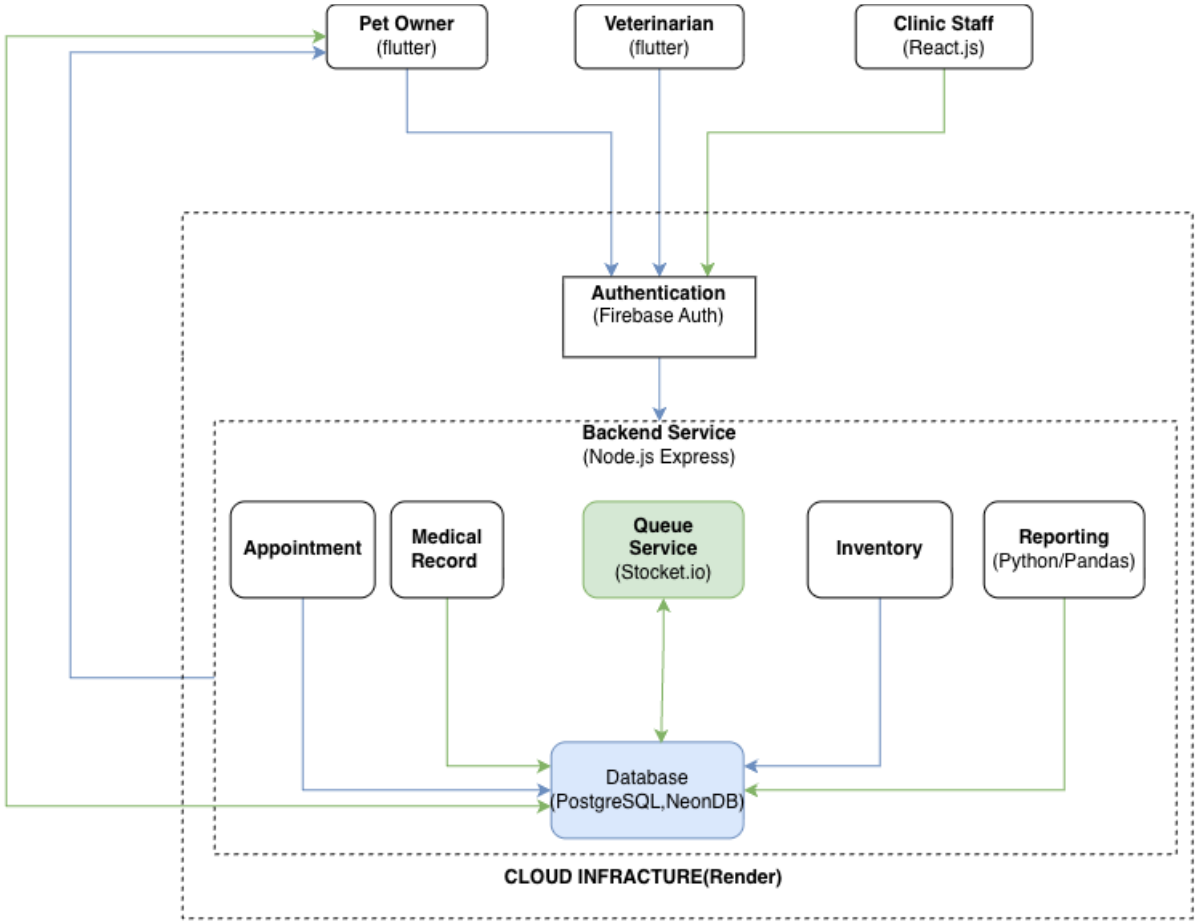



*Fig : 48 System Architecture Diagram*

The system follows a layered client-server architecture consisting of the presentation layer, application logic layer, and data layer. The architecture connects the user interfaces with the backend services, authentication service, database, and cloud infrastructure. The system architecture diagram shows Flutter interfaces for Pet Owners and Veterinarians, a React.js interface for Clinic Staff, Firebase Authentication, a Node.js and Express backend, PostgreSQL/NeonDB, and supporting services.

## Presentation Layer

The presentation layer provides the user interface for the different system roles. The Pet Owner and Veterinarian applications are developed using Flutter, while the Clinic Staff interface is developed using React.js. The interfaces allow users to access functions according to their assigned roles, such as appointments, medical records, queue information, inventory, and management features.

## Application Logic Layer

The application logic layer is handled by the Node.js and Express backend. It receives requests from the client applications, processes system logic, and coordinates the main services. These services include appointments, medical records, queue management,

inventory, and reporting. Socket.io is used for the queue service to support real-time communication.

### Data Layer

The data layer uses PostgreSQL through NeonDB as the main database shown in the system architecture. It stores and manages system information required by the application, including user information, pet data, appointments, medical records, inventory data, and other operational records. Python and Pandas are used for reporting and data processing.

### Authentication and Cloud Infrastructure

Firebase Authentication provides authentication for system users. The backend and database services are deployed within the cloud infrastructure using Render. This structure allows the client applications to communicate with the backend while the backend communicates with the database and supporting services.

Overall Architecture:

- Pet Owner / Veterinarian → Flutter
- Clinic Staff → React.js
- Flutter / React.js → Firebase Authentication
- Flutter / React.js → Node.js + Express Backend
- Backend → Appointment | Medical Record | Queue | Inventory | Reporting
- Backend Services → PostgreSQL / NeonDB
- System → Render Cloud Infrastructure

## 4.4. Technology Used

| Component | Technology | Purpose |
|---|---|---|
| Frontend | Flutter | Development of the Pet Owner and Veterinarian interfaces. |
| Staff Frontend | React.js | Development of the Clinic Staff web interface. |
| Authentication | Firebase Authentication | User authentication and account access. |
| Backend | Node.js + Express.js | API services, application logic, request processing, and communication. |
| Database | PostgreSQL / NeonDB | Storage and management of clinic and operational data. |
| Real-time Service | Socket.io | Real-time queue communication and updates. |
| Reporting | Python / Pandas | Reporting and data processing. |
| Cloud Infrastructure | Render | Cloud deployment and hosting of system services. |
| UI/UX Design | Figma | Interface design, wireframes, prototypes, and visual planning. |

*Table : 42 Technology Used Table*

The selected technologies support the different parts of the Nway's Love Vet Clinic Management System. Flutter provides the main cross-platform interfaces for Pet Owners and Veterinarians, while React.js is used for the Clinic Staff interface. Node.js with Express.js provides the backend services and application logic. Firebase Authentication manages user authentication, and PostgreSQL through NeonDB provides the main data storage. Socket.io supports real-time queue updates, while Python and Pandas are used for reporting and data processing. Render provides the cloud infrastructure for deployment. Figma is used during the UI/UX design stage to plan interfaces, layouts, and prototypes.

Flutter

Flutter is used to develop the Pet Owner and Veterinarian interfaces and provide a consistent cross-platform application experience.

React.js

React.js is used to develop the Clinic Staff web interface and provide access to clinic management functions.

Firebase Authentication

Firebase Authentication is used to manage user authentication and account access within the system.

Node.js and Express.js

Node.js and Express.js are used for the backend of the system. They handle application requests, system logic, and communication between the client applications and database services.

PostgreSQL / NeonDB

PostgreSQL through NeonDB is used as the main database for storing and managing system information and operational records.

Socket.io

Socket.io is used to support real-time communication for the queue service, allowing queue information to be updated between the system components.

Python / Pandas

Python and Pandas are used for reporting and data processing within the system.

Render

Render provides the cloud infrastructure used to deploy and host the system services.

Figma

Figma is used during the UI/UX design process to create and organize interface designs, wireframes, prototypes, and navigation flows before implementation.

## 4.5 Site / Navigation Structure

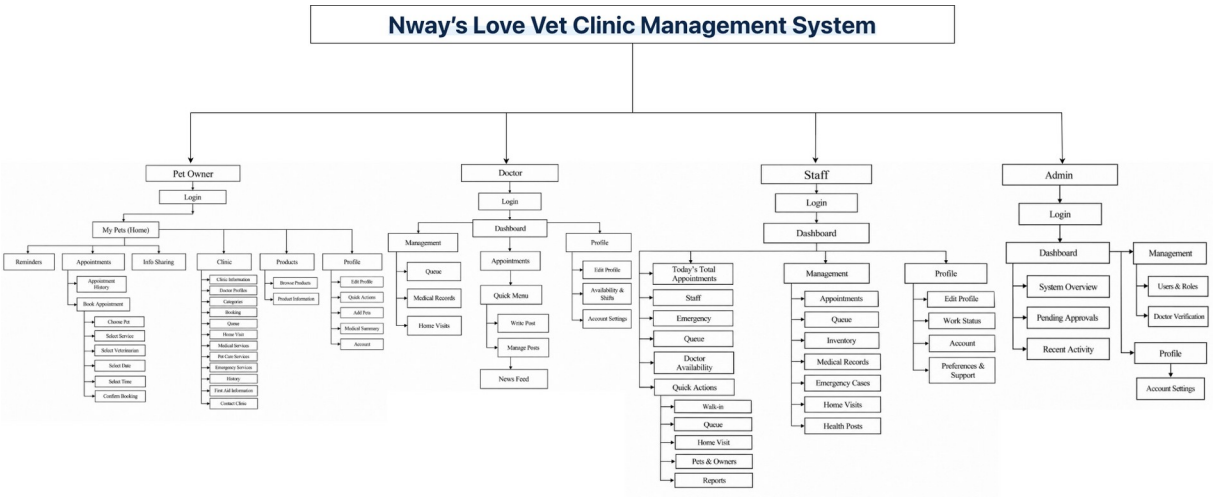



*Fig : 49 Sitemap Diagram*

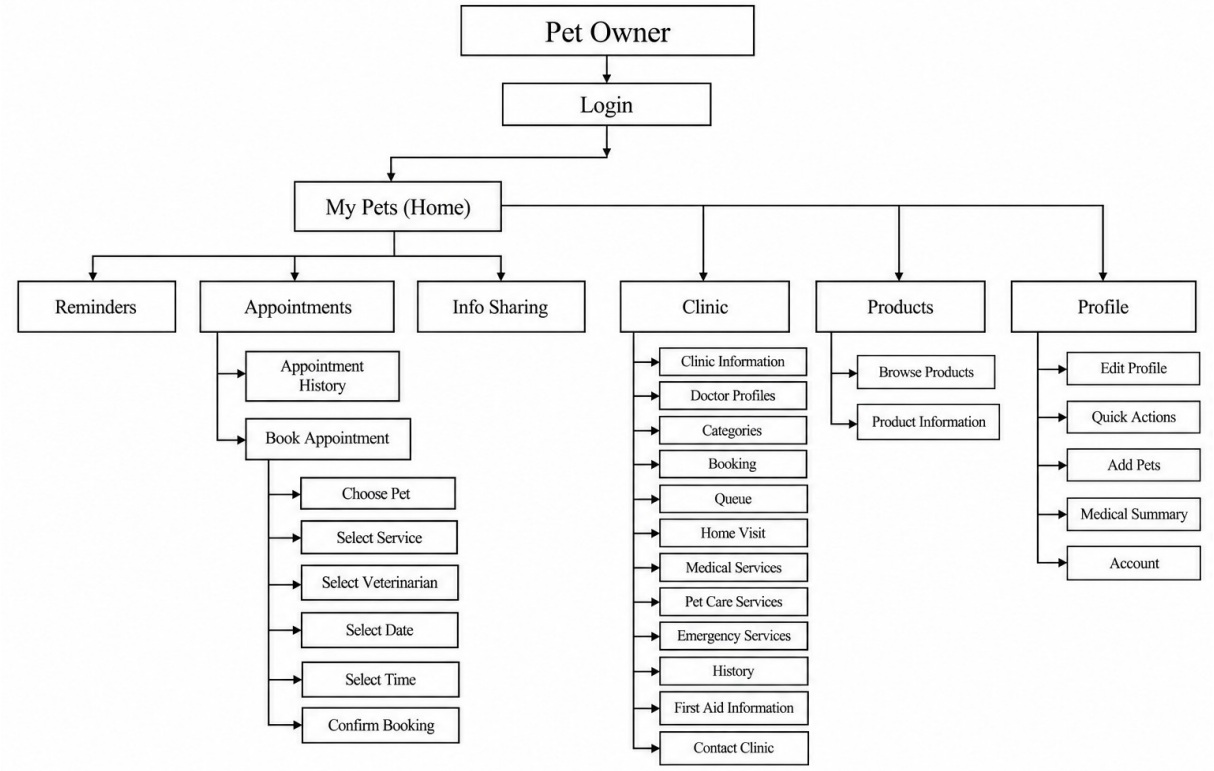



*Fig : 50 Sitemap Diagram Detail 1 (Pet Owner)*

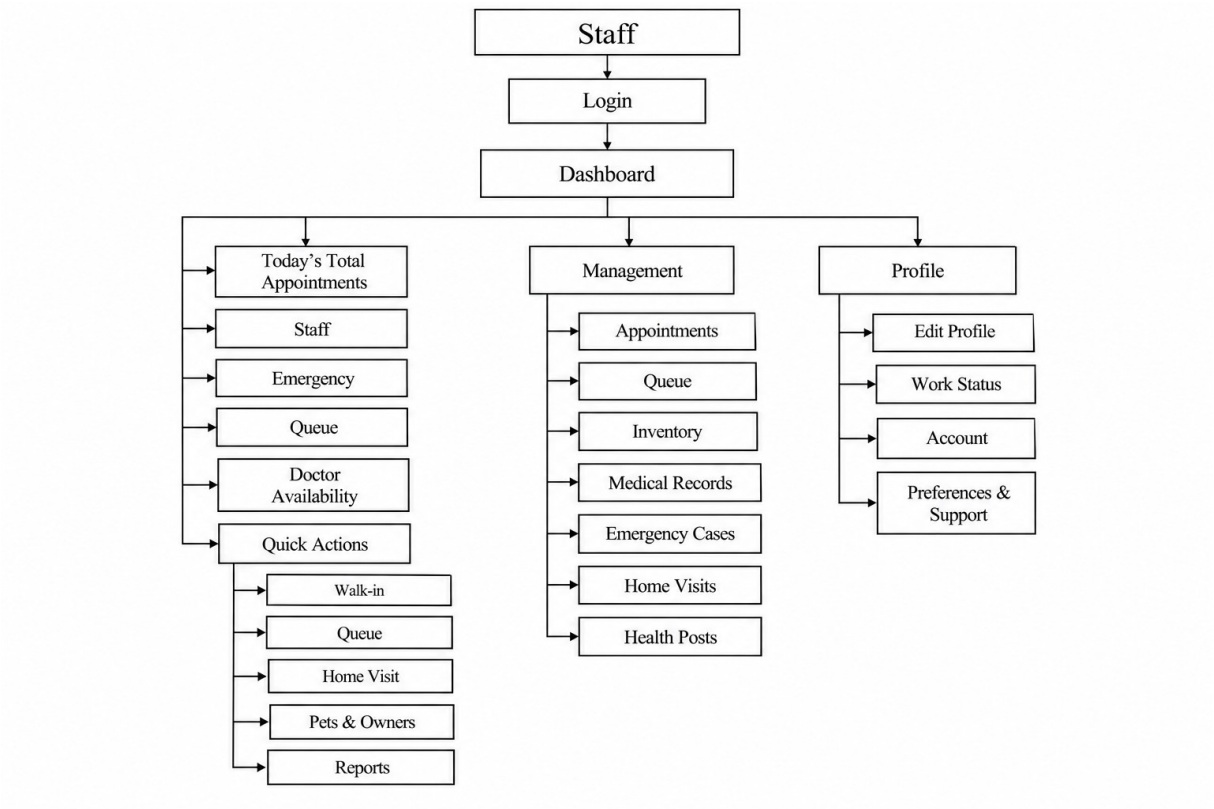



*Fig : 51 Sitemap Diagram Detail 2 (Staff)*

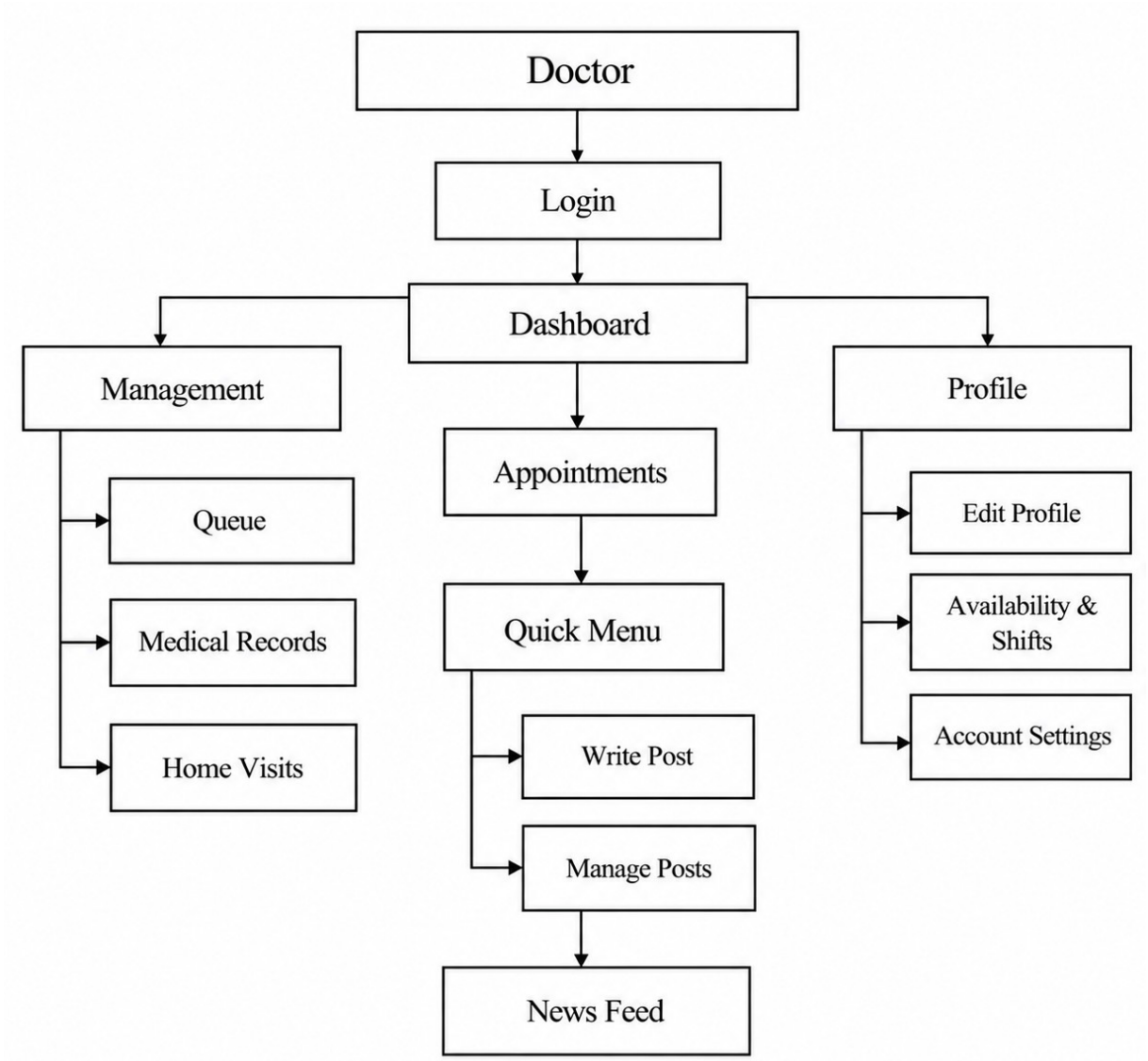



*Fig : 52 Sitemap Diagram Detail 3 (Doctor)*

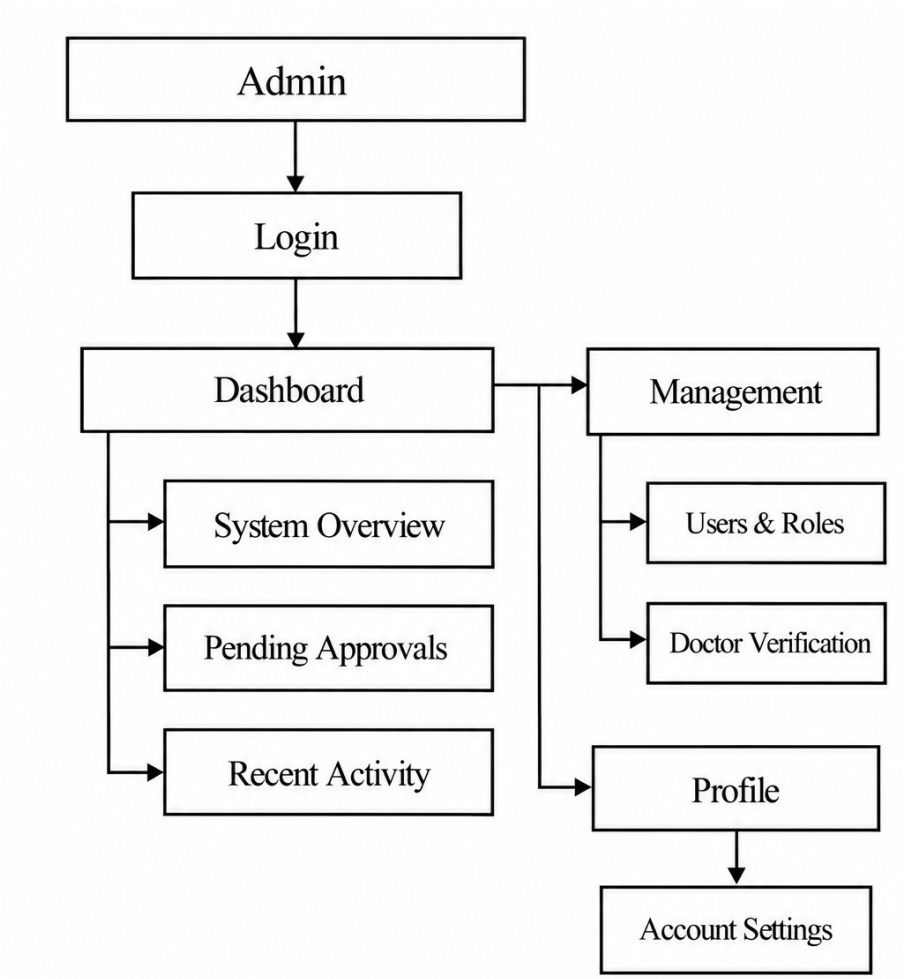


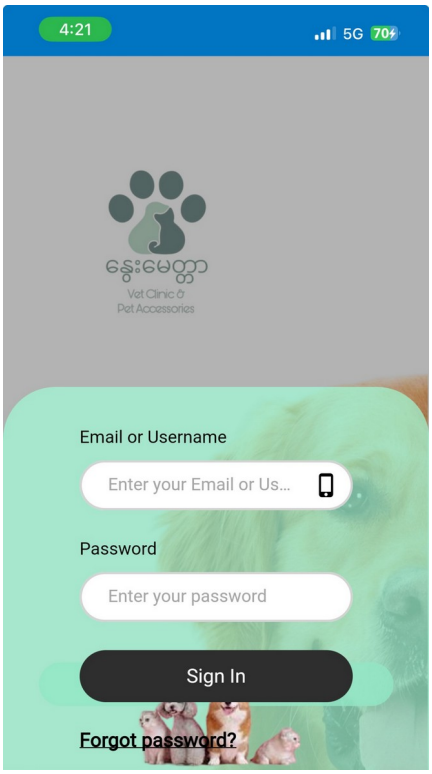

*Fig : 53 Sitemap Diagram Detail 4 (Admin)*

## 4.6 UX/UI Design

The UX/UI design focuses on making the system simple, organized, and easy to use for different types of users.

The design separates functions according to user roles so that each user can quickly access the features relevant to their responsibilities.

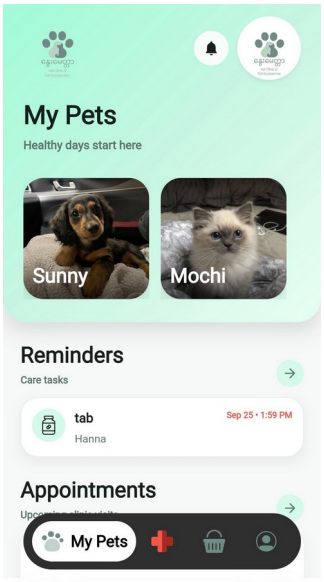
### 4.6.1 Login

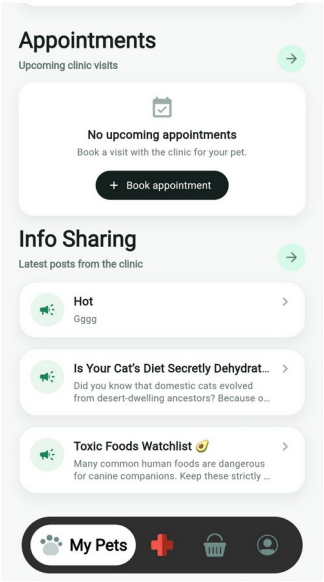



The system provides secure user authentication to ensure that only registered users can access the application. Users are required to log in using their registered credentials, while new users can create an account through the registration process. The system also implements role-based access control, which provides different permissions and features based on the user's assigned role, such as pet owner, staff, doctor, or administrator. This helps protect sensitive information and ensures that each user can only access the functions relevant to their responsibilities.

## 4.6.2 Main UI Design of Pet Owners Side

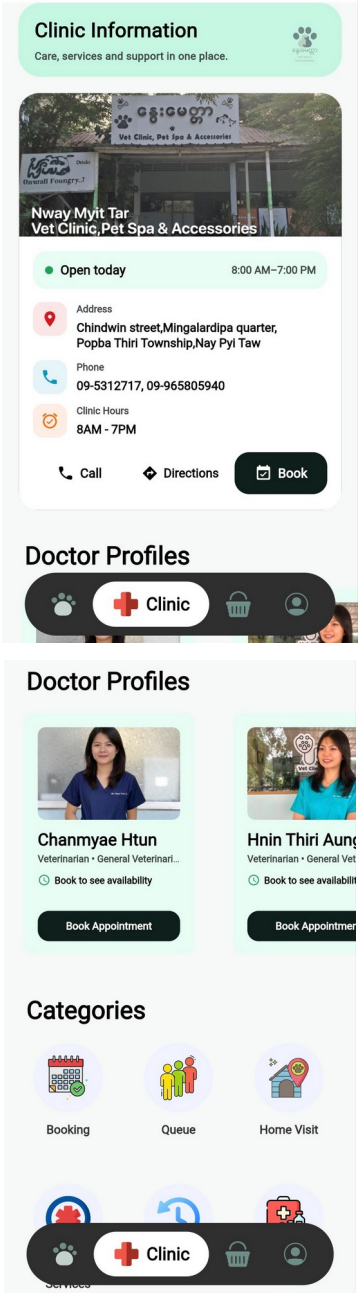
### 4.6.2.1 My pets



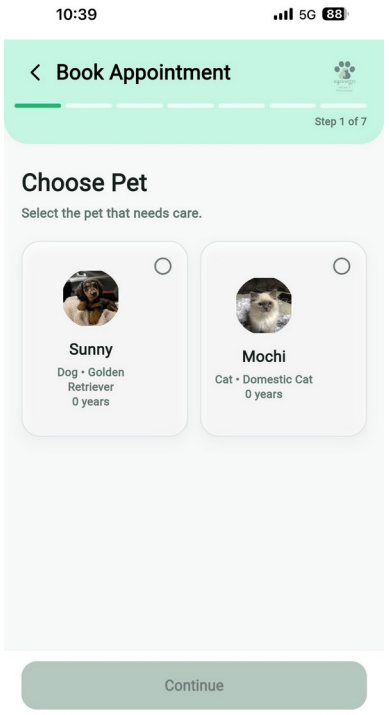

The My Pets (Home) section is the main area for pet owners. It provides access to Reminders, Appointments, and Info Sharing. Under Appointments, pet owners can view their Appointment History or book a new appointment.

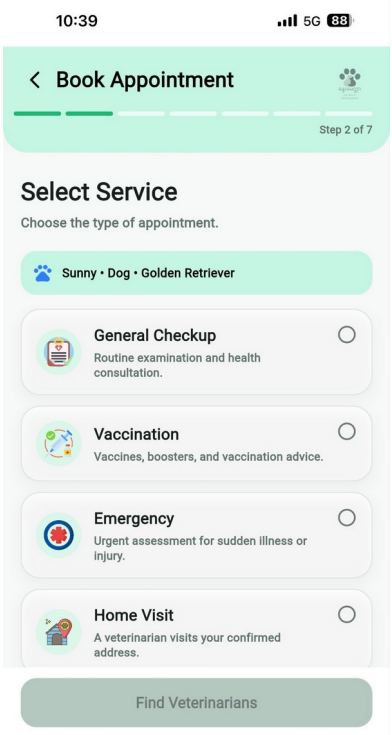
## 4.6.2.2 Clinic Page

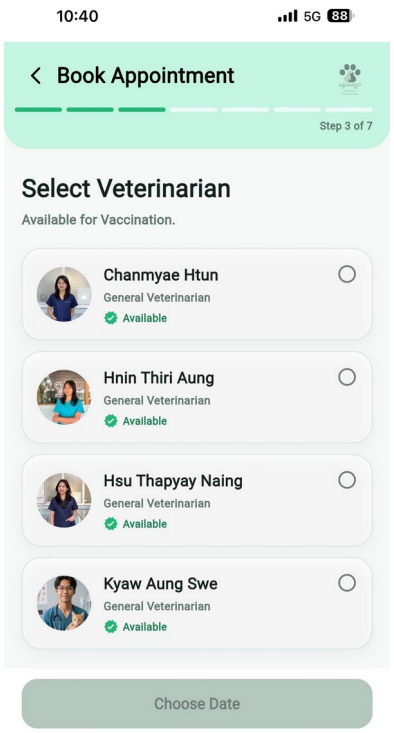


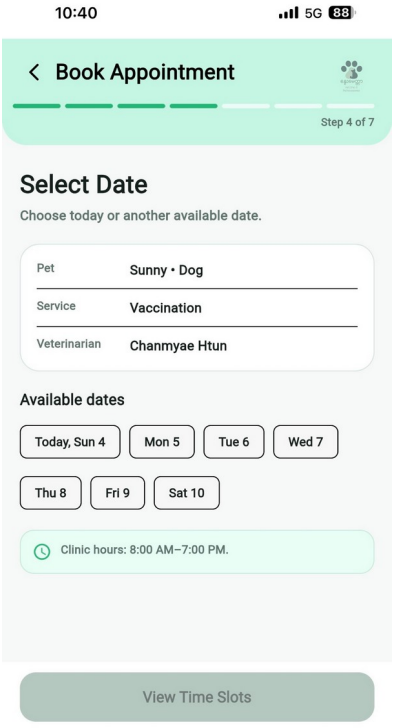

The **Clinic Information** section provides pet owners with important information about Nway Myit Tar Vet Clinic, including the clinic's location, contact numbers, opening hours, and available services. Users can also view **Doctor Profiles**, where they can see veterinarian information and book an appointment with a selected doctor. The section includes different service categories such as **Booking, Queue, Home Visit**, and other clinic services. This allows pet owners to easily access clinic information, learn about available veterinarians and services, and make appropriate decisions before booking an appointment.

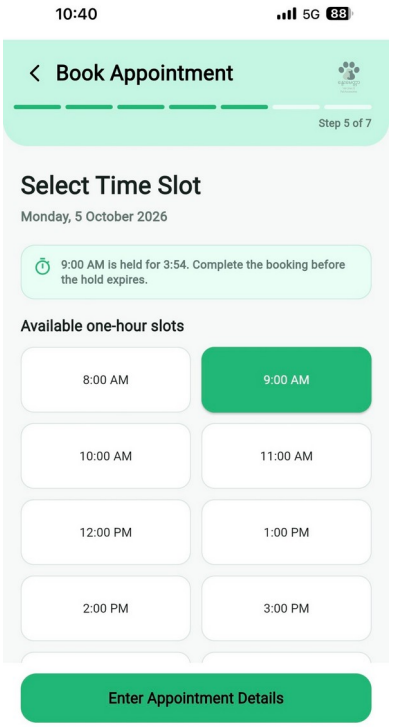
## 4.6.2.3 Booking Processes

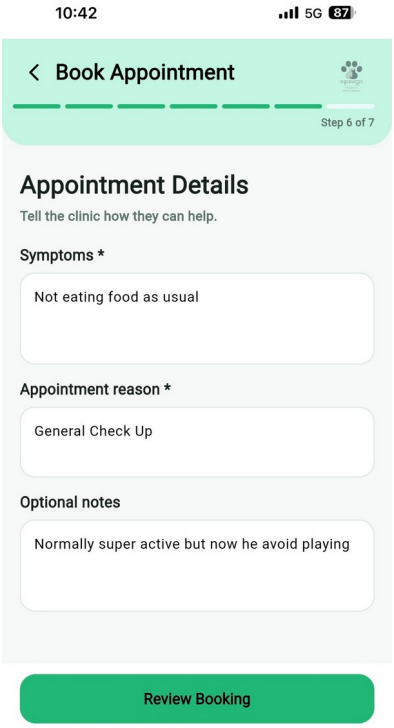


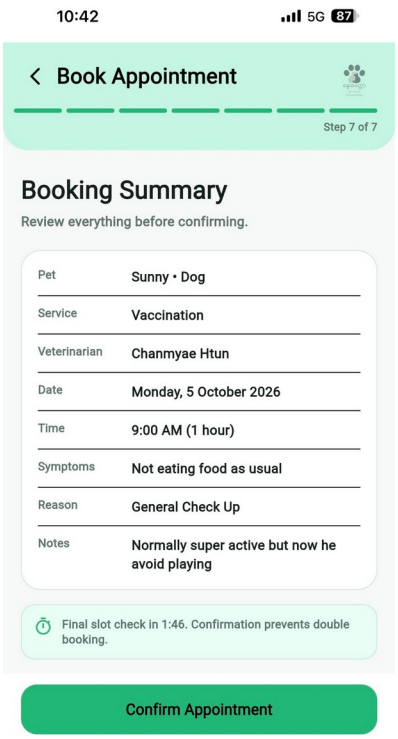



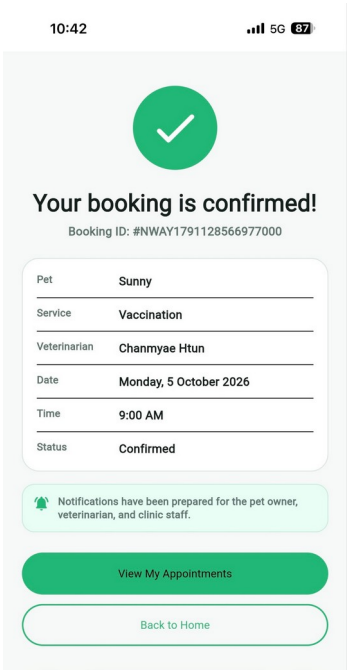

10:40
5G 88
Book Appointment
Step 3 of 7
Select Veterinarian
Available for Vaccination.
Chanmyae Htun
General Veterinarian
Available
Hnin Thiri Aung
General Veterinarian
Available
Hsu Thapyay Naing
General Veterinarian
Available
Kyaw Aung Swe
General Veterinarian
Available
Choose Date


10:40
5G 88
Book Appointment
Step 4 of 7
Select Date
Choose today or another available date.
Pet
Sunny • Dog
Service
Vaccination
Veterinarian
Chanmyae Htun
Available dates
Today, Sun 4
Mon 5
Tue 6
Wed 7
Thu 8
Fri 9
Sat 10
Clinic hours: 8:00 AM–7:00 PM.
View Time Slots


10:40
5G 88
Book Appointment
Step 5 of 7
Select Time Slot
Monday, 5 October 2026
9:00 AM is held for 3:54. Complete the booking before the hold expires.
Available one-hour slots
8:00 AM
9:00 AM
10:00 AM
11:00 AM
12:00 PM
1:00 PM
2:00 PM
3:00 PM
Enter Appointment Details

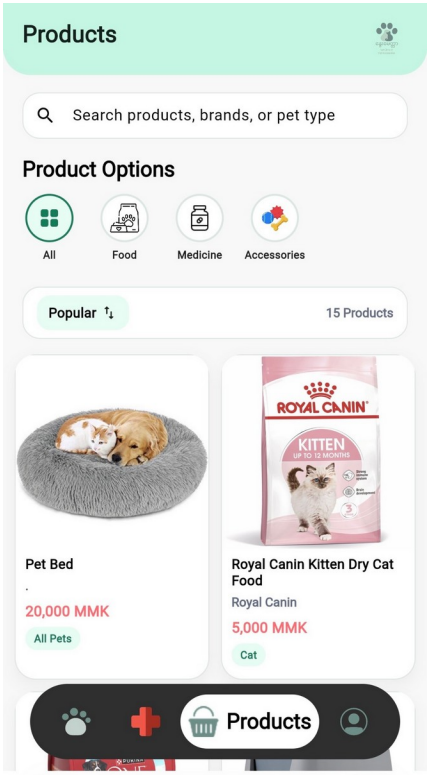

10:42
5G 87
Book Appointment
Step 6 of 7
Appointment Details
Tell the clinic how they can help.
Symptoms *
Not eating food as usual
Appointment reason *
General Check Up
Optional notes
Normally super active but now he avoid playing
Review Booking

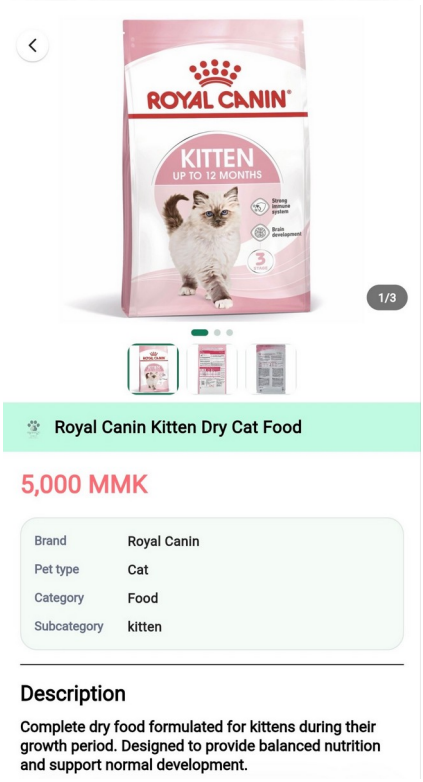


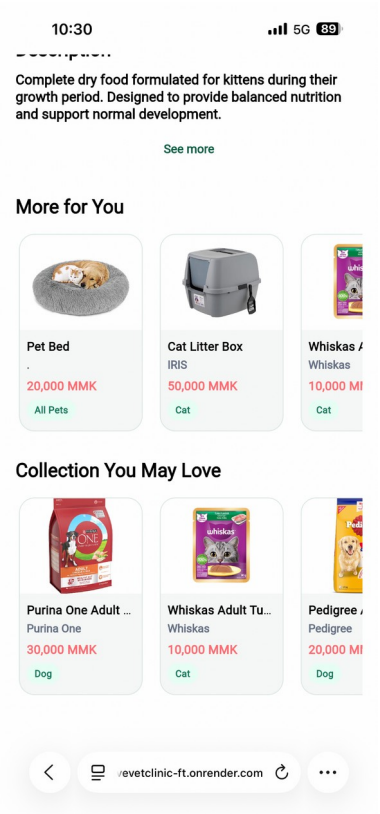




The booking process allows them to choose a pet, select a service, select a veterinarian, choose a date and time, and confirm the booking. This structure makes it easier for pet owners to manage their pets, appointments, and clinic-related information from one central location.

## 4.6.2.4 Product Page (Pet Owners)

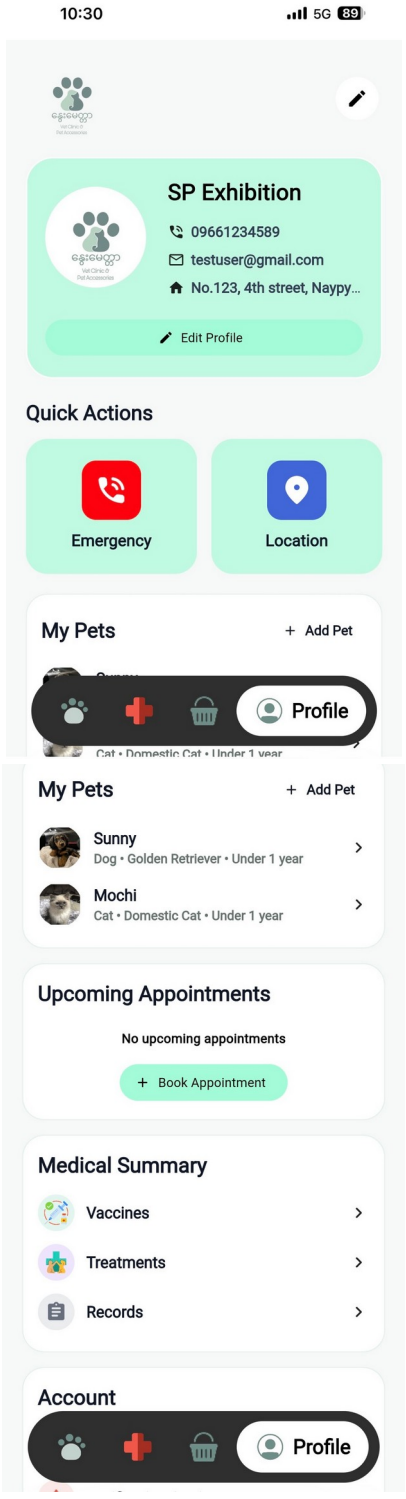


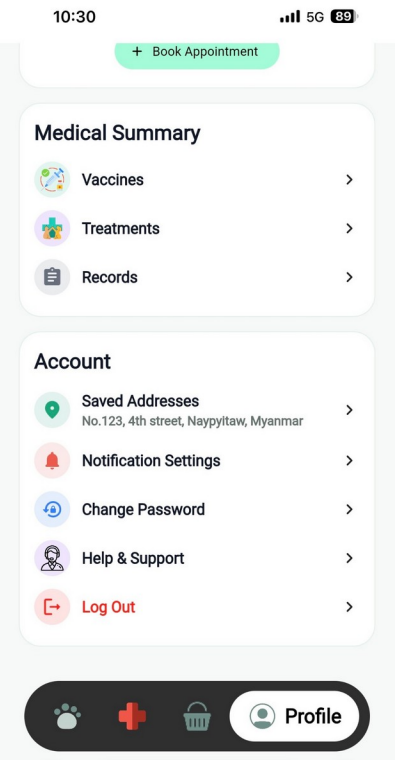






The **Products** section allows pet owners to browse and search for pet-related products available through the clinic system. Products are organized into categories such as **Food, Medicine, and Accessories**, with an option to view all products or search by product name, brand, or pet type. Users can select a product to view detailed information, including its **price, brand, pet type, category, subcategory, product images, and description**. The system also provides related product recommendations through sections such as **"More for You"** and **"Collection You May Love,"** making it easier for pet owners to discover other suitable products for their pets.

## 4.6.2.5 Profile Page (Pet Owners)

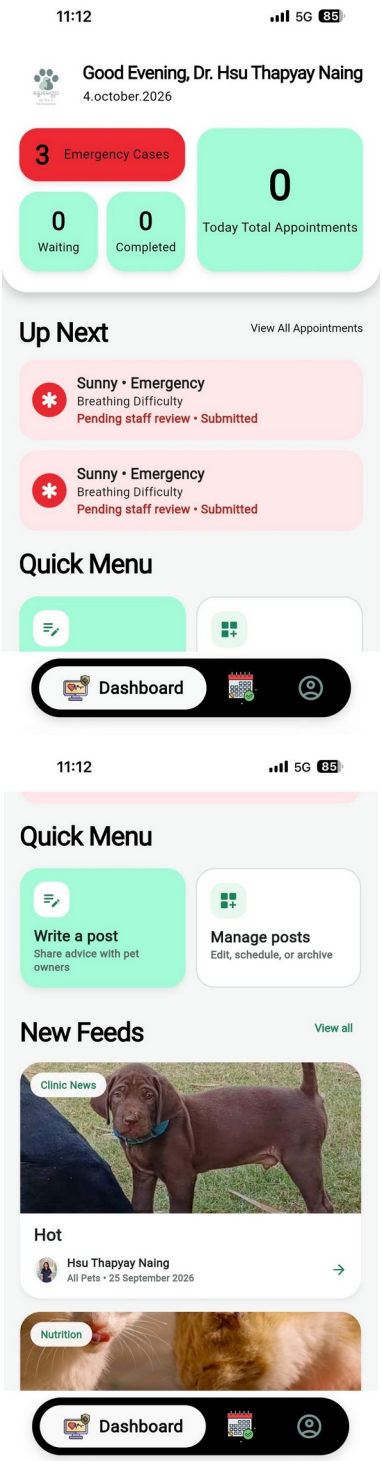





The **Profile** section allows pet owners to manage their personal information, pets, appointments, medical information, and account settings. Users can edit their profile information and access quick actions such as **Emergency** and **Location**. The **My Pets** section displays registered pets and allows users to add new pets. Users can also view **Upcoming Appointments**and book an appointment when needed. The **Medical Summary** provides access to **Vaccines, Treatments, and Records** for managing important pet health information. In addition, the **Account** section includes **Saved Addresses, Notification Settings, Change Password, Help & Support, and Log Out**, giving users control over their account and system preferences.

## 4.6.3 Main UI Design of Doctor Side

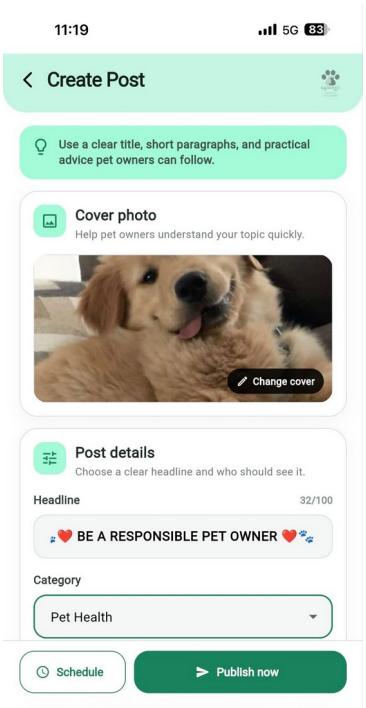
### 4.6.3.1 Dashboard

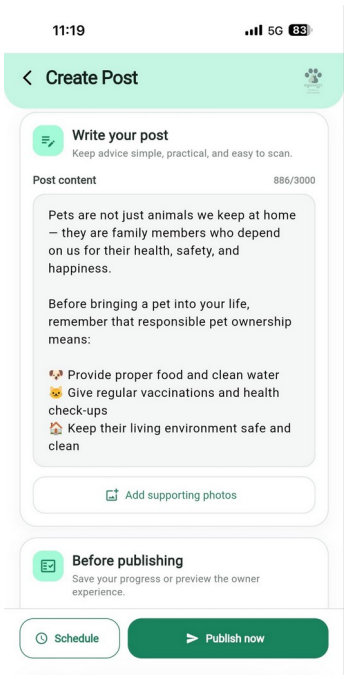


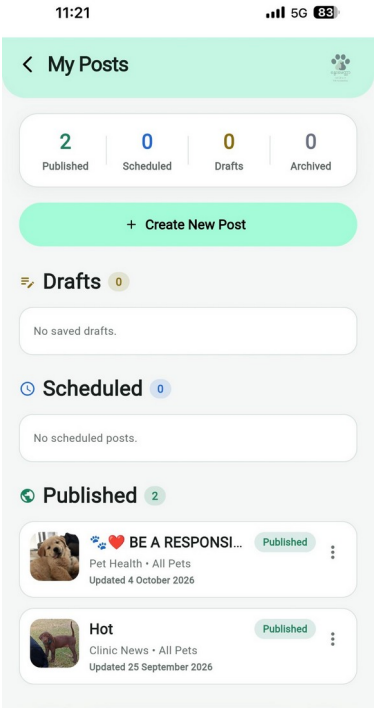

The **Doctor Dashboard** provides veterinarians with an overview of daily clinic activities and important patient information. The dashboard displays the number of **emergency cases, waiting patients, completed cases, and total appointments** for the day. The **Up Next** section highlights upcoming emergency cases and their current status, allowing doctors and staff to respond quickly to urgent situations. The **Quick Menu** provides access to functions

such as creating and managing posts, while the **New Feeds** section allows clinic staff to share pet health advice, clinic news, and other useful information with pet owners.

### 4.6.3.2 Write a Post and Manage Posts

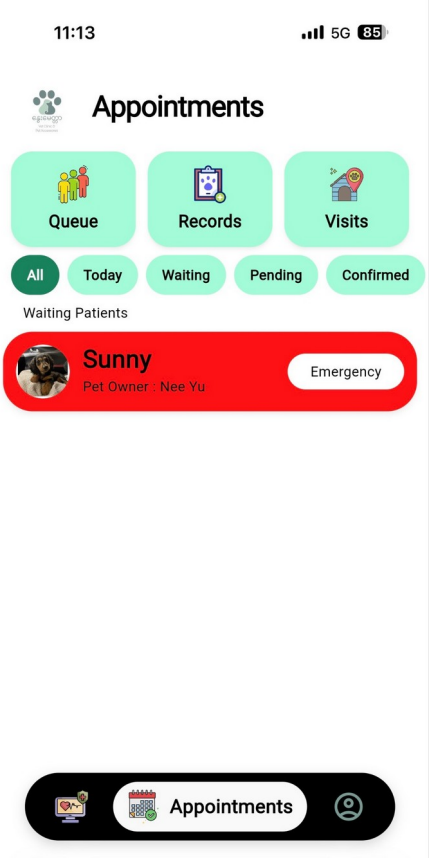




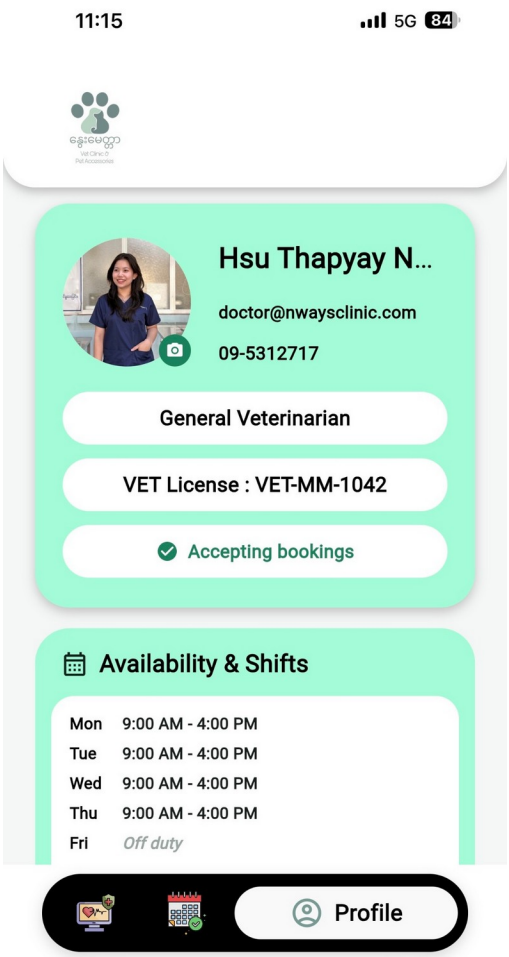

The system also includes a **Post Management** function. Doctors can create educational posts by adding a cover photo, headline, category, and post content. Posts can be published immediately or scheduled for a later time. The **My Posts** section allows staff to manage published, scheduled, draft, and archived posts. This helps the clinic share useful pet-care information, announcements, and health advice with pet owners through the system.

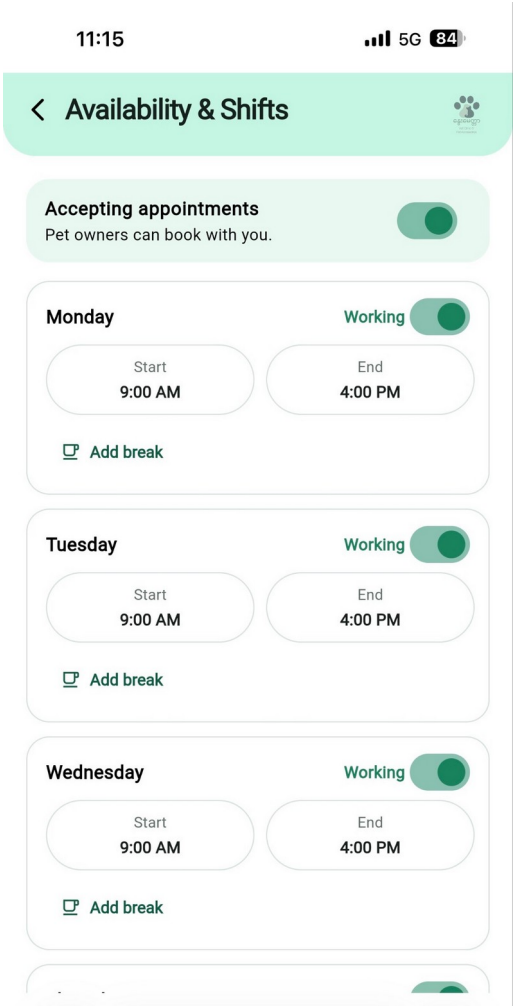
### 4.6.3.3 Appointments




The **Appointments** section allows doctors to manage patient appointments through different filters, including **All, Today, Waiting, Pending, and Confirmed**. It also provides access to **Queue, Records, and Visits**, helping staff organize patient care and appointment information.

## 4.6.3.4 Profile

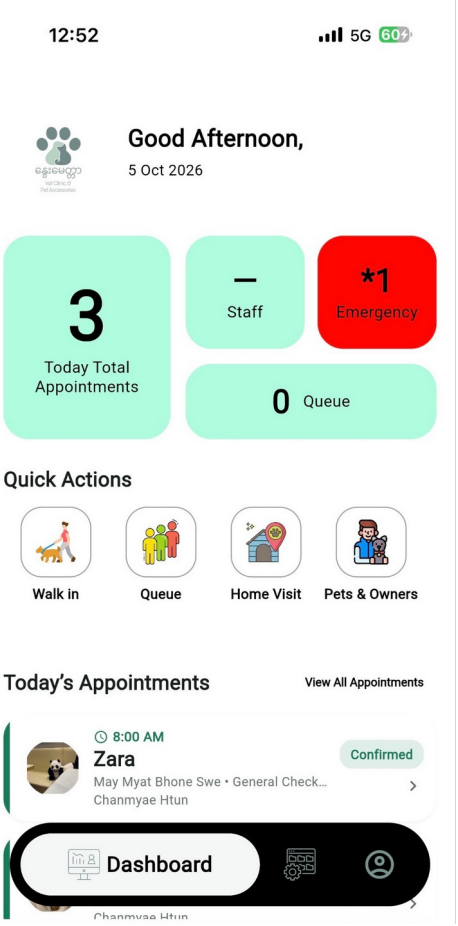


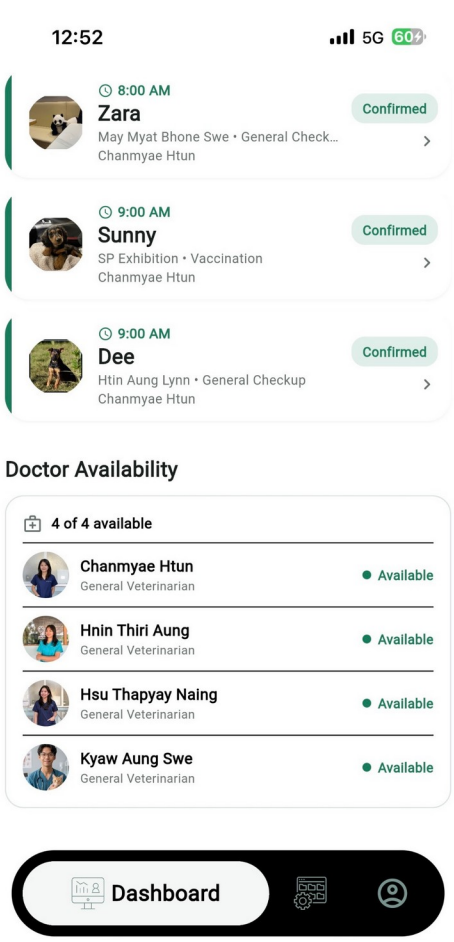



The **Doctor Profile** section displays the veterinarian's professional information, including their specialization, veterinary license, contact details, booking status, and working schedule. Through **Availability & Shifts**, doctors can set whether they are accepting appointments and configure their working hours for each day.

## 4.6.4 Main UI Design of Staff Side

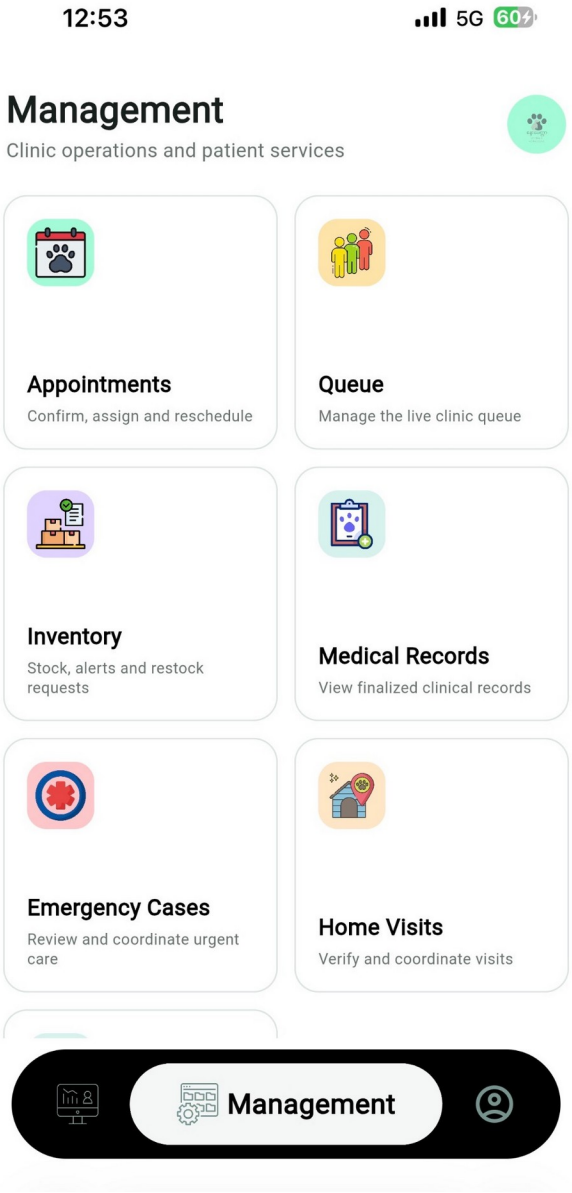
### 4.6.4.1 Dashboard





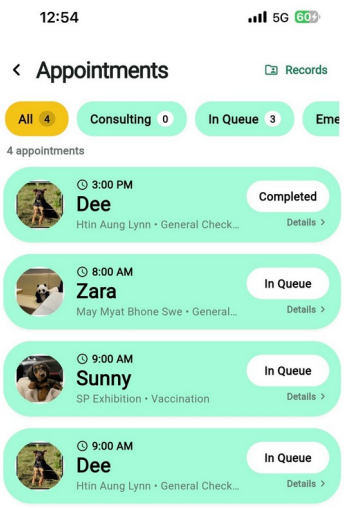

The dashboard provides staff with an overview of daily clinic activities, including appointments, emergency cases, queue status, and doctor availability. Quick actions allow staff to access common functions such as walk-ins, home visits, and pet information.

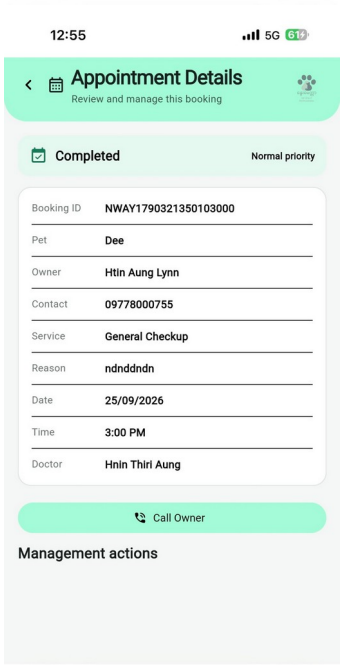
## 4.6.4.2 Management

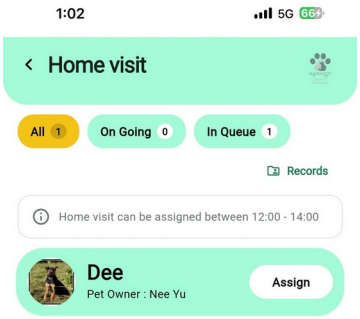


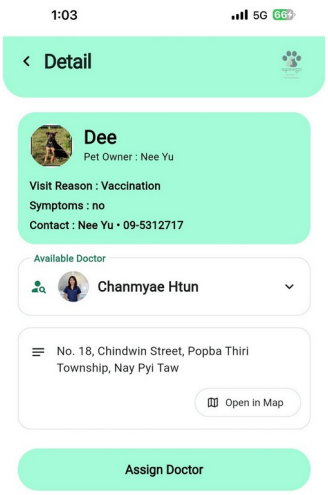

The Management section provides centralized access to appointments, queue management, inventory, medical records, emergency cases, and home visits, helping staff efficiently manage daily clinic operations.

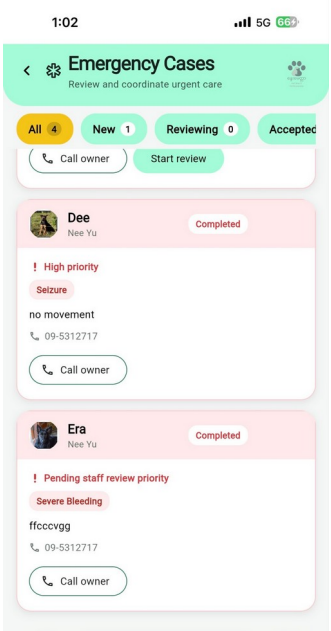
# 4.6.4.3 Appointments, Home Visit & Emergency Cases



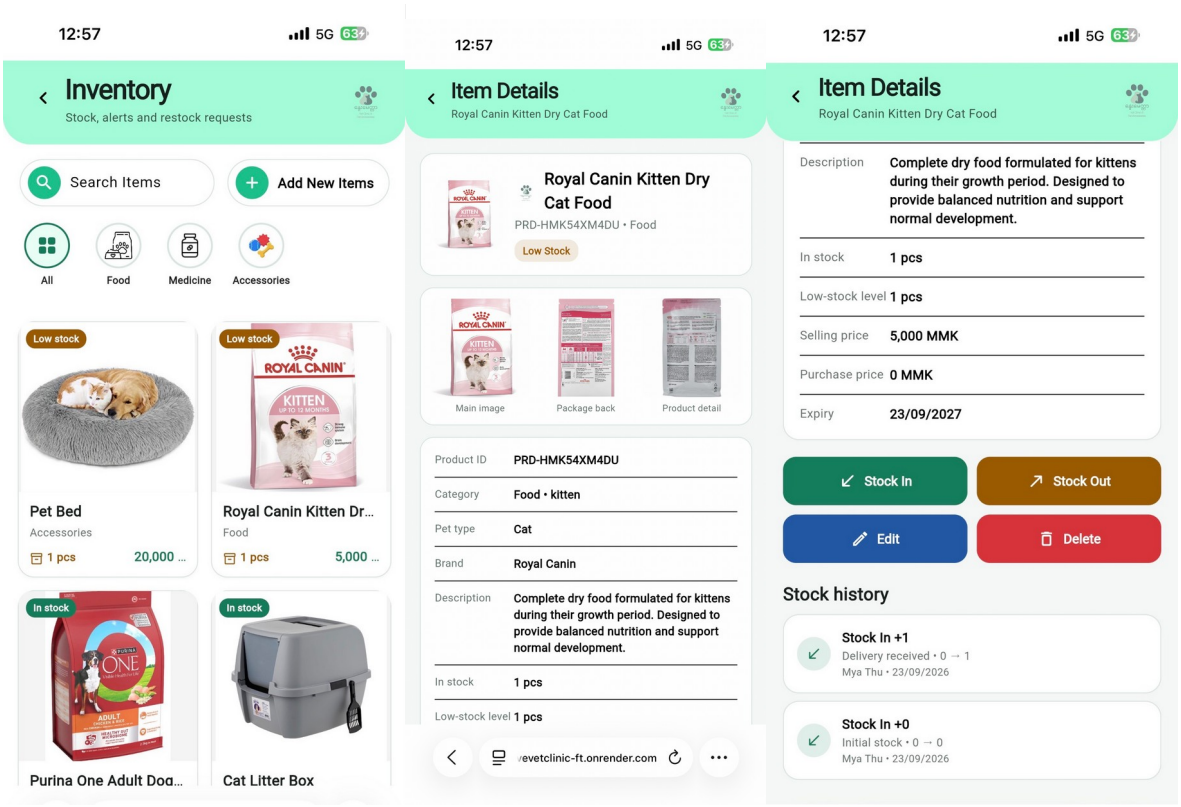

1:02
5G
Home visit
All 1
On Going 0
In Queue 1
Records
Home visit can be assigned between 12:00 - 14:00
Dee
Pet Owner : Nee Yu
Assign

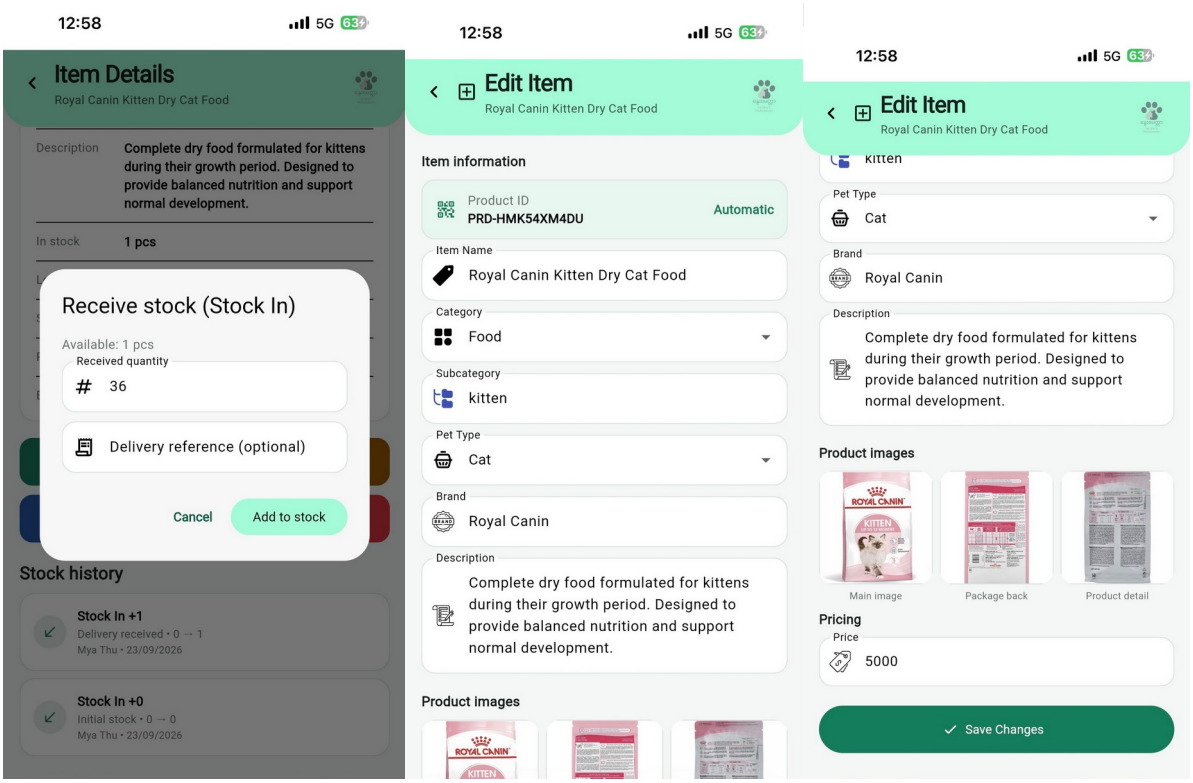

1:03
5G
Detail
Dee
Pet Owner : Nee Yu
Visit Reason : Vaccination
Symptoms : no
Contact : Nee Yu • 09-5312717
Available Doctor
Chanmyae Htun
No. 18, Chindwin Street, Popba Thiri Township, Nay Pyi Taw
Open in Map
Assign Doctor

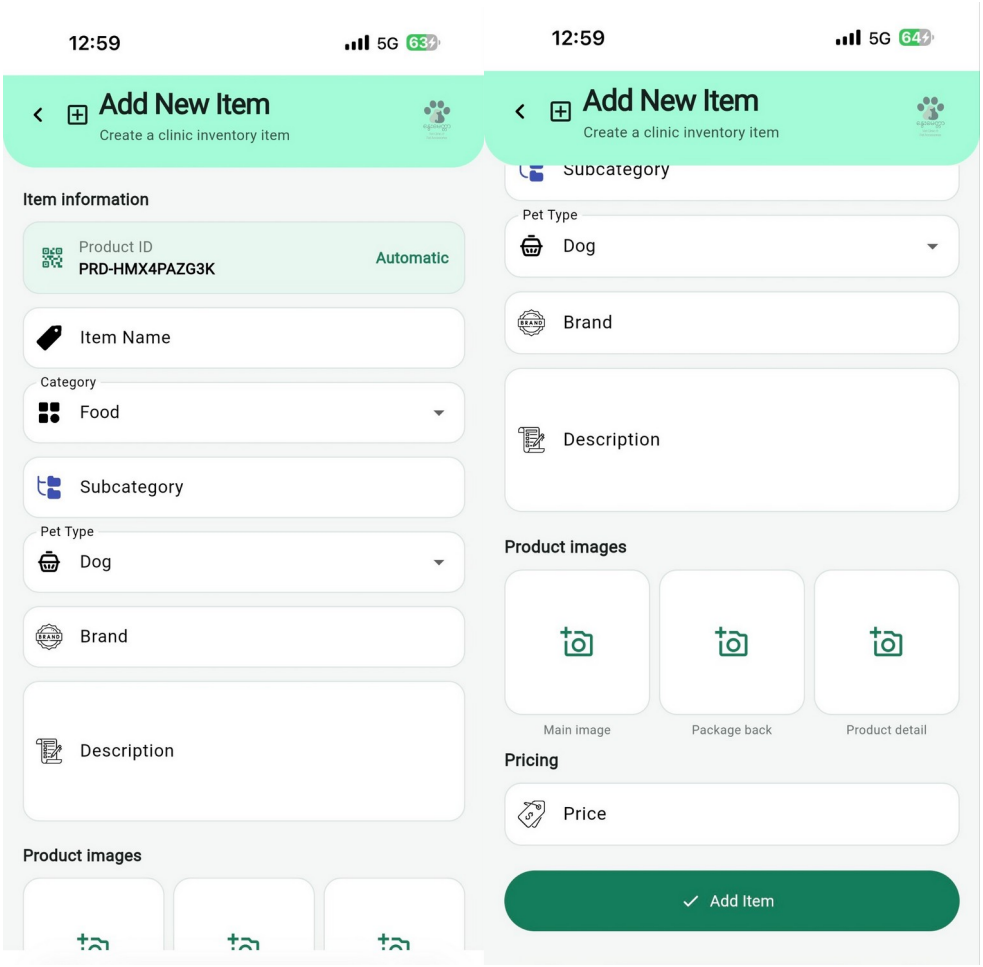



Staff can manage appointments, assign doctors, coordinate home visits, and review emergency cases. Emergency cases are prioritized and clearly highlighted to support faster response and appropriate veterinary care.

# 4.6.4.4 Inventory

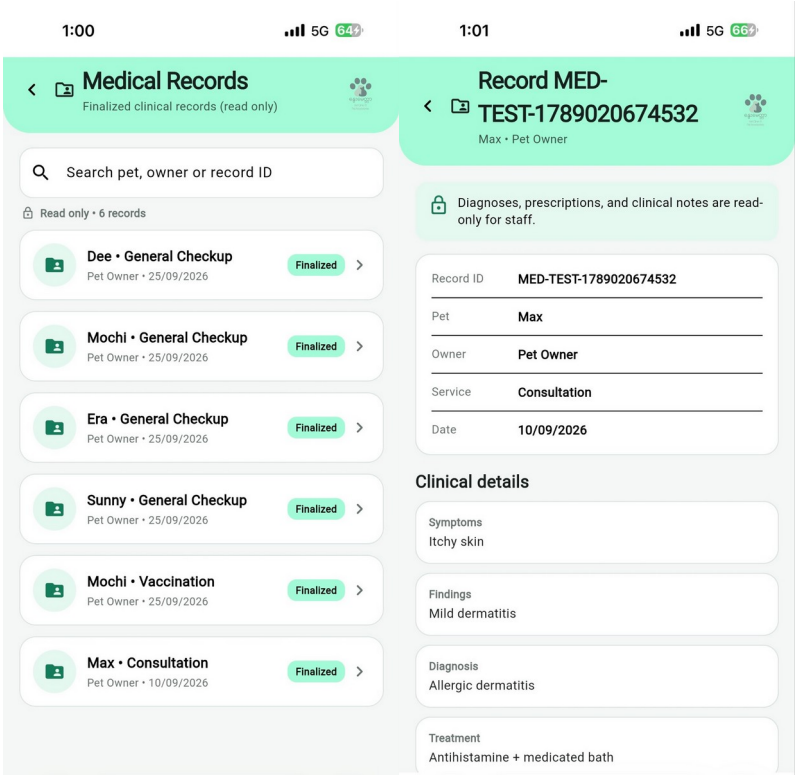





The Inventory module allows staff to add, edit, and monitor products, including food, medicine, and accessories. Staff can manage stock-in and stock-out transactions, view stock history, and identify low-stock items.

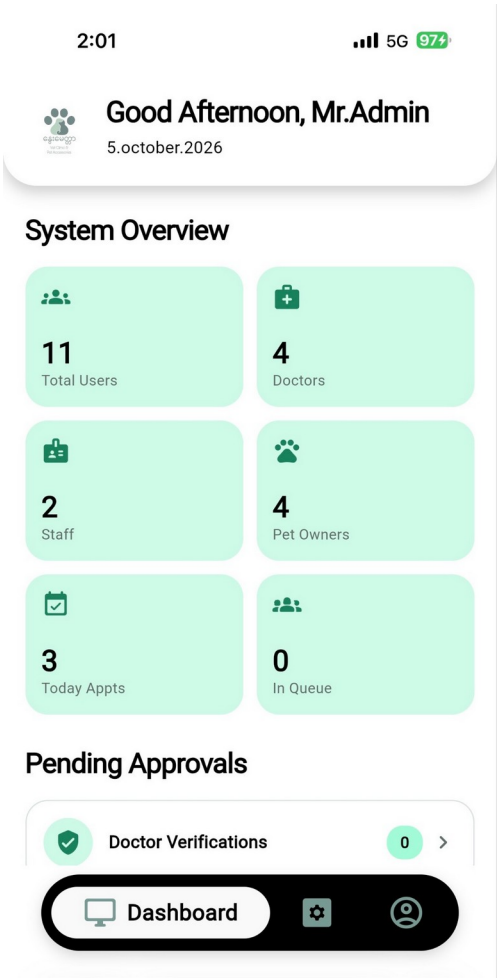
## 4.6.4.5 Medical Records

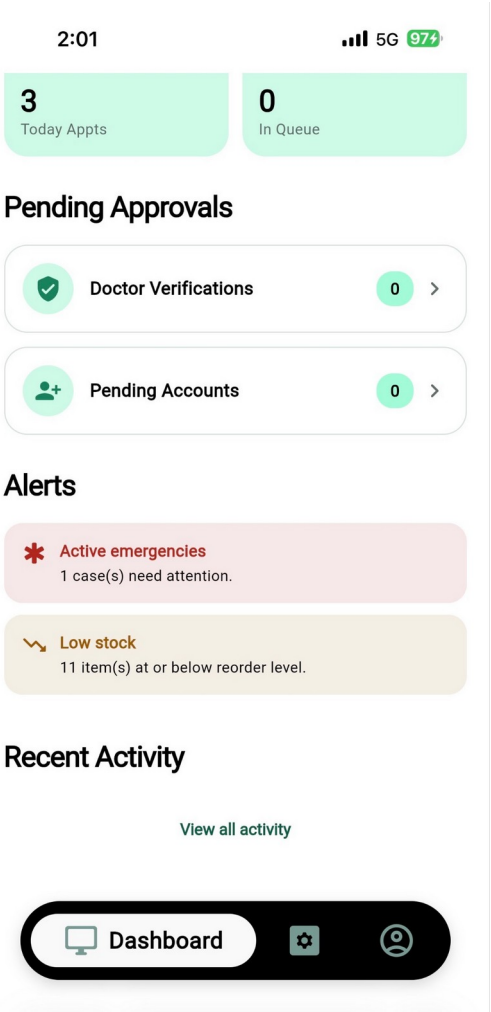



The Medical Records module allows staff to search and view finalized clinical records. Records contain patient information, symptoms, findings, diagnosis, and treatment details and are maintained as read-only information.

## 4.6.5 Main UI Design of Admin Side

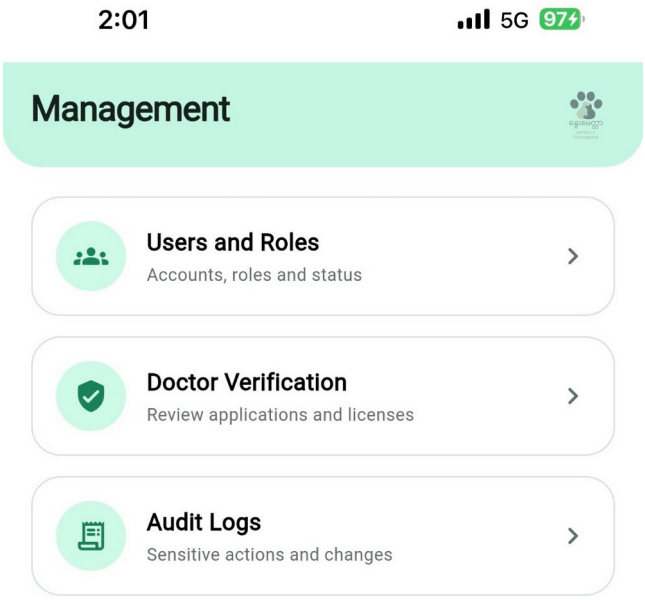
### 4.6.5.1 Dashboard






The Admin Dashboard provides an overview of system activities, including total users, doctors, staff, pet owners, appointments, and queue status. It also displays pending approvals, active emergencies, low-stock alerts, and recent system activity.

### 4.6.5.2 Management

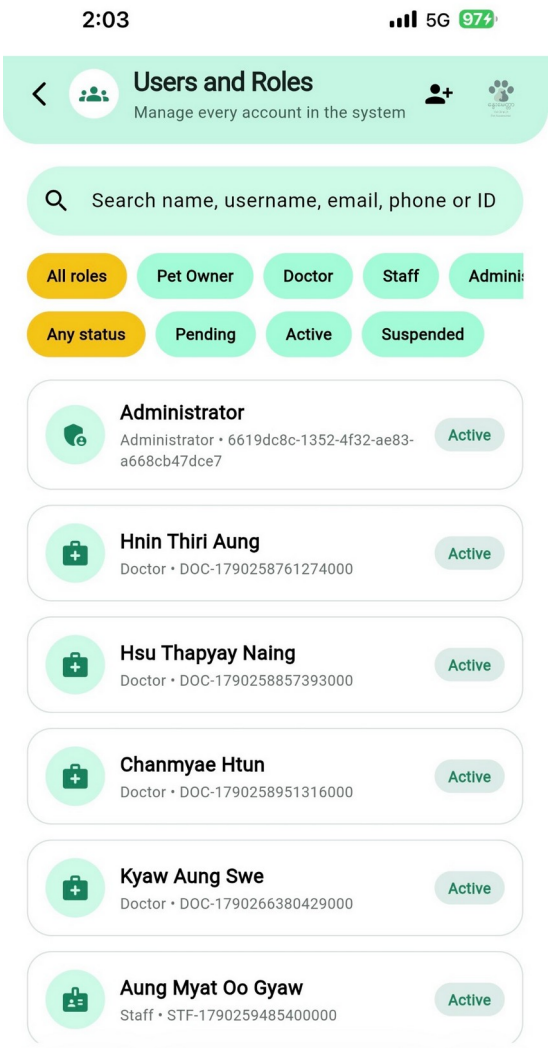


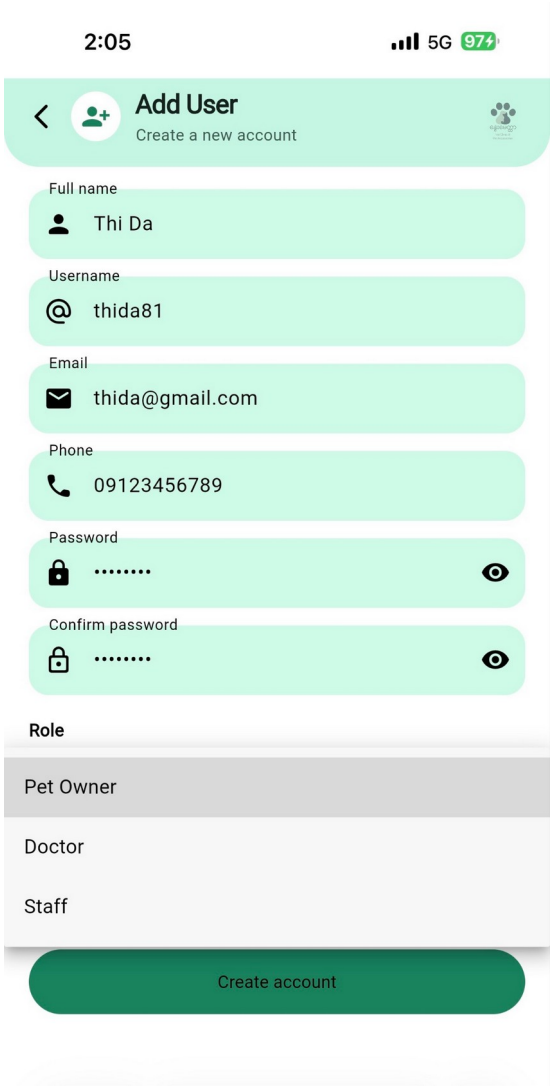




The Management section allows administrators to manage user accounts and roles, verify doctor applications and licenses, and review audit logs. These functions help maintain system security, user access control, and accountability.

## 4.6.5.3 Users and Roles






The Users and Roles module allows administrators to manage system accounts by searching and filtering users based on their roles and account status. Administrators can also create new accounts by entering personal information, login credentials, and assigning roles such as Pet Owner, Doctor, or Staff.

# Chapter 5: Result and System Demonstration

## 5.1 Functionality Overview

The Nway's Love Vet Clinic Management System is designed to provide a centralized platform for managing the main activities of the veterinary clinic. The system supports different functions according to the user's role, including pet owner, veterinarian, clinic staff, and administrator.

The main functions demonstrated in the system include user authentication, pet management, appointment booking, medical records, vaccination tracking, reminders, queue management,

home visits, billing, product sales, inventory management, reporting, and administrative management.

The following sections demonstrate the main views and functions of the system.

## 5.2 Authentication View

The Authentication View provides the login function for users.

Users can enter their account information to access the system. After successful authentication, users are directed to the appropriate interface according to their role.

The authentication function helps protect clinic information by restricting access to authorized users.

## 5.3 Pet Owner Home View

The Pet Owner Home View provides pet owners with access to the main services of the clinic.

The home interface provides access to:

- Pet profiles
- Appointments
- Medical records
- Vaccination information
- Reminders
- Invoices
- Announcements

The dashboard is designed to allow pet owners to access important information without needing to contact clinic staff for every basic request.

## 5.4 Appointment View

The Appointment View allows users to manage veterinary appointments.

Pet owners can select an available date and time and request an appointment based on veterinarian availability. The system can also support appointment cancellation and rescheduling.

Clinic staff can manage appointment information and monitor the daily appointment schedule.

## 5.5 Pet Profile and Medical Record View

The Pet Profile View stores important information about each pet, including:

- Pet name
- Breed
- Age
- Weight
- Owner information
- Medical history

The Medical Record View allows authorized clinic personnel to record and access diagnosis, treatment, and other relevant medical information.

This provides a more organized alternative to maintaining medical information through paper records.

## 5.6 Vaccination and Reminder View

The Vaccination View allows the clinic to record and monitor a pet's vaccination information.

The system can store vaccination details and the next vaccination date. Reminder functions can notify pet owners about upcoming appointments or vaccinations.

This helps reduce the possibility of missed vaccination schedules and supports better follow-up care.

## 5.7 Queue and Emergency View

The Queue View helps the clinic organize patients waiting for consultation.

Normal cases can follow a first-come-first-served order, while emergency cases can receive priority when necessary.

The system provides clinic staff and veterinarians with clearer information about the current queue and appointment status.

## 5.8 Home Visit View

The Home Visit View supports the management of veterinary services provided at the pet owner's location.

Staff can record information related to the pet, owner, service, date, and visit.

Home visits are intended for appropriate treatments and services, while operations are excluded from this service according to the clinic's requirements.

## 5.9 Billing and Payment View

The Billing and Payment View provides digital billing and invoice management.

The system can record charges such as:

- Consultation fees
- Treatment fees
- Product purchases
- Other applicable service charges

A digital invoice provides a clearer record of the customer's transaction and helps clinic staff manage billing more efficiently.

## 5.10 Product and Inventory View

The Product and Inventory View allows clinic staff to manage products sold by the clinic, including medicine, pet food, and accessories.

Product information can include:

- Product name
- Category
- Price
- Stock quantity
- Expiry date

The system can update stock after a sale and provide low-stock information to help staff monitor inventory.

## 5.11 Reporting View

The Reporting View provides organized information about clinic activities.

The system can support reports related to:

- Daily appointments
- Emergency cases
- Vaccination due lists
- Services
- Product sales
- Inventory

## 5.12 Veterinarian View

The Veterinarian View provides veterinarians with functions related to patient care and appointment management.

Veterinarians can:

- View their appointments
- Check pet information
- Access medical records
- Record diagnosis and treatment
- Manage vaccination information
- Handle emergency cases
- View their schedules

The interface is designed to provide veterinarians with the information required during consultations.

## 5.13 Clinic Staff Dashboard

The Clinic Staff Dashboard provides access to the main operational functions required by clinic staff.

Staff can manage:

- Appointments
- Patient queues
- Home visits
- Billing
- Products
- Inventory
- Vaccination records
- Announcements
- Reports

The dashboard brings these functions together so that staff do not need to rely on separate manual records for each activity.

## 5.14 Admin Dashboard

The Admin Dashboard provides administrative control over the system.

The administrator can manage system users, staff access, clinic information, system data, reports, and settings.

The dashboard provides an overview of the system and supports the overall management of the clinic's digital platform.

## 5.15 Expected Benefits

The implementation of the Nway's Love Vet Clinic Management System is expected to provide several benefits to the clinic.

**Operational Benefits**

- Reduces paperwork and manual record keeping.
- Reduces repetitive data entry.
- Makes appointment and queue management more organized.
- Provides faster access to pet medical records.
- Improves vaccination tracking and reminders.
- Provides better inventory visibility.
- Makes billing and invoice management more organized.
- Makes reports easier to retrieve and manage.

**Benefits for Pet Owners**

- Easier appointment management.
- Easier access to pet information.
- Better vaccination reminders.
- Clearer billing information.
- Improved communication with the clinic through announcements.
- More convenient access to clinic services.

**Benefits for Clinic Staff and Veterinarians**

- Easier access to appointment schedules.
- More organized medical information.
- Better coordination between clinic roles.
- Reduced manual administrative work.
- Improved access to inventory and billing information.

Overall, the system is expected to improve the efficiency and organization of Nway's Love Vet Clinic while providing a more convenient service experience for pet owners.

# Chapter 6: Conclusion

## 6.1 Achievements

The Nway's Love Vet Clinic Management System provides a centralized approach to managing the clinic's main activities.

The project addresses the problems associated with manual appointment booking, paper-based medical records, vaccination tracking, queue management, inventory management, billing, and reporting.

The proposed system brings these activities together in one platform and provides different functions for pet owners, veterinarians, clinic staff, and administrators.

The project also establishes a structured system architecture using Flutter, Node.js, and MySQL, with Figma used for the UI/UX design process.

## 6.2 Future Work

Future development can improve the system by adding or expanding several functions.

Potential future improvements include:

- More advanced online appointment booking.
- SMS and push notification services.
- More detailed analytics and dashboards.
- Additional digital payment methods.
- Expanded laboratory and inpatient management.
- More advanced inventory and supplier management.
- More detailed financial reporting.
- Improved data backup and recovery.
- Additional security features.
- Additional communication functions if required by the clinic.

These improvements could further expand the system and provide additional support for the clinic's future operations.

## 6.3 Conclusion

The Nway's Love Vet Clinic Management System is designed to provide a centralized digital solution for managing veterinary clinic operations. It addresses several limitations of manual processes by organizing appointments, pet information, medical records, vaccination tracking, queues, billing, inventory, and reporting within one system.

The system provides different interfaces and functions according to the responsibilities of pet owners, veterinarians, clinic staff, and administrators. By reducing dependence on paper records and repetitive manual processes, the system is expected to improve operational efficiency, information management, and service quality.

Overall, the project provides a foundation for Nway's Love Vet Clinic to move toward a more organized and efficient digital management process while allowing the system to be expanded with additional features in the future.

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